

The block below carries no module guard, so docstrip copies it into EVERY generated file. That is what we want for the TeX files and wrong for the latexmkrc, which is Perl and does not understand lines opening with two percent signs – hence the negated guard around it.

The guard lines themselves are hidden behind a false conditional. docstrip reads a guard line either way, but ltxdoc, which typesets this file, would otherwise set the literal text of the guard as a stray paragraph in the body of the manual.

Nothing in this area may name a TeX conditional, not even in prose: outside the meta-comment at the top of the file, every documentation line here IS LaTeX source and is executed, and a conditional named in passing opens or closes a real one. That is not hypothetical – an earlier draft of this very comment swallowed the rest of the manual.

Three conditional primitives are deliberately absent from the lists below: the ones that close or branch a conditional. `\DoNotIndex` scans its argument with TeX’s own conditional machinery live, so a bare one of them closes the wrong conditional. The manual ended its run with “Too many }’s” and exited with status 1, which failed the docs step of the harness even though the PDF had been written. Leaving them out of the lists means they get indexed, which is harmless. Note that they cannot be named here either, not even inside `...` shorthand split across a line.

The coppe document class

Version 4.0

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Abstract

In this work, it is described the `coppe` document class as well as other files distributed by the COPPETEX project. This class is suitable for writing academic dissertations, thesis and qualifying exams according to the formatting rules of the Alberto Luiz Coimbra Institute for Graduate Studies and Research in Engineering. The minimalist set of macro commands allows its users to concentrate most of their efforts on text composition rather than on the document layout.

1 Introduction

Writing documents in L^AT_EX may be a laborious task when the authors have to prepare their manuscripts rigorously respecting formatting rules imposed by publishers. Regardless of difficulty, a lot of thesis presented to the Coordination of Graduate Studies and Research in Engineering of the Federal University of Rio de Janeiro (COPPE/UFRJ) is typesetted in L^AT_EX. This demand motivated the creation of the COPPETEX project, which tries to facilitate and encourage the use of L^AT_EX within the COPPE/UFRJ scope.

The `coppe` document class is the main product of COPPETEX. It was designed to be clear and succinct. It enables the creation of dissertations, qualifying exams and thesis in a simple and automatic way. The main goal of the `coppe` class is to maintain authors strictly focused on text composition without worrying about margins sizes, line spacing, paper size, vertical and horizontal alignment, etc. The COPPETEX project comprehends also BibTEX and MakeIndex style files for creating lists of references, symbols and abbreviations. Although there aren't official guidelines to write qualifying exams, we provide this option just for convenience, as this exam is a requisite to obtain the DSc degree and, for some of the programs, the MSc degree.

In which follows, it is described the user interface of the `coppe` class. Some details about using the style files cited above are also given. We use the term *thesis* to generally refer to dissertation, qualifying exam, and thesis itself.

2 License

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copying, distributing or modifying the source code, among other acts covered by this license.

To see the full text of the GNU GPL license, go to the `COPYING` file attached to this package.

3 Support

We maintain a WhatsApp group where users can send questions, comments, and bugs to. More details can be found [here](#).

However, as the project maintenance was transferred to a new repository, bug reports, as well as new feature requests, should be directed to <https://github.com/COPPE-UFRJ/CoppeTeX>.

4 User interface

`\frontmatter` A thesis to be approved by the Academic Registry at COPPE/UFRJ must contain three-parts: *front*, *main* and *back* matters (COPPE/UFRJ Academic Registry guidelines). Each one of these parts is started by calling its corresponding macro `\frontmatter`, `\mainmatter` or `\backmatter`. The front matter of a thesis consists of front cover and face, cataloging page, dedication, acknowledgments, abstracts, table of contents, and lists of tables, algorithms, symbols and abbreviations. The main matter is just composed by chapters, while the back matter usually consists of bibliographic references, appendices and index.

You must invoke the `\frontmatter` macro immediately after the `\maketitle` one. The `\mainmatter` command comes right before the first chapter, and `\backmatter` must be typed before the list of references.

Cover (capa)

The ABNT cover suggested in the manual's Annex A. The `\maketitle` command builds it automatically, ahead of the title page: the university, the full COPPE name, and the graduate program (taken from `\department`), followed by the author, the title, and the city and year, all centred and in uppercase. The UFRJ and COPPE logos head the page. The capa is not counted.

Front face

The front face is unnumbered. There, it is not allowed to use hyphenation (COPPE/UFRJ Academic Registry guidelines). It is constructed by calling `\maketitle`. Next, it is described the commands used to enter the information required to create it.

`\author` The `\author` command was redefined. Here, it takes two arguments: the author's first names and surname, e.g., `\author{First Names}{Surname}`. The words should be typed with only first letters in uppercase.

`\title` The macros `\title` and `\foreigntitle` are used to enter the titles of your monograph in Brazilian Portuguese and English respectively (the two built-in languages of CoppeTeX 4.x). For documents whose main language is neither

brazilian nor english – e.g. spanish, french, italian, or a user-supplied language –, also call `\titlein{<lang>}{<text>}` to register the title in that language; the cover and folha de rosto print whichever title matches the main language declared in the class option. See the section “Multilingual support” below for the complete picture. The `babel` package is loaded automatically by `coppe.cls`, so you do not need to load it yourself.

`\advisor` Every COPPE student is coordinated by at least one advisor. M.Sc. and D.Sc. students can have at most 2 and 3 advisors, respectively. Their names must be provided by issuing the command `\advisor` as below:

```
\advisor{Title}{Advisor's Name}{Surname}{Degree}
\advisor{Title}{Second Advisor's Name}{Surname}{Degree}
\advisor{Title}{Third Advisor's Name}{Surname}{Degree}
```

The advisors are not necessarily members of the thesis examination board. Thus, it is required to enter the names of all examiners using the `\examiner` macro. The examiners' names are entered differently:

```
\examiner{Title}{First Examiner's Name Surname}{Degree}
\examiner{Title}{Second Examiner's Name Surname}{Degree}
...
\examiner{Title}{N-th Examiner's Name Surname}{Degree}
```

Remember that all names must be given before calling `\maketitle`.

`\department` The Alberto Luiz Coimbra institute is divided into 12 academic units: Biomedical Engineering (PEB), Civil Engineering (PEC), Electrical Engineering (PEE), Mechanical Engineering (PEM), Metallurgical and Materials Science Engineering (PEMM), Nuclear Engineering (PEN), Ocean Engineering (PENO), Energy Planning (PPE), Production Engineering (PEP), Chemical Engineering (PEQ), Systems Engineering and Computer Science (PESC), and Transportation Engineering (PET). You must specify your department using one of the above abbreviations, e.g., `\department{PEC}`.

`\date` This macro is used to set the month and year of defense. This information is required to create the front face, cataloging details page and abstracts. For example, October 2007 should be entered as `\date{10}{2007}`.

`\keyword` The keywords should describe the concentration areas of your work. You must provide them as follows:

```
\keyword{First Keyword}
\keyword{Second Keyword}
...
\keyword{N-th Keyword}
```

Usually, six words are enough.

Folha adicional: Coleta CAPES

Since August 2026 every thesis and dissertation must carry a *folha adicional* immediately after the folha de rosto, holding the fields the CAPES/MEC collection requires plus the ficha catalogafica (SiBI manual, 9th rev. ed. 2026, 3.1.2.1.2

and Annex H). The sheet is filled in by the graduate programme, and is neither counted nor numbered.

The macros below feed that sheet. They are institutional data, so the sheet is always typeset in Portuguese regardless of the main language of the work, as the COPPE norm requires of institutional identity.

`\areaconcentracao` The area of concentration, e.g. `\areaconcentracao{Engenharia de Sistemas e Computacao}`. It appears on the folha de rosto, on the folha de aprovacao and on the folha adicional.

`\linhapesquisa` The research line, required by the CAPES model of the folha de rosto.

`\tipoproducao` The kind of intellectual output, one of `bibliografica`, `artistica`, `tecnologica` or `tecnica`. Anything else raises a warning.

`\projetovinculado` Whether the work belongs to a research project (`sim` or `nao`) and, if so, its

`\nomeprojeto` name.

`\agenciafomento` A funding agency, given as full name and acronym; repeat once per agency:

```
\agenciafomento{Conselho Nacional de Desenvolvimento
                  Cientifico e Tecnologico}{CNPq}
\agenciafomento{Fundacao Carlos Chagas Filho de Amparo a
                  Pesquisa do Estado do Rio de Janeiro}{FAPERJ}
```

Cataloging details

This page contains cataloging information useful for librarians. Fortunately, it is automatically generated from the data you entered at the time you call `\maketitle`. It is not needed in qualifying exams, though.

Dedication (optional)

`\dedication` This macro was added for convenience. The input text is placed at the right bottom of a blank page. It is emphasized and in normal size.

Abstracts

`abstract (env.)` As stated by the COPPE/UFRJ Academic Registry, abstracts must be in one

`foreignabstract (env.)` page each, with at most 250 words. We recommended that they should be only one paragraph long. They must be defined inside the environments `abstract` and `foreignabstract`.

Lists of symbols and abbreviations (optional)

`\abbrev` The lists of symbols and abbreviations are optional, although highly recommended.

`\syml` It is a good practice to define a symbol/abbreviation in its first occurrence in the text. To define a symbol use `\syml[alphabetic symbol]{Symbol}{Symbol Definition}`, and for abbreviations `\abbrev[alphabetic symbol]{Abbreviation}{Abbreviation Definition}`. These commands are called *dummy*, since they don't output anything at the place they are executed, just an entry in the correspondent list.

`\makeloabbreviations` These lists are lexicographically sorted by using the MakeIndex program, which

`\makelosymbols` is part of any L^AT_EX implementation. For `\syml`, if the optional parameter is

`\printloabbreviations` provided, it will used as sort key. This was later, in 2024, implemented also for

`\printlosymbols`

`\abbrev`, otherwise `Symbol`, or `Abreviation` will be used as sort key, what can result in an undesirable order if it contains $\LaTeX$ commands, mathematical symbols, or mix of uppercase and lowercase. `MakeIndex` needs two commands to create a final sorted list: one which generates a list of entries and the other that indicates the position where the list will be printed out. To generate the lists of symbols and abbreviations, the `coppe` class provides the commands `\makeloabbreviations` and `\makelosymbols`, respectively. They must be called in the document preamble. The commands `\printlosymbols` and `\printloabbreviations` have to be invoked at the point where you want these lists appear, e.g., following the list of tables as showed in the example. Once you call `latex`, it will be created two files with extensions `abx` and `syx`, which contain `MakeIndex` input data. They must be processed with `makeindex` in order to get the lists correctly produced, redirecting the output to files with extension `lab` and `los` respectively:

```
makeindex -s coppe.ist -o example.lab example.abx
makeindex -s coppe.ist -o example.los example.syx
```

Note the `-s` option for specifying the style `coppe.ist`. Now, rerun `latex` twice to get the references solved and you are done.

References

`COPPE $\TeX$` produces citations and the reference list with `biblatex` and the `biber` backend, through its own bibliography style implementing the post-2020 ABNT NBR 6023 rules. The class loads this style automatically: there is no `\bibliographystyle` to set and no `BIB $\TeX$` .bst file to choose (the former `coppe-plain` and `coppe-unsrt` `BIB $\TeX$` styles have been retired).

By default, citations follow the author–date style, e.g., “Chartier (2002)”. Passing the `numbers` class option switches to the numeric style, in which citations are bracketed numbers and the reference list is numbered in order of citation (unsorted).

Declare your bibliography database in the preamble with `\addbibresource` and print the list where it should appear with `\printbibliography`:

```
\addbibresource{example.bib} % in the preamble
...
\printbibliography           % where the list goes
```

Entry types and fields may be written in English (`@book`, `title`, `author`) or with their unaccented Portuguese synonyms (`@livro`, `titulo`, `autor`).

Run in sequence `LaTeX`, `biber`, and twice again `LaTeX` to resolve the citations and references. The style is `natbib` compatible (via the `biblatex natbib=true` option), so you may freely issue `\citet` and `\citep`, as well as the other `natbib` commands.

Reference entry types

Every ABNT document category maps to a `biblatex` entry type, each with an unaccented Portuguese synonym (a worked example of every type is in the sample manuscript’s reference list):

Academic @book (@livro); @article (@artigo); @artigodejornal (newspaper); @incollection (@partedelivro); @inbook (@emlivro); @inreference (@verbete); @proceedings (@anais); @inproceedings (@trabalhodeevento); @thesis (@tese/@dissertacao, with the presets @dissertacaomestrado, @tesedoutorado, @tesephD, @tcc); @report (@relatorio); @periodical (@periodico); @manual; @misc (@diverso); @online.

Special (NBR 6023 §4.2.5–4.2.14) @patent (@patente); @legislation (@legislacao/@atonormat @jurisdiction (@jurisprudencia, with a relator field); @legal (@documentojuridico); @standard (@norma); @music (@partitura); @map (@mapa); @image (@iconografico); @artwork (@obradearte); @video (@audiovisual); @movie (@filme); @letter (@correspondencia); @dataset (@conjuntodedados).

Games and TV @game (@jogo); @boardgame (@jogodetabuleiro); @videogame (@jogoeletronico); @tvshow (@programadetv).

When an entry has no author or editor, it is entered by title, with the first significant word (and a leading article) in capitals.

Appendix and Annex (Optional)

`\appendix` Appendices and annexes are optional chapters that are part of the back matter.

`\annex` The `\appendix` command is a standard L^AT_EX command used before all appendices. COPPET_{EX} introduces the `\annex` command, which should be used only in the back matter (i.e., after the `\backmatter` command) and only after all existing `\appendix` chapters. This restriction is due to implementation constraints.

Therefore, the order for backmatter is:

```
\backmatter
\printbibliography
\appendix
\chapter{An Appendix}
\chapter{Another Appendix}
\annex
\chapter{First Annex}
\chapter{Second Annex}
```

Annex was introduced in v3.5

Final colophon page (`\coppetexfinalpage`)

`\coppetexfinalpage` Optional closing page modelled on the printer's *colophon*: it records, in Portuguese, the tooling and edition used to typeset the thesis. Call it once, after every other piece of content, immediately before `\end{document}`. The macro

- forces a verso (even) page, so in a two-sided print the colophon sits opposite the rear cover;
- types the body at the foot of the page (`\vfill`);
- renders the text in Portuguese unconditionally (Brazilian babel), even when the thesis main language is English, Spanish, French or Italian — the colophon is institutional metadata, not body content.

Three slot macros are user-overrideable. Redefine them with `\renewcommand` before `\coppetexfinalpage` if the document deviates from the defaults:

`\coppefinalengine` the T_EX engine identifier. Default: auto-detected via `iftex` (`\hologo{pdfLaTeX}`, `\hologo{LuaLaTeX}` or `\hologo{XeLaTeX}`).

`\coppefinalfont` the text body font. Default: *Latin Modern* (`\lmodern`), which is what the class loads. Redefine if you loaded `mathptmx`, an OpenType math font, etc.

`\coppefinalmanual` the reference manual title (typeset in italics). Default: the current COPPE/UFRJ norm. Redefine if your work follows an earlier edition.

The macro also prints the COPPE_TE_X class version (captured into `\coppe@version` at class-load time, so it survives any later package that may redefine the global `\fileversion`) and the GitHub repository URL, <https://github.com/COPPE-UFRJ/CoppeTeX>.

5 Class options

There are some options users can specify in order to customize the appearance of the output produced by the `coppe` class. These options can be passed to `coppe` as follows: `\documentclass[option1, option2]{coppe}`. In which follows, we give a brief description of all supported options.

dsc, msc, dsceexam, msceexam The `coppe` class is able to produce thesis, dissertations, and qualifying exams, which are enabled by the `dsc`, `msc`, `msceexam`, and `dsceexam` options, respectively.

doublespacing The default line spacing is one-and-a-half. For enabling double spacing between lines, use the `doublespacing` option.

numbers The default citation style is the author–date scheme. For numbered citations, specify the `numbers` option to the `coppe` class; the reference list is then numbered in order of citation (unsorted). The matching `biblatex` style is selected automatically in either case.

brazilian, english, spanish, french, italian Pick the document’s *main* language. The class loads Babel with that language as the main one (the last in the Babel option list); inside the document `\selectlanguage{...}` and `\begin{otherlanguage}{...}` keep working as usual. The default, when no language option is given, is **brazilian**.

- **brazilian** and **english** are *built-in*: their strings ship inside `coppe.cls` and need no extra file.
- **spanish, french, italian** are shipped as *language packs*: two files per language (`coppe-lang-<lang>.def` and `<lang>-coppe.lbx`) that must sit next to `coppe.cls` (or in your project directory). They are auto-loaded by the class. Picking one of them as the main language without the matching pack files installed raises a clear class error.

- Any other Babel-known language can be plugged in by writing the same two files and either passing the language name as a class option or calling `\usecoppelanguage{<lang>}` in the preamble.

See the section “Multilingual support” below for the three-slot architecture, the `\titlein / \brazilianabstract` API, and how to create a new language pack.

pdfa Produce the deposited file as **PDF/A-2b**, which 2.2(d) of the manual requires of the digital version. The class loads `pdfx` at level `a-2b` and writes the XMP metadata from the document’s own `\title`, `\author`, `\keyword` and `\department`. Level `a-2b` rather than `a-1b`: `a-1b` forbids transparency, which `tcolorbox` and several graphics packages produce. The metadata reaches the PDF on the SECOND run, because `pdfx` reads it from a file the class writes at the end of the first one – which is no burden, since three runs are needed for cross-references anyway.

assinaturas Give every member of the board a signature rule of its own on the folha de aprovacao, as Annex D of the manual shows, instead of the compact list. The spacing adapts to the size of the board.

coorientador List the coadvisors on the three abstract pages, under the advisors, in the same shape the folha de rosto uses. Off by default: the abstract page listing only the advisors is a COPPE tradition, and 3.1.2.1.4 does not require either form.

rascunhoficha While writing, fill the folha adicional with a ficha catalografica typeset by the class itself, so the page is not an empty frame. **Never valid for deposit:** the real ficha comes from the SiBI generator or the Programa’s library, and goes in with `\fichacatalografica`.

twoside, oneside The 2026 edition dropped the verso margins and the deposit is digital, so the class sets one-sided pages with a 3 cm left margin on every folha. **twoside** restores the mirrored layout for anyone who is going to PRINT and bind the work; **oneside** is the default and is there to be explicit.

semmorewrites, morewrites T_EX has sixteen output streams and a work that asks for every list runs out of them; the failure, “No room for a new `\write`”, lands on whichever list came last and explains nothing. Under pdfT_EX the class therefore loads the `morewrites` package by itself when it is installed, which removes the ceiling. **semmorewrites** turns that off, for the rare case where the package disagrees with something else in the document – it buffers what is written until shipout, so a package that writes a file and reads it back in the same run can misbehave. **morewrites** means “I insist”: a missing package becomes an error instead of a warning. Under LuaLaT_EX, which has 128 streams, none of this is loaded and none of it is needed.

listasnosumario Put the pre-textual lists – figures, tables, quadros, programas, abreviaturas, simbolos – back into the sumario. **Contrary to 3.1.2.1.6**, which tells the author to follow the manual’s own sumario, and that one opens at “1 INTRODUCAO”: the lists are pre-textual elements and come BEFORE the sumario, so listing them inside it is listing what the reader

has already passed. The option exists because every CoppeTeX document written before this release had them there, and a work in its final weeks should not have its sumario change under it.

numeraisromanos Number the pre-textual folhas in lowercase roman numerals.

Contrary to 2.7, which has the count running from the folha de rosto with nothing printed until the textual part. Kept only so that documents written under the older norm still compile; do not use it for a new work.

5.1 Changing document identification

\freeconfig The user could *optionally* use the command **freeconfig** to modify the parameters that print the document identification. The command **freeconfig** needs all those parameters, which are degree initials, degree name, title, foreign title, local doctype, and foreign doctype as in the following example:

```
\freeconfig{Dr.}{Philosophiae Doctor}{PhD}{Doutor}{Dissertation}{Tese}
```

5.2 Multilingual support

CoppeTeX 4.x reserves three fixed language slots:

main the language of the body, captions, TOC labels and the **abstract** environment. Picked by the class option (**brazilian** by default; one of **english**, **spanish**, **french**, **italian** also accepted out of the box; any other Babel language possible with a user-supplied pack).

foreign the language of the **foreignabstract** environment. **english** by default for every main language; switches automatically to **brazilian** when **main = english**.

third optional. Loaded on demand with **\usecoppelanguage{<lang>}** in the preamble, useful when the body quotes or cites in a third language (e.g. a Spanish-main thesis citing a French source). Does not have a class option of its own.

\titlein
brazilianabstract (env.)
\usecoppelanguage

Two pieces of user-facing API come with the multilingual model:

- **\titlein{<lang>}{<text>}** records the title in any language. **\title{...}** still writes the Brazilian-Portuguese title and **\foreigntitle{...}** still writes the English one (back-compat); for a main language outside the **brazilian/english** pair, also call **\titlein{<lang>}{...}**. The cover and folha de rosto print whichever title matches **\coppe@mainlang**.
- The **brazilianabstract** environment is the optional third abstract used when the main language is neither **brazilian** nor **english** – typically a Spanish/French/Italian thesis that also wants a Portuguese resumo for UFRJ examiners. It wraps its body in **\begin{otherlanguage}{brazilian}** and uses the Portuguese template strings. A document with **main = brazilian** can simply keep using **abstract**; **brazilianabstract** is redundant then.
- **\usecoppelanguage{<lang>}** loads an extra language pack in the preamble without making it the main or foreign language.

Institutional names stay Portuguese

The cover, folha de rosto and ficha catalográfica are a Brazilian institutional template (NBR 14724 + COPPE manual). The strings `universityname`, `cityname`, `statename`, `countryname`, the departments declared with `\department` and the surrounding “apresentada/submetida ao...” wording are deliberately kept in Portuguese for every main language. Only the title (printed via `\coppe@selecttitle`) follows the main language.

Writing a new language pack

A language pack is just two files dropped next to `coppe.cls` (or anywhere on the `TEXINPUTS` path):

`coppe-lang-<lang>.def` class-level strings. Populate the dispatcher table with `\copperdefstring{<lang>}{<key>}{<value>}` for every key already populated for `brazilian/english` in `coppe.cls` (`appendix`, `annex`, `frame`, `listframe`, `source`, `program`, `listprogram`, `references`, `in:`, `art`, `platform`, `directedby`, `producedby`, `listabbreviation`, `listsymbol`, `glossary`, `advisor`, `advisors`, `dept`, `approvedmale`, `approvedfemale`, `universityname`, `cityname`, `statename`, `countryname`, `monthname1...monthname12`, `algorithm2eopt`, `babelname`). Finish with

```
\AtBeginDocument{\DeclareLanguageMapping{<lang>}{<lang>-coppe}}
```

so the `biblatex` mapping is registered after the bibliography engine is loaded by `coppe.cls` (which happens *after* the language pack is read).

`<lang>-coppe.lbx` `biblatex` localization strings. `\InheritBibliographyExtras{<lang>}` then `\DeclareBibliographyStrings` for the keys `availablefrom`, `urlseen`, `in`, `editor`, `editors`, `depositor`, `coppescale`, `mscdiss`, `dscthesis`, `phdthesis`, `tcc`. The shipped `spanish-coppe.lbx`, `french-coppe.lbx` and `italian-coppe.lbx` are the reference templates to copy from.

Tip: the `algorithm2eopt` key holds whatever `algorithm2e` calls the language. Mind the spelling: `algorithm2e` ships `portuguese`, `english`, `spanish`, `french` and `italiano` – not `italian`.

Example: a Spanish-main thesis

```
\documentclass[spanish,dsc]{coppe}
\addbibresource{example.bib}
\begin{document}
  \title{T'\i tulo em portugu\~es} % Brazilian abstract slot
  \foreigntitle{English title} % English abstract slot
  \titlein{spanish}{T'\i tulo en espa\~nol} % cover/folha
  \author{Juan}{P\~erez}
  \advisor{Prof.}{Mar'\i a}{Garc'\i a}{D.Sc.}
  \department{PESC}\date{01}{2026}
  \keyword{multilenguaje}
  \maketitle
  \frontmatter
```

```

\begin{abstract}          Resumen principal en espa\~nol.      \end{abstract}
\begin{foreignabstract} English foreign abstract.              \end{foreignabstract}
\begin{brazilianabstract} Resumo em portugu\~es para a banca. \end{brazilianabstract}
\tableofcontents
\mainmatter
\chapter{Introducci\~on} Cuerpo en espa\~nol\dots
\end{document}

```

6 Files in your project folder

In a normal install where CoppeTeX is part of your `TEXINPUTS` path (e.g. inside the local `texmf` tree of MiKTeX or TeX Live), you do not need to think about this list at all — the files travel with the class. This section is for two situations where you do:

- shipping the class together with your manuscript (a self-contained submission package); or
- installing CoppeTeX by hand, dropping the files next to `coppe.cls` in the project directory.

Always required

These files must be findable at every compilation, regardless of the main language or any class option:

`coppe.cls` the document class itself.

`coppe.dbx` custom `biblatex` datamodel (declares the ABNT-specific entry types `@game`, `@videogame`, `@boardgame`, `@tvshow` and `@map`, plus a few custom fields).

`coppe.bbx` and `coppe.cbx` `biblatex` bibliography and citation styles for the author–date scheme (the default).

`brazilian-coppe.lbx` and `english-coppe.lbx` `biblatex` localisation tables for Portuguese and English. Both are required *even when neither language is the main one*, because `coppe.bbx` registers

```

\DeclareLanguageMapping{brazilian}{brazilian-coppe}
\DeclareLanguageMapping{english}{english-coppe}

```

unconditionally, so `biblatex` tries to read both LBX files at every compilation.

Conditional on class options

`coppe-numeric.bbx` and `coppe-numeric.cbx` only when the `numbers` class option is given (`\documentclass[numbers]{coppe}`). Without it, `biblatex` never tries to read them.

`coppe.ist` only when the glossary, list-of-symbols, or list-of-abbreviations features are used. It is the `makeindex` style file invoked by the `makeindex -s coppe.ist ...` step described in subsection *Lists of symbols and abbreviations* above.

Conditional on the main language

The main language is chosen by the matching class option (`brazilian`, `english`, `spanish`, `french`, `italian`). The two built-in languages ship their strings inside `coppe.cls` and so need no extra file. The three language packs add two files each — one class-level (`coppe-lang-<lang>.def`) and one `biblatex`-level (`<lang>-coppe.lbx`):

Main language	Extra files needed
<code>brazilian</code> (default)	— (built into <code>coppe.cls</code>)
<code>english</code>	— (built into <code>coppe.cls</code>)
<code>spanish</code>	<code>coppe-lang-spanish.def</code> , <code>spanish-coppe.lbx</code>
<code>french</code>	<code>coppe-lang-french.def</code> , <code>french-coppe.lbx</code>
<code>italian</code>	<code>coppe-lang-italian.def</code> , <code>italian-coppe.lbx</code>

Loading a third language at preamble time via `\usecoppelanguage{<lang>}` brings in the same pair of files for that language. A user-supplied language pack follows the same scheme: drop `coppe-lang-<lang>.def` and `<lang>-coppe.lbx` next to `coppe.cls`.

Cover assets

The institutional title page produced by `\maketitle` needs two PDF logo files at compile time:

`ufrj-logo.pdf` and `coppe-logo.pdf` the UFRJ shield and the COPPE/UFRJ wordmark. Both ship with the CoppeTeX distribution; they must be findable from the project directory (or on `TEXINPUTS`).

7 Required L^AT_EX packages

The class is built on the standard `book` class (`12pt`, `a4paper`, `oneside`) and brings in the stock CTAN packages listed below with `\RequirePackage`. On a full T_EX Live or MiK_TE_X install they are all present already, so this list matters only for a minimal install — or when you simply want to know which commands are already in scope. You do *not* need to `\usepackage` any of these yourself; every command and environment they provide is available in your manuscript. Options in brackets are the ones the class passes.

Engine and encoding

`iftex` detects the running engine, so the class can branch its setup between `pdfTEX` and the Unicode engines (LuaT_EX/XeT_EX).

`inputenc (utf8)` UTF-8 input; loaded *only under pdfT_EX* (the Unicode engines read UTF-8 natively).

`fontenc (T1)` T1 font encoding, for correctly composed accented glyphs and hyphenation.

Mathematics and symbols

`amsmath` the AMS mathematics environments.

`amssymb` the AMS symbol fonts; loaded *only under pdfTeX*. Under a Unicode engine load `unicode-math` (or `amssymb`) yourself — the class warns at end of document if neither is present.

Page layout and spacing

`geometry` A4 one-sided margins (3/3/2/2 cm) and header height.

`fancyhdr` running headers and footers.

`titlesec` chapter and section title formatting.

`setspace` one-and-a-half line spacing (double with the `doublespacing` option).

Lists, tables and rules

`enumitem` customisable list spacing and labels.

`tabularx` tables with automatic-width columns.

`booktabs` professional horizontal rules (`\toprule`, `\midrule`, `\bottomrule`).

`longtable` tables that break across pages.

Floats, graphics and captions

`graphicx` image inclusion.

`float` float placement control (e.g. the `[H]` specifier).

`caption` caption typography (the ABNT caption-on-top layout).

`subcaption` subfigures and subtables.

Code, verbatim and algorithms

`listings` source-code listings.

`fancyvrb` extended verbatim.

`algorithm2e` (`linesnumbered`, `lined`, `algochapter`, `ruled`) typeset algorithms; the `natural-language` option tracks the main language.

Bibliography, quotations and language

`babel` multilingual typesetting; the language options are computed from the main and foreign language slots.

`csquotes` context-sensitive quotation marks (the recommended companion of `biblatex`).

`biblatex` (`backend=biber`) the bibliography engine and the ABNT `coppe` styles (`coppe` or, with the `numbers` option, `coppe-numeric`). The backend is `biber`, not `bibTeX`.

Hyperlinks

`hyperref` clickable cross-references, PDF bookmarks and document metadata.

Low-level utilities

`etoolbox` e-TeX programming toolkit.

`ltxcmds` low-level command helpers (`\ltx@ifpackageloaded, \dots`).

`ifthen` the `\ifthenelse` conditional.

`hyphenat` hyphenation control (`\nohyphens`, used on the cover).

`lastpage` `\pageref{LastPage}` for the cataloging page count.

`hologo` TeX-family logos (`\hologo{LaTeX}, \dots`).

`xcolor` colour support (used by `listings` and `hyperref`).

8 Quick, useful tips

Pictures. The default picture format of L^AT_EX is the Encapsulated PostScript (EPS). If you use pdfL^AT_EX, the default format becomes the PDF, but you can equally load PNG files. For such, you must enter the name of your image file without extension, e.g., `\includegraphics{filename}`, and pdfL^AT_EX will firstly look for a file called `filename.pdf` and after for file `filename.png`. For producing high quality pictures with embedded fonts we recommend the Ipe drawing software available [here](#).

Fonts. The default font in L^AT_EX is the Computer Modern. If you would like to try its enhanced version, consider using the `lmodern` package. To use Times, it is recommended to load the package `mathptmx`, rather than the deprecated `times`. There is also an enhanced Times version available with the `tgtermes` package. You can still use the Arial font face with the package `uarial`.

Hyperref. When working with PDF's, there is the possibility to add extra information to the file as the author's name, document title, subject, keywords, etc. This is easily done with the `hyperref` package, which is loaded by the `coppe` class itself (since v4.0): the user must *not* load it again in the manuscript preamble. Author, title, subject and keyword metadata are filled in automatically by `\maketitle`.

Math (amssymb vs. unicode-math). The class always loads `amsmath`. The companion AMS symbol package `amssymb`, however, is loaded *only* under pdfL^AT_EX: it is a legacy 8-bit package built around the `msam/msbm` Type 1 fonts and clashes with `unicode-math`, the modern OpenType-math package used on Unicode engines (LuaL^AT_EX, XeL^AT_EX). On those engines the class skips `amssymb` and schedules an *end-of-document* check: if by then the author has loaded neither `unicode-math`

nor `amssymb` (which they could have loaded by hand, though it is discouraged), a `ClassWarning` reminds them to add

```
\usepackage{unicode-math}
\setmathfont{Latin Modern Math} % or STIX Two Math, ...
```

to the manuscript preamble. The class deliberately does *not* auto-load `unicode-math`, because picking an OpenType math font is a typographic choice that belongs to the author. **Never load `amssymb` together with `unicode-math`:** it would re-establish the 8-bit AMS fonts on top of the OpenType math font and split symbols across two incompatible encodings.

Printing. To get your work correctly printed, you must ensure that any page scaling option (e.g., fit or shrink to printable area) isn't enabled. This kind of option often comes in print dialogs of document visualization softwares.

longquote Quotation A quotation longer than three lines is INDENTED BY 4 cm from the left margin – not set with a 4 cm left margin, which is a different and much narrower thing. 2.2(b) of the manual says “recuo de 4 cm da margem esquerda”, so on an A4 page with a 3 cm margin the quoted text starts 7 cm from the edge of the paper. It carries no quotation marks and is set in the smaller font, single spaced. The `coppe` class provides the `longquote` environment, which does all of that; the indentation is `\recuolongquote`, should a Programa ever need to change it.

9 A simple example

```
1 \*example>
2 \documentclass[dsc]{coppe}
3
4 \usepackage{rotating}           % rodando coisas, como tabelas
5 \usepackage[most]{tcolorbox} % caixas de texto
6 \usepackage{tikz}              % gráficos vetoriais (o tcolorbox já o traz)
7 % Sob LuaLaTeX/XeLaTeX, descomente as duas linhas a seguir para obter os
8 % símbolos AMS (\mathbb, \hbar, \varnothing, ...) via unicode-math. Sob
9 % pdfLaTeX, a classe já carrega amssymb automaticamente; deixe comentado.
10 %\usepackage{unicode-math}
11 %\setmathfont{Latin Modern Math} % ou STIX Two Math, TeX Gyre Termes Math...
12 \DeclareMathOperator{\RMSE}{RMSE} % operadores próprios (cap. de matemática)
13 \DeclareMathOperator{\diag}{diag}
14 % A classe coppe já carrega: amsmath, booktabs, longtable, subcaption,
15 % hyperref, biblatex, listings, fancyvrb, algorithm2e -- não recarregue esses
16 % pacotes aqui (em particular, hyperref deve ficar a cargo da classe para que
17 % a ordem com setspace e biblatex permaneça correta).
18
19 \addbibresource{example.bib}% base de referências (biblatex/biber)
20
21 \makelossymbols
22 \makeloabbreviations
23 % Algumas coisas opcionais que não cabem no coppetex.sty
24 \renewcommand{\familydefault}{\sfdefault} % para usar fontes SEM serifa no default
25 \renewcommand{\familydefault}{\rmdefault} % para usar fontes COM serifa no default
```



```

26
27
28 \begin{document}
29   \title{Titulo da Tese}
30   \foreigntitle{Thesis Title}
31   \author{Nome do Autor}{Sobrenome}
32   \advisor[UFRJ]{Prof.}{Nome do Primeiro Orientador}{Sobrenome}{D.Sc.}
33   \advisor[UFRJ]{Prof.}{Nome do Segundo Orientador}{Sobrenome}{Ph.D.}
34   % A classe aceita de um a tres \advisor; se houver apenas um, o rotulo na
35   % capa e na folha de rosto sai como "Orientador" (singular), com dois ou
36   % mais sai como "Orientadores" (plural). Para o caso 1/3 orientadores,
37   % veja tests/test_brazilian_one_advisor.tex e tests/test_brazilian_three_advisors.tex.
38
39   \examiner[UFRJ]{Prof.}{Nome do Primeiro Examinador Sobrenome}{D.Sc.}
40   \examiner[UFRJ]{Prof.}{Nome do Segundo Examinador Sobrenome}{Ph.D.}
41   \examiner[UFF]{Prof.}{Nome do Terceiro Examinador Sobrenome}{D.Sc.}
42   \examiner[UNICAMP]{Prof.}{Nome do Quarto Examinador Sobrenome}{Ph.D.}
43   \examiner[USP]{Prof.}{Nome do Quinto Examinador Sobrenome}{Ph.D.}
44   \department{PESC}
45   \date{06}{2026}
46
47   \keyword{Primeira palavra-chave}
48   \keyword{Segunda palavra-chave}
49   \keyword{Terceira palavra-chave}
50
51   % Dados da folha adicional (Coleta CAPES), obrigatoria desde agosto de 2026.
52   % A folha e preenchida pelo Programa; estes campos apenas a compoem.
53   \areaconcentracao{Engenharia de Sistemas e Computa\c c\~ao}
54   \linhapesquisa{Engenharia de Dados e Conhecimento}
55   \tipoproducao{bibliografica} % bibliografica | artistica | tecnologica | tecnica
56   \projetovinculado{sim} % sim | nao
57   \nomeprojeto{Nome do Projeto de Pesquisa ao qual o trabalho se vincula}
58   \agenciafomento{Conselho Nacional de Desenvolvimento Cientifico e
59   Tecnologico}{CNPq}
60   \agenciafomento{Fundacao Carlos Chagas Filho de Amparo a Pesquisa do
61   Estado do Rio de Janeiro}{FAPERJ}
62   % A ficha oficial vem de fichacatalografica.sibi.ufrj.br; tendo o arquivo,
63   % descomente a linha abaixo e a moldura vazia da folha adicional some.
64   % \fichacatalografica{ficha.pdf}
65
66   \maketitle
67
68   \frontmatter
69   \dedication{A alguém cujo valor é digno desta dedicatória.}
70
71   \chapter*{Agradecimentos}
72
73   Gostaria de agradecer a todos os que contribu\iram para este trabalho:
74   orientadores, colegas, familiares e \as ag\~encias de fomento. Lorem ipsum
75   dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt
76   ut labore et dolore magna aliqua.
77
78   \begin{abstract}
79

```

```

80 Apresenta-se, nesta tese, um exemplo de resumo. Lorem ipsum dolor sit amet,
81 consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et
82 dolore magna aliqua. Ut enim ad minim veniam, quis nostrud exercitation
83 ullamco laboris nisi ut aliquip ex ea commodo consequat. Duis aute irure dolor
84 in reprehenderit in voluptate velit esse cillum dolore eu fugiat nulla
85 pariatur. Excepteur sint occaecat cupidatat non proident, sunt in culpa qui
86 officia deserunt mollit anim id est laborum. Sed ut perspiciatis unde omnis
87 iste natus error sit voluptatem accusantium doloremque laudantium, totam rem
88 aperiam, eaque ipsa quae ab illo inventore veritatis et quasi architecto beatae
89 vitae dicta sunt explicabo, nemo enim ipsam voluptatem quia voluptas sit
90 aspernatur aut odit aut fugit.
91
92 \end{abstract}
93
94 \begin{foreignabstract}
95
96 In this work, we present an example abstract. Lorem ipsum dolor sit amet,
97 consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et
98 dolore magna aliqua. Ut enim ad minim veniam, quis nostrud exercitation
99 ullamco laboris nisi ut aliquip ex ea commodo consequat. Duis aute irure dolor
100 in reprehenderit in voluptate velit esse cillum dolore eu fugiat nulla
101 pariatur. Excepteur sint occaecat cupidatat non proident, sunt in culpa qui
102 officia deserunt mollit anim id est laborum. Sed ut perspiciatis unde omnis
103 iste natus error sit voluptatem accusantium doloremque laudantium, totam rem
104 aperiam, eaque ipsa quae ab illo inventore veritatis et quasi architecto beatae
105 vitae dicta sunt explicabo, nemo enim ipsam voluptatem quia voluptas sit
106 aspernatur aut odit aut fugit.
107
108 \end{foreignabstract}
109
110 % Ordem pre-textual (NBR 14724/6027, R55): listas (ilustracoes, tabelas,
111 % abreviaturas e siglas, simbolos) ANTES do sumario; sumario por ultimo.
112 \listoffigures
113 \listoftables
114 \listofquadtros
115 \listofprogramas
116 \listofalgorithms
117 \printloabbreviations
118 \printlosymbols
119 \tableofcontents
120
121 \mainmatter
122 \chapter{Introdução}
123
124 Este é um documento exemplo para o uso da classe CoppeTeX, destinado a ajudar os alunos do
125
126 A classe \verb|coppe| foi criada originalmente por Vicente Helano e George Ainsworth, poré
127 Em 2026, Geraldo Xexéo, com forte ajuda das ferramentas Claude 4.7 Opus e Max e 4.8 Opus M
128
129 Provavelmente Geraldo Xexéo, professor do Programa de Engenharia de Sistemas e Computação
130
131 A versão mais atual dessa classe é mantida no GitHub, no repositório \url{https://github.c
132
133 Esse documento segue a norma de formatação de teses e dissertações da COPPE. Ele também po

```

134

135 Há um grupo de WhatsApp para perguntas e discussões do estilo: `\href{https://github.com/COPPE-UFRJ/CoppeTeX/}`

136

137 Bugs devem ser reportados na `\href{https://github.com/COPPE-UFRJ/CoppeTeX/issues}{página d`

138

139 Esse documento é usado como exemplo de coisas que podem ser feitas. Ele está configurado p

140

141 É importante de notar que essa classe não foi construída sobre a classe `\LaTeX \` para a A

142

143 Essa classe implementa citações ABNT modernas e vários tipos de entrada descritos na norma

144

145 Este documento não substitui, mas complementa, o documento que descreve a classe.

146

147

148

149 `\chapter{Configurações Iniciais}`

150

151 A primeira coisa a fazer é escolher o tipo de documento. Isso é feito como uma opção no c

152

153 Como pode ser visto nesse documento, muita coisa pode ser configurada, o que gerará o tra

154

155 Recomendo ler o documento “The `\verb*|coppe|` document class” para entender melhor todas

156

157 `\section{Linguagem principal do texto}`

158

159 Essa classe considera que o texto principal está em português e algumas partes específicas

160 `\begin{itemize}`

161 `\item` A opção `\verb|english|` deve ser usada no comando `\verb|\documentclass|`.

162 `\item` A bibliografia segue automaticamente o idioma do Babel (português ou inglês); não é p

163 `\end{itemize}`

164

165 A variação de linguagem, em inglês ou português apenas, já é suportada pela classe `\CoppeT`

166

167

168 `\section{Por que usar o \LaTeX}`

169

170 Há uma grande discussão entre usuário de Word e `\LaTeX`, principalmente, quanto ao uso dess

171

172 Nós escolhemos o `\LaTeX` por alguns motivos: grande facilidade de seguir um estilo sem se p

173

174 As principais desvantagens são: idiossincrasias que podem gastar tempo, pouco controle sob

175

176 `\section{Como e onde usar o \LaTeX}`

177

178 Existem muitos tutoriais de `\LaTeX`, mas basicamente, em 2024, ele é usado em dois ambiente

179 `\begin{enumerate}`

180 `\item` Na sua máquina, instalando uma versão completa como o `\hologo{MiKTeX}`, típico do Win

181 `\item` Usar um ambiente na rede, como o `Overleaf`.

182 `\end{enumerate}`

183

184 Em todo caso, recomendo fortemente que, ao mesmo tempo, mantenha versões no Git e faça o b

185

186

187 `\chapter{Algumas Regras da COPPE}`

188

189 Todas abreviaturas e símbolos devem ser definida antes de utilizada. Isso é facilmente fei

190

191 É imprescindível definir os símbolos, tal como o

192 conjunto dos números reais $\mathbb{R}$ e o conjunto vazio $\emptyset$.

193 `\syml{\mathbb{R}}{Conjunto dos números reais}`

194 `\syml{\emptyset}{Conjunto vazio}`. Usamos esse exemplo aqui justamente para mostrar como

195

196 Para as listas de abreviaturas e símbolos funcionarem no Overleaf é necessário rodar o `\ve`

197

198 Como as listas de símbolos e de abreviaturas usamo o mesmo comando usado para criar índice

199

200 `\section{Citações}`

201 `\label{sec:citacoes}`

202

203 Citações curtas podem ser feitas `\enquote{o comando quote}` ou direto com “duas crases e do

204

205 `\begin{longquote}`

206 Um exemplo de citação longa nas regras da ABNT (4cm de recuo e fonte menor)

207 feita com o ambiente `\verb*|longquote|` The primary objective of this

208 investigation was to determine the feasibility of detecting corrosion in

209 aluminum Naval aircraft components with neutron radiographic interrogation

210 and the use of standard corrosion penetrameters. Secondary objectives

211 included the determination of the effect of object thickness on image quality,

212 the defining of minimum levels of detectability and a preliminary investigation

213 of a means whereby the degree of corrosion could be quantified with neutron

214 radiographic data. `\citep{article-example}`

215 `\end{longquote}`

216

217 Citações devem apontar as referências. Para isso, está disponível o ótimo pacote `\verb*|na`

218

219 Em todo caso, `\textbf{deve se tomar enorme atenção com as citações, para evitar ocorrer em`

220

221 Quando se cita uma obra por meio de outra (`\textit{apud}`), pela ABNT entra na lista de ref

222

223 `\chapter{Floats}`

224

225 Grande parte dos problemas de iniciantes, e veteranos, em $\text{\LaTeX}$ é da localização dos `\textit`

226

227 A regra geral de posicionamento é que uma figura ou quadro só pode aparecer a partir da mesm

228

229 `\textbf{Pela norma atual (manual2024), a legenda}` `\verb|\caption|` `\textbf{de toda ilustração`

230

231 Quadros são ilustrações cujos dados são delimitados por linhas em `\emph{todas}` as bordas (ao

232

233

234 `\section{Tabelas e Figuras Padrão}`

235

236 Vamos ver uma tabela padrão, como a `\autoref{tab:exemplo_numeros}`.

237

238 `\begin{table}[ht]`

239 `\centering` % Centraliza a tabela

240 `\caption{Exemplo de Tabela de Números}`

241 `\label{tab:exemplo_numeros}`

```

242 \begin{tabular}{ccc} % Define a quantidade de colunas
243 \hline % Linha superior
244 \textbf{Col. 1} & \textbf{Col. 2} & \textbf{Col. 3} \\ % Cabeçalhos
245 \hline % Linha média
246 1 & 2 & 3 \\ % Primeira linha de dados
247 \hline
248 4 & 5 & 6 \\ % Segunda linha de dados
249 \hline
250 7 & 8 & 9 \\ % Terceira linha de dados
251 \hline
252 10 & 11 & 12 \\ % Quarta linha de dados
253 \hline % Linha inferior
254 \end{tabular}
255 \fonte{Elabora\c{c}\~ao pr\'opria.} % fonte obrigatoria, abaixo do float
256 \end{table}
257
258
259
260 Já a \autoref{fig:exemplo_figura} é uma figura padrão, com controle da largura.
261
262 \begin{figure}[ht]
263 \centering % Centraliza a figura
264 \caption{Exemplo de Figura com Legenda Acima} % Legenda em cima (norma atual)
265 \label{fig:exemplo_figura} % Etiqueta para referência cruzada
266 \includegraphics[width=0.5\textwidth]{coppe-logo.pdf} % Inclui a imagem com metade da largura
267 \source{COPPE/UFRJ.} % fonte obrigatoria, abaixo da figura
268 \end{figure}
269
270 Quadros têm a mesma regra (legenda acima, fonte abaixo), mas com linhas em todas as bordas.
271
272 \begin{quadro}[ht]
273 \centering
274 \caption{Exemplo de Quadro}
275 \label{qua:exemplo}
276 \begin{tabular}{|l|l|}
277 \hline
278 \textbf{Ilustração} & \textbf{Delimitação} \\
279 \hline
280 Tabela & Linhas apenas no topo e na base. \\
281 \hline
282 Quadro & Linhas em todas as bordas. \\
283 \hline
284 \end{tabular}
285 \fonte{Elabora\c{c}\~ao pr\'opria.}
286 \end{quadro}
287
288
289
290
291
292 \section{Tabelas mais elegantes}
293
294 Atualmente a tendência é usar tabelas mais leves, como \autoref{tab:exemplo_numerosbom}. Iss
295

```

```

296 \begin{table}[ht]
297 \centering % Centraliza a tabela
298 \caption{Exemplo de Tabela de Números mais elegantes}
299 \label{tab:exemplo_numerosbom}
300 \begin{tabular}{ccc} % Define a quantidade de colunas
301 \toprule % Linha superior
302 \textbf{Col. 1} & \textbf{Col. 2} & \textbf{Col. 3} \\ % Cabeçalhos
303 \midrule % Linha média
304 1 & 2 & 3 \\ % Primeira linha de dados
305 4 & 5 & 6 \\ % Segunda linha de dados
306 7 & 8 & 9 \\ % Terceira linha de dados
307 10 & 11 & 12 \\ % Quarta linha de dados
308 \bottomrule % Linha inferior
309 \end{tabular}
310 \fonte{Elabora\c{c}\~ao pr'opria.}
311 \end{table}
312
313 \section{Tabelas Longas ou Largas}
314
315 Se sua tabela é muito longa ou larga, existem várias opções.
316 \begin{itemize}
317 \item alterar o tamanho da letra
318 \item Usar o longtable
319 \item rodar a tabela, fazendo ela em \textit{landscape}
320 \item fazer a tabela dentro de um minibox
321 \end{itemize}
322
323
324 \subsection{Tabelas largas demais}
325
326 É comum em teses que as tabelas sejam largas demais. Há várias formas de resolver isso.
327
328 A \autoref{tab:tabela_largafns} é larga demais, e nela isso é resolvido diminuindo a fonte p
329
330 \begin{table}[ht]
331 \centering % Centraliza a tabela
332 \caption{Exemplo de Tabela Larga com Fonte Menor}
333 \label{tab:tabela_largafns}
334 \footnotesize % Aplica uma fonte menor para a tabela
335 \begin{tabular}{ccccccc} % Aumente o número de colunas conforme necessário
336 \toprule
337 \textbf{Col. 1} & \textbf{Col. 2} & \textbf{Col. 3} & \textbf{Col. 4} & \textbf{Col. 5} & \textbf{Col. 6} & \textbf{Col. 7} \\
338 \midrule
339 Dado 1.1 & Dado 1.2 & Dado 1.3 & Dado 1.4 & Dado 1.5 & Dado 1.6 & Dado 1.7 & Dado 1.8 \\
340 Dado 2.1 & Dado 2.2 & Dado 2.3 & Dado 2.4 & Dado 2.5 & Dado 2.6 & Dado 2.7 & Dado 2.8 \\
341 Dado 3.1 & Dado 3.2 & Dado 3.3 & Dado 3.4 & Dado 3.5 & Dado 3.6 & Dado 3.7 & Dado 3.8 \\
342 \bottomrule
343 \end{tabular}
344 \fonte{Elabora\c{c}\~ao pr'opria.}
345 \end{table}
346
347 O comando \verb|\resizebox{width}{height}{content}| permite ajustar o tamanho de qualquer co
348
349 \begin{table}[ht]

```

```

350 \centering
351 \caption{Exemplo de Tabela Redimensionada}
352 \label{tab:examplerb}
353 \resizebox{\textwidth}{!}{%
354 \begin{tabular}{llll}
355 \toprule
356 Coluna 1 & Coluna 2 & Coluna 3 & Coluna 4 \\
357 \midrule
358 Dados 1 & Dados 2 & Dados 3 & Dados 4 \\
359 Dados 5 & Dados 6 & Dados 7 & Dados 8 \\
360 \bottomrule
361 \end{tabular}}%
362 }
363 \fonte{Elabora\c{c}\~ao pr\'opria.}
364 \end{table}
365
366
367 Para rodar uma tabela muito larga em 90 graus no LaTeX, voc\ea pode usar o pacote \verb*|rotat
368
369 Aqui est\aa um exemplo de como usar o ambiente \verb*|sidewaystable| para girar uma tabela. Pr
370
371 \begin{sidewaystable}
372 \centering
373 \caption{Sua Legenda Aqui}
374 \label{tab:sua_tabela}
375 \begin{tabular}{llll}
376 \toprule
377 Coluna 1 & Coluna 2 & Coluna 3 \\
378 \midrule
379 Item 1 & Item 2 & Item 3 \\
380 Item 4 & Item 5 & Item 6 \\
381 \bottomrule
382 \end{tabular}
383 \fonte{Elabora\c{c}\~ao pr\'opria.}
384 \end{sidewaystable}
385
386 Se a tabela for muito longa, o ambiente \verb|longtable| \ea o ideal. Ele fornece comandos par
387
388 % Exemplo de tabela longa que se estende por v\arias p\aginas
389 \begin{longtable}{|c|c|c|}
390 % primeiro cabe\c{a}lho (\ea o caption)
391 \caption{Exemplo de Tabela Longa}\label{tab:longa} \\
392 \hline \textbf{Col. 1} & \textbf{Col. 2} & \textbf{Col. 3} \\ \hline
393 \endfirsthead
394 % cabe\c{a}lho normal
395 \multicolumn{3}{c}%
396 {\table\ \thetable} -- continua\c{a}o da p\agina anterior} \\
397 \hline \textbf{Col. 1} & \textbf{Col. 2} & \textbf{Col. 3} \\ \hline
398 \endhead
399 % p\ea normal
400 \hline \multicolumn{3}{|r|}{Continua na pr\oxima p\agina} \\ \hline
401 \endfoot
402 \hline
403 % \c{u}ltimo p\ea

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404 \multicolumn{3}{|r|}{{Continua na próxima página}}\\
405 \hline \hline
406 \endlastfoot
407
408 % Conteúdo da tabela
409 1 & 2 & 3 \\
410 4 & 5 & 6 \\
411 1 & 2 & 3 \\
412 4 & 5 & 6 \\
413 1 & 2 & 3 \\
414 4 & 5 & 6 \\
415 1 & 2 & 3 \\
416 4 & 5 & 6 \\
417 1 & 2 & 3 \\
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431 4 & 5 & 6 \\
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447 4 & 5 & 6 \\
448 1 & 2 & 3 \\
449 4 & 5 & 6 \\1 & 2 & 3 \\
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456 1 & 2 & 3 \\
457 4 & 5 & 6 \\

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458 1 & 2 & 3 \\
 459 4 & 5 & 6 \\
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 506 1 & 2 & 3 \\
 507 4 & 5 & 6 \\
 508 1 & 2 & 3 \\
 509 4 & 5 & 6 \\
 510 1 & 2 & 3 \\
 511 4 & 5 & 6 \

```

512 1 & 2 & 3 \\
513 4 & 5 & 6 \\
514 1 & 2 & 3 \\
515 4 & 5 & 6 \\
516
517 % Repetir linhas semelhantes conforme necessário para estender a tabela por 3 páginas
518 \end{longtable}
519 \fonte{Elabora\c{c}\~ao pr\'opria.}% a longtable não é float: a fonte vem após o ambiente
520
521 \section{Subfiguras (subcaption)}
522
523 Para colocar várias imagens (ou tabelas) lado a lado, cada uma com sua própria legenda, use
524
525 \begin{figure}[ht]
526   \centering
527   \begin{subfigure}[b]{0.42\textwidth}
528     \centering\includegraphics[width=\linewidth]{example-image-a}
529     \caption{Primeira vista}\label{fig:sub-a}
530   \end{subfigure}
531   \hfill
532   \begin{subfigure}[b]{0.42\textwidth}
533     \centering\includegraphics[width=\linewidth]{example-image-b}
534     \caption{Segunda vista}\label{fig:sub-b}
535   \end{subfigure}
536   \caption{Figura com duas subfiguras}
537   \label{fig:subs}
538   \fonte{Elabora\c{c}\~ao pr\'opria.}
539 \end{figure}
540
541 Referencie a figura inteira com \verb|\autoref{fig:subs}|. Cada parte tem
542 rótulo próprio: \verb|\autoref{fig:sub-a}| leva à \autoref{fig:sub-a}.
543
544 \section{Imagens (graphicx)}
545
546 A classe já carrega o \verb|graphicx|. Insira imagens com \verb|\includegraphics|; as opções
547 \begin{itemize}
548   \item \verb|[width=0.8\textwidth]| ou \verb|[scale=0.5]| --- tamanho;
549   \item \verb|[angle=90]| --- rotação;
550   \item \verb|[trim=l b r t, clip]| --- recorte (corta as bordas \emph{esquerda, baixo, direi}
551 \end{itemize}
552 Formatos aceitos com \hologo{pdfLaTeX}/\hologo{LuaLaTeX}: \texttt{pdf}, \texttt{png}, \texttt{t}
553
554 \section{Introdução ao tikz}
555
556 0 \verb|tikz| (disponível também via \verb|tcolorbox|) desenha gráficos vetoriais no próprio
557
558 \begin{figure}[ht]
559   \centering
560   \begin{tikzpicture}
561     \node[draw, rounded corners] (a) at (0,0) {Início};
562     \node[draw, rounded corners] (b) at (4,0) {Fim};
563     \draw[->, thick] (a) -- (b) node[midway, above] {processo};
564   \end{tikzpicture}
565   \caption{Diagrama simples em tikz}

```

```

566 \label{fig:tikz}
567 \fonte{Elabora\c{c}\~ao pr\'opria.}
568 \end{figure}
569
570 Para gráficos de dados (curvas, barras), veja o \verb|pgfplots|, construído sobre o \verb|tikz|
571
572 \section{Float ou direto no texto?}
573
574 Colocar a tabela ou imagem dentro de \verb|figure|/\verb|table| (um \emph{float}) deixa o \verb|caption|
575
576 \chapter{Using BibLaTeX}
577 \label{cap:biblatex}
578
579 Esta classe gera as referências com BibLaTeX e o \emph{backend} \texttt{biber}, no padrão
580
581 \textbf{É} obrigatório compilar com \texttt{biber} (e não com \hologo{BibTeX}). A sequência
582
583 Os comandos do \texttt{natbib} (\verb|\citet|, \verb|\citep|) continuam disponíveis porque
584
585 \begin{table}[h]
586 \caption{Exemplos de cita\c{c}\~oes utilizando o comando padr\~ao}
587 \texttt{\textbackslash cite} do \LaTeX\ e
588 o comando \texttt{\textbackslash citep},
589 disponível pela opção \texttt{natbib} do BibLaTeX.
590 \label{tab:citation}
591 \centering
592 {\footnotesize
593 \begin{tabular}{|c|c|c|}
594 \hline
595 Tipo da Publicação & \verb|\cite| & \verb|\citep|\\
596 \hline
597 Livro & \cite{book-example} & \citep{book-example}\\
598 Artigo & \cite{article-example} & \citep{article-example}\\
599 Relatório & \cite{techreport-example} & \citep{techreport-example}\\
600 Relatório & \cite{techreport-exampleIn} & \citep{techreport-exampleIn}\\
601 Anais de Congresso & \cite{inproceedings-example} & \\
602 & \citep{inproceedings-example}\\
603 Séries & \cite{incollection-example} & \citep{incollection-example}\\
604 Em Livro & \cite{inbook-example} & \citep{inbook-example}\\
605 Dissertação de mestrado & \cite{mastersthesis-example} & \\
606 & \citep{mastersthesis-example}\\
607 Tese de doutorado & \cite{phdthesis-example} & \citep{phdthesis-example}\\
608 \hline
609 \end{tabular}}
610 \fonte{Elabora\c{c}\~ao pr\'opria.}
611 \end{table}
612
613 \begin{table}[h]
614 \caption{Exemplos de cita\c{c}\~oes utilizando o comando padr\~ao}
615 \texttt{\textbackslash cite} do \LaTeX\ e
616 o comando \texttt{\textbackslash citet},
617 disponível pela opção \texttt{natbib} do BibLaTeX. Além disso, usando o booktabs.}
618 \label{tab:citation1}
619 \centering

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```

620 {\footnotesize
621 \begin{tabular}{ccc}
622 \toprule
623 Tipo da Publicação & \verb|\cite| & \verb|\citet|\
624 \midrule
625 Livro & \cite{book-example} & \citet{book-example}\
626 Artigo & \cite{article-example} & \citet{article-example}\
627 Relatório & \cite{techreport-example} & \citet{techreport-example}\
628 Relatório & \cite{techreport-exampleIn} & \citet{techreport-exampleIn}\
629 Anais de Congresso & \cite{inproceedings-example} &
630 \citet{inproceedings-example}\
631 Séries & \cite{incollection-example} & \citet{incollection-example}\
632 Em Livro & \cite{inbook-example} & \citet{inbook-example}\
633 Dissertação de mestrado & \cite{mastersthesis-example} &
634 \citet{mastersthesis-example}\
635 Tese de doutorado & \cite{phdthesis-example} & \citet{phdthesis-example}\
636 \bottomrule
637 \end{tabular}}
638 \fonte{Elabora\c{c}\~ao pr\'opria.}
639 \end{table}
640
641 \section{Comandos de citação}
642
643 Os principais comandos são: \verb|\cite| (referência simples), \verb|\citet| (autor por ex
644
645 \subsection{Parâmetro opcional: página e “tradução nossa”}
646
647 Todos os comandos aceitam um parâmetro opcional para a localização (página, seção) e obser
648
649 \subsection{Citação de citação (\emph{apud})}
650
651 Para citar uma obra através de outra, use \verb|\citetapud| e \verb|\citepapud| (demonstra
652
653 \subsection{Múltiplas bibliografias}
654
655 Para separar as referências (por tipo, capítulo ou tema), use o ambiente \verb|refsection|
656
657 \section{Tipos de entrada bibliográfica}
658
659 Cada tipo de documento da ABNT corresponde a um tipo de entrada do BibLaTeX, com um sinôni
660
661 \subsection{Tipos acadêmicos principais}
662 \begin{itemize}
663 \item Livro / monografia --- \verb|@book| (\verb|@livro|)
664 \item Artigo de periódico --- \verb|@article| (\verb|@artigo|); matéria de jornal --- \verb|@journal|
665 \item Capítulo de livro --- \verb|@incollection| (\verb|@partedelivro|), \verb|@inbook|
666 \item Evento no todo (anais) --- \verb|@proceedings| (\verb|@anais|); trabalho em evento
667 \item Tese/dissertação --- \verb|@thesis| (\verb|@tese|, \verb|@dissertacao|); com tipo
668 \item Relatório --- \verb|@report| (\verb|@relatorio|, \verb|@relatoriotecnico|); periód
669 \item Manual --- \verb|@manual|; diversos --- \verb|@misc| (\verb|@diverso|); página/sit
670 \end{itemize}
671
672 \subsection{Tipos especiais (ABNT §4.2.5--4.2.14)}
673 \begin{itemize}

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674 \item Patente --- \verb|@patent| (\verb|@patente|)
675 \item Legislação / ato normativo --- \verb|@legislation| (\verb|@legislacao|, \verb|@ato
676 \item Norma técnica --- \verb|@standard| (\verb|@norma|); partitura --- \verb|@music| (\
677 \item Documento cartográfico (mapa) --- \verb|@map| (\verb|@mapa|, \verb|@cartografico|)
678 \item Audiovisual --- \verb|@video| (\verb|@audiovisual|); filme --- \verb|@movie| (\verb
679 \end{itemize}
680
681 \subsection{Jogos e TV (tipos próprios da CoppeTeX)}
682 \begin{itemize}
683 \item Jogo --- \verb|@game| (\verb|@jogo|); de tabuleiro --- \verb|@boardgame| (\verb|@j
684 \item Programa de TV / série --- \verb|@tvshow| (\verb|@programadetv|, \verb|@serietv|)
685 \end{itemize}
686
687 Os campos também aceitam sinônimos em português (sem acentos): \verb|autor|, \verb|titulo|
688
689 \chapter{Alguns outros exemplo úteis}
690
691 \begin{tcolorbox}[title=Meu Textbox]
692 Este é o conteúdo do meu textbox. Você pode adicionar qualquer texto aqui, bem como incluir
693 \end{tcolorbox}
694
695 \begin{tcolorbox}
696 Este é o conteúdo do meu textbox sem título. Você pode adicionar qualquer texto aqui, bem co
697 \end{tcolorbox}
698
699 \begin{figure}[ht]
700 \centering
701 \begin{tikzpicture}
702 \node[anchor=south west,inner sep=0] (image) at (0,0) {\includegraphics[width=0.5\textwidth]{
703 \begin{scope}[x={(image.south east)},y={(image.north west)}]
704 % Definindo o textbox dentro da figura
705 \node[anchor=north west, text width=0.3\textwidth, fill=white, opacity=0.7, text=
706 \begin{tcolorbox}[colback=red!5!white,colframe=red!75!black,title=Textbox de
707 Este textbox fala sobre como inserir um textbox dentro de uma figura usa
708 \end{tcolorbox}
709 };
710 \end{scope}
711 \end{tikzpicture}
712 \caption{Figura com Textbox}
713 \label{fig:figura_com_textbox1}
714 \fonte{Elabora\c{c}\~ao pr\'opria.}
715 \end{figure}
716
717
718 \begin{figure}[ht]
719 \centering
720 \begin{tcolorbox}
721 Este é o conteúdo do meu textbox sem título. Você pode adicionar qualquer texto aqui, bem co
722 \end{tcolorbox}
723 \caption{Figura com Textbox simples}
724 \label{fig:figura_com_textbox}
725 \fonte{Elabora\c{c}\~ao pr\'opria.}
726 \end{figure}
727

```

```

728 \section{Listagens de código e algoritmos}
729
730 O CoppeTeX traz embutido o suporte a listagens de programas (pacote \verb|listings|) e a alg
731
732 \begin{python}[caption={Exemplo de programa em Python},label=prog:exemplo]
733 def saudacao(nome):
734     # imprime uma saudação acentuada
735     print(f"Olá, {nome}!")
736
737 saudacao("COPPE")
738 \end{python}
739 \source{Elaboração própria.}
740
741 Para inserir código no meio do texto use \verb|\pythoninline|, como em \pythoninline{saudaca
742
743 \begin{Verbatim}[frame=lines,numbers=left,numbersep=0.5em]
744 Olá, COPPE!
745 \end{Verbatim}
746
747 Há estilos prontos também para \verb|java|, \verb|xml|, \verb|html| e \verb|prolog|.
748 Há também \verb|\pythoninline| e \verb|\javainline| para código no meio do
749 texto, e o ambiente \verb|Verbatim| (com as opções acima) para a saída de programas. Para al
750
751 \begin{algorithm}[H]
752 \caption{Soma dos elementos de um vetor}
753 \label{alg:soma}
754 \Dados{vetor  $v$  com  $n$  elementos}
755 \Resultado{soma  $s$  dos elementos de  $v$ }
756  $s \leftarrow 0$ ;
757 \Para{ $i \leftarrow 1$  \Ate{ }  $n$ }{
758      $s \leftarrow s + v[i]$ ;
759 }
760 \Retorna  $s$ ;
761 \end{algorithm}
762
763 \chapter{Escrevendo matemática}
764
765 A classe carrega \texttt{amsmath} em qualquer motor. Sob \hologo{pdfLaTeX} ela também carr
766
767 \section{Alinhamento (align)}
768 O ambiente \verb|align| alinha equações pelo \verb|&| e numera cada linha (\verb|\\| quebr
769 \begin{align}
770 f(x) &= (x+1)^2 \\
771      &= x^2 + 2x + 1 .
772 \end{align}
773
774 \section{Matrizes (matrix)}
775 O \texttt{amsmath} traz \verb|pmatrix|, \verb|bmatrix|, \verb|vmatrix|, entre outros:
776 \begin{equation}
777 A = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} \\
778 \quad \quad \quad \\
779 \det A = \begin{vmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{vmatrix} \\
780 \end{equation}
781

```

```

782 \section{Subequações (subequations)}
783 Para numerar um grupo de equações sob o mesmo número, com letras (a), (b):
784 \begin{subequations}
785 \begin{align}
786   a &= b + c , \\
787   d &= e - f .
788 \end{align}
789 \end{subequations}
790
791 \section{Estruturas alinhadas (array)}
792 O \verb|array| monta estruturas dentro do modo matemático, como definições por partes:
793 \begin{equation}
794   |x| = \left\{ \begin{array}{l}
795     x, \quad \text{se } x \geq 0 , \\
796     -x, \quad \text{caso contrário.}
797   \end{array} \right.
798 \end{equation}
799
800 \section{Operadores (\textbackslash DeclareMathOperator)}
801 Para operadores com a tipografia correta (romano, espaçamento adequado), declare-os no pre
802 \begin{verbatim}
803 \usepackage{amsmath}
804 \DeclareMathOperator{\RMSE}{RMSE}
805 \DeclareMathOperator{\diag}{diag}
806 \end{verbatim}
807 Assim:
808 \begin{itemize}
809   \item \verb|$\RMSE(y,\hat{y})$| produz  $\RMSE(y,\hat{y})$ ;
810   \item \verb|$\diag(\lambda_1,\lambda_2)$| produz
811      $\diag(\lambda_1,\lambda_2)$ .
812 \end{itemize}
813
814 \section{Unicode e LuaLaTeX (unicode-math)}
815 \label{sec:unicode-math}
816 Ao migrar para \hologo{LuaLaTeX} (ou \hologo{XeLaTeX}), \textbf{não carregue} \texttt{amss
817 \begin{verbatim}
818 \usepackage{amsmath}           % a classe já carrega
819 \usepackage{unicode-math} % NÃO use \usepackage{amssymb} junto
820 \setmathfont{Latin Modern Math} % ou STIX Two Math
821 \end{verbatim}
822 O código clássico continua valendo ( $\mathbb{R}$ ,  $\int_a^b f(x)dx$ ,  $\alpha$ ) e você ai
823
824 \chapter{Referências cruzadas, notas e índices}
825 \label{cap:refs}
826
827 Este capítulo reúne os recursos para \emph{apontar} para coisas: rótulos, notas de rodapé,
828
829 \section{Rótulos e referências cruzadas}
830
831 Marque um elemento com \verb|\label{tipo:nome}| e aponte para ele com \verb|\ref| (só o nú
832
833 \section{Notas de rodapé}
834
835 Use \verb|\footnote{texto}| \footnote{Como esta nota, que a classe imprime em espaçamento s

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836
837 \section{hyperref e marcadores}
838
839 A classe carrega o \verb|hyperref|: o sumário, as citações e os \verb|\ref|/\verb|\autoref|
840
841 \section{URLs}
842
843 Para endereços, use \verb|\url{https://exemplo.org}|: o \verb|hyperref| cria o link e quebra
844 use \verb|\href{https://exemplo.org}{texto}|. Evite digitar URLs como texto comum --- elas não
845
846 \section{Índice remissivo}
847
848 Marque os termos com \verb|\index{termo}| e imprima o índice com \verb|\printindex| (carregado
849
850 \section{Traduções (NBR 10520)}
851
852 Ao citar um trecho que você mesmo traduziu, coloque a tradução no corpo do texto e escreva
853
854 \chapter{Truques de \LaTeX}
855
856 \section{Comentários mágicos (\texttt{\% !TeX})}
857 Linhas \verb|\% !TeX| no topo do arquivo dizem ao editor (\hologo{TeX}studio, Overleaf, \hologo{TeX}Live)
858 \begin{verbatim}
859 % !TeX program = lualatex
860 % !TeX encoding = UTF-8
861 % !TeX spellcheck = pt_BR
862 % !BIB program = biber
863 \end{verbatim}
864 Com isso o documento compila igual em qualquer ambiente. O \texttt{latexmkrc} cumpre papel
865
866 \section{Espaço engolido}
867 Um comando sem argumento engole o espaço seguinte: \verb|\LaTeX é bom| sai como “\LaTeX é bom”
868
869 \section{Escopo (grupos)}
870 Chaves \verb|{...}| (ou \verb|\begingroup| \dots \verb|\endgroup|) criam um escopo: mudanças locais
871
872 \section{Descarregar figuras: \texttt{\textbackslash clearpage} × \texttt{\textbackslash newpage}}
873 \verb|\newpage| apenas quebra a página e leva os floats pendentes adiante, acumulando-os.
874
875 \section{\texttt{latexmkrc}, perl e Overleaf}
876 O \verb|latexmk| automatiza a compilação (roda \LaTeX, \verb|biber| e \verb|makeindex| quando necessário)
877
878 \chapter{Método Proposto}
879 \chapter{Resultados e Discussões}
880
881 \section{Algumas Demonstrações}
882
883 A Lista de Símbolos precisa usar comandos específicos. Aqui vamos usar os símbolos $\alpha$, $\beta$, $\gamma$, $\delta$, $\epsilon$, $\zeta$, $\eta$, $\theta$, $\iota$, $\kappa$, $\lambda$, $\mu$, $\nu$, $\xi$, $\omicron$, $\pi$, $\rho$, $\sigma$, $\tau$, $\upsilon$, $\phi$, $\chi$, $\psi$, $\omega$, $\Gamma$, $\Delta$, $\Theta$, $\Lambda$, $\Xi$, $\Omicron$, $\Pi$, $\Sigma$, $\Upsilon$, $\Phi$, $\Psi$, $\Omega$, $\alpha$, $\beta$, $\gamma$, $\delta$, $\epsilon$, $\zeta$, $\eta$, $\theta$, $\iota$, $\kappa$, $\lambda$, $\mu$, $\nu$, $\xi$, $\omicron$, $\pi$, $\rho$, $\sigma$, $\tau$, $\upsilon$, $\phi$, $\chi$, $\psi$, $\omega$, $\Gamma$, $\Delta$, $\Theta$, $\Lambda$, $\Xi$, $\Omicron$, $\Pi$, $\Sigma$, $\Upsilon$, $\Phi$, $\Psi$, $\Omega$
884 \symbl[beta]{Beta}{A palavra Beta mais e corrigida}
885 \symbl[zzbeta]{\beta}{A letra $\beta$ corrigida}
886 \symbl{beta}{A palavra beta}
887 \symbl{alpha}{A palavra alpha}
888 \symbl[alpha]{Alpha}{A palavra Alpha}
889 \symbl[zzalpha]{\alpha}{A letra $\alpha$ corrigida}

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890 \syml[marco]{Marco}{A palavra Marco corrigida}
891
892 A Lista de Abreviações segue, a partir de 2024, a mesma regra, e aqui seguem alguns exemplos
893 \abbrev{GoT}{Game of Thrones}
894 \abbrev[GOT]{GoT}{Game of Thrones ordenado como GOT}
895 \abbrev[iot]{IoT}{IoT ordenado como iot}
896 \abbrev[IoT]{IoT}{IoT ordenado como IoT}
897 \abbrev[IOT]{IoT}{IoT ordenado como IOT}
898 \abbrev{IoT}{IoT com ordenação default}
899 \abbrev[ITU]{ITU}{ITU mesmo}
900
901 \newsigla{ufrj}{UFRJ}{Universidade Federal do Rio de Janeiro}
902 \newsigla{cpgp}{CPGP}{Comissão de Pós-Graduação e Pesquisa de Engenharia}
903
904 As siglas (manual2024, seção 2.8) são expandidas na primeira ocorrência e registradas automaticamente
905
906
907
908
909 \chapter{Conclusões}
910
911 \backmatter
912
913 % Tipos exóticos da ABNT (NBR 6023): patente, legislação, norma, mapa,
914 % partitura --- declarados via sinônimos em português em example.bib.
915 \nocite{patente-example,legislacao-example,norma-example,mapa-example,partitura-example}
916 \nocite{jurisprudencia-example}
917 \nocite{videogame-example,filme-example,semautor-example,semautorart-example}
918 \nocite{jornalcaderno-example,jornal-example}
919 \printbibliography
920
921
922
923 \appendix
924
925 \chapter{Um apêndice}
926
927 Segundo a norma da ABNT (Associação Brasileira de Normas Técnicas), a definição e utilização de apêndices
928
929 Apêndice: O apêndice é um texto ou documento elaborado pelo autor do trabalho com o objetivo de complementar
930
931
932 \chapter{Outro apêndice}
933
934 \annex
935
936
937 \chapter{Um Anexo}
938 Segundo a norma da ABNT (Associação Brasileira de Normas Técnicas), a definição e utilização de anexos
939
940
941
942 Anexo: O anexo, por sua vez, consiste em um texto ou documento não elaborado pelo autor, que complementa
943

```

```

944 No modelo \CoppeTeX os anexos devem obrigatoriamamente vir depois dos apêndices e usam o comand
945
946
947 \chapter{Outro Anexo}
948
949
950 \coppetexfinalpage% colofão institucional (página par, face do contracapa)
951
952 \end{document}
953
954 </example>

```

10 Implementation

10.1 The ‘coppe.cls’ file

```

955 <*class>
956 \def\filename{coppe.dtx}
957 \def\fileversion{v4.1}
958 \def\filedate{2026/09/09}
959 \NeedsTeXFormat{LaTeX2e}[1995/12/01]
960 \ProvidesClass{coppe}[\filedate\ \fileversion\ COPPE Dissertations and Thesis]
961 % Save the class version and date into private macros immediately, before any
962 % required package can redefine the very common \fileversion / \filedate
963 % global names (LaTeX has no convention reserving them for the host class).
964 % \coppe@version / \coppe@date are what user-facing macros (e.g.
965 % \coppetexfinalpage) should reference.
966 \let\coppe@version\fileversion
967 \let\coppe@date\filedate
968 % Document date stored in private slots so \date can set them without ever
969 % touching TeX's \month/\year primitives (which would break \today). They
970 % default to the current system month/year, so a document without \date
971 % still renders sensibly on the cover and folha de rosto.
972 \edef\coppe@docmonth{\the\month}
973 \edef\coppe@docyear{\the\year}
974 \LoadClass[12pt,a4paper,oneside]{book}
975 % setspace, hyperref and the bibliography engine (biblatex+biber) are loaded at
976 % the END of the class (after every class hook is defined), in that order:
977 % setspace -> hyperref -> biblatex. hyperref must come late so all the macros
978 % it patches (TOC, refs, lists) are already defined, and biblatex must come
979 % after hyperref so its citation links and back-references work.
980 \RequirePackage{hyphenat}
981 \RequirePackage{lastpage}
982 \RequirePackage{ifthen}
983 \RequirePackage{graphicx}
984 \RequirePackage{tabularx}
985 \RequirePackage{booktabs} % regras de tabela profissionais (toprule/midrule/bottomrule)
986 \RequirePackage{longtable} % tabelas que quebram entre páginas
987 \RequirePackage{etoolbox}
988 \RequirePackage{ltxcmds}
989 \RequirePackage{hologo} % TeX-family logos: \hologo{TeX}, \hologo{LaTeX}, ...
990 % Render \TeX, \LaTeX and friends via hologo (\CoppeTeX is a custom logo, kept).
991 \def\TeX{\hologo{TeX}}
992 \def\LaTeX{\hologo{LaTeX}}

```

```

993 \def\LaTeXe{\hologo{LaTeXe}}
994 \def\BibTeX{\hologo{BibTeX}}
995 \def\pdfLaTeX{\hologo{pdfLaTeX}}
996 \def\pdfTeX{\hologo{pdfTeX}}
997 \def\LuaLaTeX{\hologo{LuaLaTeX}}
998 \def\XeLaTeX{\hologo{XeLaTeX}}
999 \def\MiKTeX{\hologo{MiKTeX}}
1000 % Engine-aware UTF-8 input: pdfLaTeX needs inputenc(utf8); LuaLaTeX (and XeTeX)
1001 % are UTF-8 native, so inputenc must NOT be loaded there (it would clash). Source
1002 % files may use accented Latin characters (âéíóú) directly. Under pdfLaTeX we also
1003 % suggest LuaLaTeX, which handles Unicode/fonts better (at the cost of speed).
1004 \RequirePackage{iftex}
1005 \ifPDFTeX
1006   \RequirePackage[utf8]{inputenc}
1007   \AtEndOfClass{\ClassWarningNoLine{coppe}{%
1008     Although slower, you should try compiling with LuaLaTeX}}
1009 \fi
1010 % lmodern MUST come with the T1 encoding. Without it, T1 falls back to the EC
1011 % fonts, which exist only as METAFONT bitmaps in most installations, and the
1012 % whole document is typeset in Type3 bitmap fonts. Measured before this change:
1013 % example.pdf carried 87 Type3 fonts against 6 Type1, and coppe.pdf 133 against
1014 % 2. The consequences are not cosmetic:
1015 %
1016 % * the text renders blurred at any zoom and prints poorly;
1017 % * copying and searching text in the PDF is unreliable;
1018 % * PDF/A -- required for deposit by 2.2(d) of the 2026 manual -- is not
1019 %   attainable with bitmap text.
1020 %
1021 % Latin Modern is the vector successor of Computer Modern, metrically and
1022 % visually near-identical, and ships with every current distribution, so the
1023 % page layout does not move.
1024 \RequirePackage{lmodern}
1025 \RequirePackage[T1]{fontenc}
1026 % Math support. amsmath is universal and works on every engine, so the class
1027 % brings it in unconditionally (it must precede hyperref, which is loaded at
1028 % the end of the class). amssymb is the legacy AMS symbol package, built
1029 % around the 8-bit Type 1 fonts msam/msbm; on a Unicode engine (LuaLaTeX or
1030 % XeLaTeX) it clashes with unicode-math, which already provides all of its
1031 % symbols through Unicode math codes. So amssymb is loaded ONLY under
1032 % pdfLaTeX; Unicode-engine users get a ClassWarning pointing them at
1033 % unicode-math (the class does not auto-load unicode-math because choosing
1034 % an OpenType math font is a typographic decision that belongs to the author).
1035 \RequirePackage{amsmath}
1036 \ifPDFTeX
1037   \RequirePackage{amssymb}
1038 \else
1039   % Unicode engine: defer the check to end of document. Only warn if the
1040   % user did NOT add either unicode-math (the modern OpenType-math route)
1041   % or amssymb (the legacy route, which they would have loaded manually).
1042   % This avoids nagging authors who already followed the documented
1043   % \usepackage{unicode-math} workflow.
1044   \AtEndDocument{%
1045     \ifpackageloaded{unicode-math}{\}%
1046     \ifpackageloaded{amssymb}{\}%

```

```

1047 \ClassWarningNoLine{coppe}{%
1048     You are compiling with a Unicode engine (LuaLaTeX or XeLaTeX) and\MessageBreak
1049     neither unicode-math nor amssymb is loaded. AMS-style symbols\MessageBreak
1050     (\string\mathbb, \string\hbar, \string\varnothing, ...) will be undefined.\MessageBreak
1051     Add to your preamble:\MessageBreak
1052     \space\space\string\usepackage{unicode-math}\MessageBreak
1053     \space\space\string\setmathfont{Latin Modern Math}\MessageBreak
1054     Do NOT load amssymb together with unicode-math -- it would clash\MessageBreak
1055     with the OpenType math font setup}}}%
1056 \fi
1057 % Margins per the SiBI manual, 9th rev. ed. (2026), 2.3: left 3cm, top 3cm,
1058 % right 2cm, bottom 2cm -- ONE-SIDED. The 2025 edition also specified verso
1059 % margins (right 3cm / left 2cm) and recommended printing the body on both
1060 % sides; the 2026 revision DELETED the verso margins entirely, following CEPG
1061 % Resolution 246/2023, which ended the printed deposit. A digital-only document
1062 % has no gutter, so the mirrored layout (previously inner=outer=2cm plus a 1cm
1063 % bindingoffset) no longer has any normative basis and is now opt-in via the
1064 % \texttt{twoside} class option, handled after \ProcessOptions below.
1065 %
1066 % headheight=15pt is given here (not patched later) so geometry's own page
1067 % arithmetic stays self-consistent: fancyhdr needs >= 14.49998pt of header
1068 % height to host the running head, and the standard 12pt default triggers
1069 % its "headheight is too small" warning. Letting geometry know up front
1070 % keeps \topmargin a derived value -- we never touch it by hand.
1071 % (geometry has no \texttt{oneside} key -- only \texttt{twoside}; with the key
1072 % absent it follows the class, which \LoadClass set to one-sided above.)
1073 % headsep is given explicitly to place the folio. 2.7 measures it the way it
1074 % measures the right margin -- to the digits themselves: "a 2cm da borda
1075 % superior, ficando o ultimo algarismo a 2cm da borda direita". The right edge
1076 % already landed at 2.02cm; the top of the digits sat at 1.87cm, too high.
1077 % geometry derives \topmargin from top - 1in - headheight - headsep, so
1078 % shortening headsep lowers the header and leaves the body top untouched:
1079 % topmargin + headheight + headsep is 13.088pt either way, and 3cm is where the
1080 % body still starts. The value is CALIBRATED, not computed -- the folio sits a
1081 % little lower than the arithmetic alone predicts, because fancyhdr's box does
1082 % not put the digits flush with the top of \headheight. Two measured points,
1083 % taken from the rendered page rather than from the PDF's font metrics:
1084 % headsep 18.07pt -> digits at 1.8669cm headsep 14.29pt -> 2.0616cm
1085 % which put 2.0000cm at 15.49pt while the folio was still set at the body size.
1086 % It is \footnotesize now (2.2(b) puts pagination in the smaller font), the
1087 % digits are shorter, and their top therefore sits lower for the same baseline:
1088 % re-measured, 15.49pt gave 2.0659cm and the value moved to 16.87pt. That is
1089 % precisely why this is a measured constant and not a computed one -- change
1090 % the folio's size or headheight and it has to be measured again, not scaled.
1091 % Verified on the rendered page: the top of the digits and their right edge come
1092 % out at the same 2.019cm, and the right margin is 2cm by construction, so that
1093 % shared 0.019cm is the ruler's own bias and not a residual error.
1094 \RequirePackage[a4paper,headheight=15pt,headsep=16.87pt,%
1095 top=3cm,bottom=2cm,left=3cm,right=2cm]{geometry}
1096 % The first paragraph of every section must be indented like any other. LaTeX's
1097 % default is the Anglo-Saxon convention -- no indent after a heading -- which is
1098 % not what Brazilian typography (nor the SiBI manual's own examples) uses.
1099 % indentfirst was mentioned only as a commented suggestion in the example
1100 % preamble, so every document built with this class came out with the first

```

```

1101 % paragraph of each section flush left. Loading it in the class fixes it for
1102 % everybody instead of asking each author to remember.
1103 %
1104 % Loaded BEFORE titlesec so titlesec's \@afterindent handling sees the patched
1105 % \@afterindentfalse.
1106 % Long unbreakable tokens -- \verb|...| in running prose, command names, URLs --
1107 % cannot be hyphenated, so TeX overfills the line rather than stretching the
1108 % spaces. example.tex alone produced 17 overfull boxes this way, the worst 86pt,
1109 % which is the "text does not always respect the margins" a recent review
1110 % reported. \emergencystretch gives TeX a final pass in which it may stretch
1111 % interword spaces instead of running into the margin. It cannot save a single
1112 % token longer than the measure, but it clears the ordinary cases.
1113 \setlength\emergencystretch{3em}
1114 \RequirePackage{indentfirst}
1115 \RequirePackage{titlesec}
1116 % --- Section heading formats (manual2024 2.6 / proposta P22) -----
1117 % chapter (numbered): "N TITLE", ALL CAPS, bold, flush left (no "Capitulo N").
1118 % APENDICE/ANEXO chapters print "APENDICE A -- Title" (R75/R76); normal chapters
1119 % keep "N Title". \appendix and \annex raise the flag (see below).
1120 \newif\if@coppeappendix \@coppeappendixfalse
1121 \titleformat{\chapter}[hang]
1122   {\normalfont\Large\bfseries}
1123   {\if@coppeappendix\MakeUppercase{\appendixname}\ \thechapter\ \textendash\else\thechapter\
1124   {1ex}\{\MakeUppercase}
1125 \titlespacing*{\chapter}{0pt}{10pt}{1.5\baselineskip}
1126 % 2.6 says the numeric indicative of a section precedes its title flush left,
1127 % and that titles WITHOUT a numeric indicative are centred -- and then lists
1128 % them, "apendice(s)" and "anexo(s)" among them. A letter is not a numeric
1129 % indicative, and the manual sets its own Annex A centred. The chapter format is
1130 % therefore re-issued when \appendix or \annex is called: same label, same
1131 % travessao, but a centred block instead of a flush-left hang. Numbered chapters
1132 % are untouched -- they DO carry a numeric indicative and stay flush left.
1133 \newcommand\coppe@appendixheadings{%
1134   \titleformat{\chapter}[block]
1135     {\normalfont\Large\bfseries\filcenter}
1136     {\MakeUppercase{\appendixname}\ \thechapter\ \textendash}
1137     {1ex}\{\MakeUppercase}%
1138   \titlespacing*{\chapter}{0pt}{10pt}{1.5\baselineskip}}
1139 \let\coppe@orig@appendix\appendix
1140 \renewcommand\appendix{%
1141   \coppe@orig@appendix\@coppeappendixtrue\coppe@appendixheadings}
1142 % Sumario (TOC) entries for APENDICE/ANEXO chapters read "APENDICE A -- Title"
1143 % / "ANEXO A -- Title" (R75/R76), not the bare "A". The label word is chosen by
1144 % which macro is written (so it is correct at .toc-render time even though the
1145 % appendixname has changed back); the chapter letter is baked in at write time.
1146 \newif\if@coppeannex \@coppeannexfalse
1147 \newcommand\coppe@tocapp[1]{%
1148   \MakeUppercase{\coppestring{appendix}}%
1149   \nobreakspace#1\nobreakspace\textendash\nobreakspace}
1150 \newcommand\coppe@tocanx[1]{%
1151   \MakeUppercase{\coppestring{annex}}%
1152   \nobreakspace#1\nobreakspace\textendash\nobreakspace}
1153 \def\@chapter[#1]#2{%
1154   \ifnum \c@secnumdepth >\m@ne

```

```

1155 \if@mainmatter
1156 \refstepcounter{chapter}%
1157 \typeout{\@chapapp\space\thechapter.}%
1158 \if@coppeappendix
1159 \if@coppeannex
1160 \addcontentsline{toc}{chapter}{\protect\coppe@tocanx{\thechapter}#1}%
1161 \else
1162 \addcontentsline{toc}{chapter}{\protect\coppe@tocapp{\thechapter}#1}%
1163 \fi
1164 \else
1165 \addcontentsline{toc}{chapter}{\protect\numberline{\thechapter}#1}%
1166 \fi
1167 \else
1168 \addcontentsline{toc}{chapter}{#1}%
1169 \fi
1170 \else
1171 \addcontentsline{toc}{chapter}{#1}%
1172 \fi
1173 \chaptermark{#1}%
1174 \addtocontents{lof}{\protect\addvspace{10\p@}}%
1175 \addtocontents{lot}{\protect\addvspace{10\p@}}%
1176 \if@twocolumn
1177 \topnewpage[\@makechapterhead{#2}]%
1178 \else
1179 \makechapterhead{#2}%
1180 \afterheading
1181 \fi}
1182 % chapter (unnumbered): centred ALL CAPS bold (errata, lists, resumo, sumario,
1183 % referencias, glossario, ... -- unnumbered headings are centred, P29-P30).
1184 \titleformat{name=\chapter,numberless}[hang]
1185 {\normalfont\Large\bfseries\filcenter}{\Opt}{\MakeUppercase}
1186 \titlespacing*{name=\chapter,numberless}{\Opt}{10pt}{1.5\baselineskip}
1187 % section: ALL CAPS, normal weight -- the \bfseries here contradicted this very
1188 % comment until v4.1, and made the secondary level as heavy as the primary one,
1189 % defeating the "destaque gradativo" of 2.6. The gradation is now
1190 % chapter = ALL CAPS bold, section = ALL CAPS regular, subsection = Title Case
1191 % bold, subsubsection = Title Case bold italic, as read from the manual.
1192 \titleformat{\section}
1193 {\normalfont\large}{\thesection}{1ex}{\MakeUppercase}
1194 % subsection: bold (author writes Title Case).
1195 \titleformat{\subsection}
1196 {\normalfont\bfseries}{\thesubsection}{1ex}{}
1197 % subsubsection: bold italic.
1198 \titleformat{\subsubsection}
1199 {\normalfont\bfseries\itshape}{\thesubsubsection}{1ex}{}
1200 % paragraph = quinary section: italic, the author capitalising only the first
1201 % word. 2.6 recommends going as deep as the quinary, and the annotation on the
1202 % 2025 manual (specs/ANOTACOES.md) describes exactly this shape.
1203 %
1204 % book makes \paragraph a RUN-IN heading, so the text continued on the same
1205 % line -- 2.6 requires "o texto deve ser iniciado em outra linha". The positive
1206 % "after" spacing in \titlespacing turns it into a displayed heading.
1207 \titleformat{\paragraph}[hang]
1208 {\normalfont\itshape}{\theparagraph}{1ex}{}

```

```

1209 \titlespacing*{\paragraph}{0pt}{2.5ex plus 1ex minus .2ex}{1.5ex plus .2ex}
1210 % book defaults secnumdepth to 2, so \subsubsection and \paragraph came out
1211 % UNNUMBERED however carefully \titleformat named \thesubsubsection. 2.6 has
1212 % the numbering following the whole chain down to the quinary, and the manual's
1213 % own sumario is four levels deep, so tocdepth follows.
1214 \setcounter{secnumdepth}{4}
1215 \setcounter{tocdepth}{4}
1216 % --- 0 sumario (2.6 e 3.1.2.1.6) -----
1217 %
1218 % Duas exigencias o governam, e a segunda tem um exemplo obrigatorio.
1219 %
1220 % A 2.6: "no sumario, as secoes devem ser grafadas conforme apresentadas no
1221 % corpo do trabalho". A gradacao do corpo -- primaria em CAIXA ALTA negrito,
1222 % secundaria em CAIXA ALTA, terciaria em negrito, quaternaria em negrito
1223 % italico, quinary em italico -- tem de reaparecer aqui. So as duas primeiras
1224 % reapareciam: da terciaria em diante o sumario saia em redondo, contradizendo
1225 % o corpo.
1226 %
1227 % A 3.1.2.1.6: os titulos "devem ser alinhados a esquerda... pela margem do
1228 % titulo do indicativo mais extenso", e "utilize como exemplo o sumario deste
1229 % trabalho". Medido no PDF do manual: o indicativo comeca em 3,00 cm nos cinco
1230 % niveis -- a margem -- e o titulo em 4,75 cm nos cinco. Uma coluna so.
1231 %
1232 % 0 \@dottedtocline do LaTeX faz o contrario. Recua progressivamente (3,00 /
1233 % 3,62 / 4,57 / 5,89 / 7,13 cm nesta classe) e da a cada nivel uma caixa de
1234 % indicativo de largura fixa. Isso empurrava o titulo quinary para 9,20 cm --
1235 % 6,2 dos 16 cm de mancha -- e, pior, a caixa fixa ESTOURA: com o indicativo
1236 % "10.10.10.10.10", que uma tese de dez capitulos alcanca, o numero entrava
1237 % 4,6 mm POR CIMA do titulo. A 2.6 quer o titulo "separado por um espaco" do
1238 % indicativo; ali nao havia espaco nenhum.
1239 %
1240 % A coluna abaixo e medida, nao arbitrada: \coppe@tocnumberline mede cada
1241 % indicativo enquanto compoe o sumario e guarda o maior; o valor vai para o
1242 % .aux e governa a passada seguinte. E literalmente "a margem do titulo do
1243 % indicativo mais extenso". Custa uma passada -- a mesma que as referencias
1244 % cruzadas ja custam.
1245 \newlength\coppe@tocnumwidth
1246 \newlength\coppe@tocnummax
1247 \newlength\coppe@tocnumsep
1248 % 0 espaco entre o indicativo e o titulo (2.6, "separado por um espaco").
1249 \setlength\coppe@tocnumsep{0.5em}
1250 % Piso: a coluna do proprio manual. Pela regra literal, um trabalho so com
1251 % capitulos teria uma coluna de meio centimetro e o titulo quase colado na
1252 % margem; o piso evita isso. A coluna nunca fica MENOR que o indicativo mais
1253 % extenso mais o espaco -- o piso so a impede de ficar estreita demais.
1254 \setlength\coppe@tocnumwidth{1.75cm}
1255 \setlength\coppe@tocnummax{\z@}
1256 \newcommand\coppe@tocnumberline[1]{%
1257   \settowidth\@tempdima{#1}%
1258   \ifdim\@tempdima>\coppe@tocnummax \global\coppe@tocnummax\@tempdima \fi
1259   \hb@xt@\coppe@tocnumwidth{#1\hfil}}
1260 % A largura medida na passada anterior chega pelo .aux, que o \document le
1261 % antes dos ganchos de \AtBeginDocument.
1262 \AtBeginDocument{%

```

```

1263 \ifundefined{coppe@tocnumsaved}{\%
1264 \setlength\@tempdima{\coppe@tocnumsaved}%
1265 \addtolength\@tempdima\coppe@tocnumsep
1266 \ifdim\@tempdima>\coppe@tocnumwidth
1267 \setlength\coppe@tocnumwidth\@tempdima
1268 \fi}}
1269 \AtEndDocument{%
1270 \ifdim\coppe@tocnummax>\z@
1271 \if@filesw
1272 \immediate\write\@auxout{\string\gdef\string\coppe@tocnumsaved
1273 {\the\coppe@tocnummax}}}%
1274 \fi
1275 \setlength\@tempdima\coppe@tocnummax
1276 \addtolength\@tempdima\coppe@tocnumsep
1277 \ifdim\@tempdima>\coppe@tocnumwidth
1278 \ClassWarning{coppe}{A coluna do sumario cresceu; rode LaTeX de novo}%
1279 \fi
1280 \fi}
1281 % Uma linha do sumario. #1 nivel, #2 fonte, #3 entrada, #4 folha. A entrada
1282 % traz \numberline{N} seguido do titulo; o \hskip-\leftskip puxa a primeira
1283 % linha de volta a margem, e o \leftskip deixa as linhas seguintes de um
1284 % titulo longo alinhadas pela coluna do titulo -- nunca debaixo do numero.
1285 % A folha herda a fonte do nivel, como no sumario do manual.
1286 \newcommand\coppe@tocline[4]{%
1287 \ifnum #1>\c@tocdepth\else
1288 \vskip\z@ \@plus.2\p@
1289 {\leftskip\z@ \rightskip\@tocrmarg \parfillskip-\rightskip
1290 \parindent\z@ \interlinepenalty\@M
1291 \leavevmode
1292 #2%
1293 \advance\leftskip\coppe@tocnumwidth
1294 \null\nobreak\hskip-\leftskip
1295 #3\nobreak
1296 \leaders\hbox{$\m@th\mkern\@dotsep mu\hbox{.}\mkern\@dotsep mu$}%
1297 \hfill\nobreak
1298 \hb@xt@\@pnumwidth{\hss #4}\par}%
1299 \fi}
1300 % O pontilhado do manual e mais fechado que o do book; \@dotsep e a distancia
1301 % entre os pontos, em unidades matematicas.
1302 \renewcommand\@dotsep{2}
1303 % Os cinco niveis. O \MakeUppercase vai so nos dois primeiros porque so eles
1304 % saem em caixa alta no corpo; ele atravessa os algoritmos sem toca-los e
1305 % preserva comandos robustos, inclusive os do hyperref.
1306 %
1307 % O nivel 1 nao ganha mais o \addvspace{1.0em} do book: o sumario do manual
1308 % tem espacamento uniforme, sem folga extra antes da secao primaria. As
1309 % penalidades ficam, para que uma entrada de capitulo nao seja abandonada
1310 % sozinha no pe da folha.
1311 \renewcommand*\l@chapter[2]{%
1312 \addpenalty{-\@highpenalty}%
1313 \coppe@tocline{0}{\bfseries}{\MakeUppercase{#1}}{#2}%
1314 \penalty\@highpenalty}
1315 \renewcommand*\l@section[2]{%
1316 \coppe@tocline{1}{\normalfont}{\MakeUppercase{#1}}{#2}}

```



```

1317 \renewcommand*\l@section[2]{%
1318   \coppe@tocline{2}{\normalfont\bfseries}{#1}{#2}}
1319 \renewcommand*\l@subsubsection[2]{%
1320   \coppe@tocline{3}{\normalfont\bfseries\itshape}{#1}{#2}}
1321 \renewcommand*\l@paragraph[2]{%
1322   \coppe@tocline{4}{\normalfont\itshape}{#1}{#2}}
1323 % --- Captions (manual2024 2.10-2.11 / proposta P6): smaller font, ABNT
1324 %   "Figura 1 -- titulo" travessao separator (sources handled by \source). ---
1325 \RequirePackage{caption}
1326 % justification=centering makes MULTI-LINE captions centre like the one-line
1327 % ones. Without it, caption's singlelinecheck centres a caption that fits on one
1328 % line and justifies one that wraps, so in a document with mixed caption lengths
1329 % a couple of tables looked left-aligned while the rest looked centred -- which
1330 % is what a recent review noticed about tables 5.1 and 5.2. 2.10 does not
1331 % prescribe the alignment; it does not tolerate two alignments either.
1332 \captionsetup{font=footnotesize,labelsep=endash,justification=centering}
1333 % subcaption (NÃO subfig/subfigure) for subfigures / subtables; load right after
1334 % caption so it picks up the ABNT captionsetup above.
1335 \RequirePackage{subcaption}
1336 % --- Floats: caption on TOP and a mandatory "Fonte:" line at the bottom.
1337 %   manual2024 2.10-2.11 (R46/R51: identification on top; R48/R52: the source
1338 %   at the bottom is mandatory). The float package's plaintop style forces the
1339 %   caption above the body for figure, table and the new quadro float. -----
1340 \RequirePackage{float}
1341 \floatstyle{plaintop}
1342 \restylefloat{figure}
1343 \restylefloat{table}
1344 % New "quadro" float -- "Quadro" (pt) / "Frame" (en) (R53) -- with its own list
1345 % file (.loq). A quadro's data is bordered on all sides (drawn by the author),
1346 % unlike a tabela (rules on top and bottom only).
1347 \providecommand{\quadroname}{\coppestring{frame}}
1348 \providecommand{\listquadroname}{\coppestring{listframe}}
1349 \newfloat{quadro}{htbp}{loq}[chapter]
1350 \floatname{quadro}{\quadroname}
1351 \providecommand{\quadroautorefname}{\quadroname}
1352 % Format the Lista de Quadros entries exactly like the Lista de Figuras ones
1353 % (float's own \l@quadro lays the number/title out incorrectly).
1354 \let\l@quadro\l@figure
1355 \newcommand\listofquadros{%
1356   \chapter*{\listquadroname}%
1357   \coppe@listtoc{\listquadroname}%
1358   \@mkboth{\MakeUppercase\listquadroname}{\MakeUppercase\listquadroname}%
1359   \@starttoc{loq}}
1360 % \source / \fonte: the illustration source (ABNT-mandatory). Typeset below the
1361 % float as a smaller, single-spaced, centred line; it does NOT advance any
1362 % caption counter. \cpsource/\cpfonte are clash-safe aliases (the rename rule);
1363 % the label is "Fonte" (pt) / "Source" (en).
1364 \newcommand{\cpsourcenamename}{\coppestring{source}}
1365 \newcommand{\cpsource}[1]{%
1366   \par\addvspace{\smallskipamount}%
1367   {\footnotesize\singlespacing\centering\cpsourcenamename: #1\par}}
1368 \let\cpfonte\cpsource
1369 \@ifundefined{source}{\let\source\cpsource}{}%
1370 \@ifundefined{fonte}{\let\fonte\cpsource}{}%

```

```

1371 % \newcoppefloat{env}{caption name}{list title}: declare an extra caption-on-top
1372 % float (e.g. "picture") that supports \source and gains a \listof<env>s command.
1373 \newcommand{\newcoppefloat}[3]{%
1374   {\floatstyle{plaintop}\newfloat{#1}{htbp}{lo#1}}%
1375   \floatname{#1}{#2}%
1376   \expandafter\newcommand\csname listof#1s\endcsname{%
1377     \chapter*{#3}\coppe@listtoc{#3}%
1378     \@mkboth{\MakeUppercase{#3}}{\MakeUppercase{#3}}%
1379     \@starttoc{lo#1}}}%
1380 % --- Footnotes (manual2024 / R16, R8, R21): single spacing and a 5 cm rule
1381 %   (filete) from the left margin; footnotesize is the book default. -----
1382 \let\coppe@orig@footnotetext\@footnotetext
1383 \long\def\@footnotetext#1{\coppe@orig@footnotetext{\singlespacing#1}}
1384 \renewcommand\footnoterule{\kern-3\p@\hrule\@width 5cm\kern 2.6\p@}
1385 % --- alineas: list keyed by lowercase letters a) b) c); subitems with "--"
1386 %   (manual2024 2.6 / proposta P24). -----
1387 \RequirePackage{enumitem}
1388 \newlist{alineas}{enumerate}{2}
1389 \setlist{alineas,1}{label=\alph*},leftmargin=1.8em,itemsep=0pt}
1390 \setlist{alineas,2}{label=--,leftmargin=1.8em,itemsep=0pt}
1391 % --- Page numbering: arabic in the top OUTER corner on textual pages
1392 %   (manual2024 2.7 / proposta P25-P26). Pre-textual numbering is set by
1393 %   \frontmatter (unnumbered by default; roman with the numeraisromanos
1394 %   class option). -----
1395 % Every list goes through \@starttoc, which allocates a stream the first time
1396 % it sees an extension. That is where the ceiling is hit in practice, so that
1397 % is where the class checks -- one \ifnum before the allocation, and an error
1398 % that names the three ways out instead of the kernel's bare "No room".
1399 %
1400 % \count17 is the write allocation counter, and 15 is the last usable stream.
1401 % Under LuaTeX there are 128 and the counter is managed differently, so the
1402 % check is skipped there; the same is true when morewrites is in charge.
1403 % The wrap waits for \begin{document} on purpose: hyperref redefines
1404 % \@starttoc as well, and it is loaded at the very end of this class, so a
1405 % \let taken here would be thrown away.
1406 \newcommand\coppe@checkwrites[1]{%
1407   \ifdefined\directlua\else
1408     \if@coppemorewrites\else
1409       \ifundefined{tf@#1}{%
1410         \ifnum\count17>14
1411           \ClassError{coppe}{Acabaram os 16 fluxos de escrita do TeX}{%
1412             Este trabalho pede mais listas do que o pdfTeX consegue manter
1413             abertas ao mesmo tempo.\MessageBreak
1414             Tres saidas, nesta ordem de preferencia:\MessageBreak
1415             (1) compile com lualatex, que tem 128 fluxos;\MessageBreak
1416             (2) passe a opcao 'morewrites' a classe coppe;\MessageBreak
1417             (3) remova uma lista que o trabalho nao usa.\MessageBreak
1418             A lista que faltou agora foi a de extensao '#1', mas o problema
1419             nao e dela: e a soma de todas.}%
1420           \fi}{}%
1421         \fi
1422       }%
1423 % The wrapping itself lives in a macro of its own rather than inside the hook:
1424 % \AtBeginDocument stores its argument by expansion, and a parameter written

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1425 % there does not survive it -- the document died with "Parameters must be
1426 % numbered consecutively" at \begin{document}. One level of nesting, one
1427 % doubling, and the hook only has to call it.
1428 \newcommand\coppe@wrapstarttoc{%
1429   \let\coppe@orig@starttoc\starttoc
1430 % \numberline mede o indicativo (ver \coppe@tocnumberline) apenas no sumario:
1431 % a lista de figuras e a de tabelas numeram ilustracoes, nao secoes, e nao tem
1432 % por que compartilhar a coluna.
1433 \def\starttoc##1{%
1434   \coppe@checkwrites{##1}%
1435   \begingroup
1436     \def\coppe@@ext{##1}\def\coppe@@toc{toc}%
1437     \ifx\coppe@@ext\coppe@@toc \let\numberline\coppe@tocnumberline \fi
1438     \coppe@orig@starttoc{##1}%
1439   \endgroup}}
1440 \AtBeginDocument{\coppe@wrapstarttoc}
1441 \RequirePackage{fancyhdr}
1442 % \headheight is set to 15pt by geometry up above (see the geometry call),
1443 % which is what fancyhdr needs (>= 14.49998pt) for the running head.
1444 % 2.2(b) lists what is set in "fonte menor e uniforme": long quotations,
1445 % footnotes, PAGINATION, and the captions of illustrations and tables. The
1446 % folio was the one item on that list still coming out at the body size of 12pt.
1447 % It is \footnotesize now, which is the same smaller size the other three
1448 % already use -- "uniforme" is part of the requirement, not just "menor".
1449 % Pre-textual lists reach the sumario only under the listasnosumario option
1450 % (see the option's declaration for the reason). Post-textual material calls
1451 % \addcontentsline directly and is unaffected.
1452 \newcommand\coppe@listtoc[1]{%
1453   \if@coppe@listastoc\addcontentsline{toc}{chapter}{#1}\fi
1454 % \coppe@glosspre marks the two PRE-textual uses of the glossary environment --
1455 % the list of abbreviations and the list of symbols -- apart from the genuine
1456 % post-textual Glossario, which shares the same environment but must appear in
1457 % the sumario.
1458 \newif\if@coppeglosspre \@coppeglossprefalse
1459 \newcommand\coppe@folio{{\footnotesize\thepage}}
1460 \fancypagestyle{coppe}{{fancyhf}{} \fancyhead[R0,LE]{\coppe@folio}%
1461   \renewcommand{\headrulewidth}{0pt}}
1462 % Blank pages inserted by \cleardoublepage (two-sided) must be empty: no page
1463 % number may leak onto them (e.g. the verso right after the cover).
1464 \renewcommand{\cleardoublepage}{%
1465   \clearpage
1466   \if@twoside
1467     \ifodd\c@page\else
1468       \hbox{}\thispagestyle{empty}\newpage
1469     \if@twocolumn\hbox{}\newpage\fi
1470   \fi
1471 }
1472 \def\CoppeTeX{{\rm C\kern-.05em{\sc o\kern-.025em p\kern-.025em
1473   p\kern-.025em e}}\kern-.08em
1474 T\kern-.1667em\lower.5ex\hbox{E}\kern-.125emX\spacefactor1000}
1475 \newboolean{maledoc}
1476 \setboolean{maledoc}{false}
1477 % maledoc diz apenas que o substantivo do tipo de documento e masculino
1478 % ("Exame de Qualificacao ... apresentado", "SUBMETIDO"): serve a concordancia,

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1479 % e nada mais. Quem NAO vai a deposito -- e portanto nao tem ficha
1480 % catalografica nem folha da Coleta CAPES -- e o exame de qualificacao, que e
1481 % outra coisa. As duas condicoes coincidem hoje por acaso, e por isso ganham
1482 % cada uma o seu booleano.
1483 \newif\if@coppeexame \@coppeexamefalse
1484 %
1485 % Class options.
1486 % Multilingual model (CoppeTeX 4.x, branch nlinguas). The class holds three
1487 % fixed language slots:
1488 % * main      -- the language of the body, captions, and the main abstract.
1489 %              Set by a class option: brazilian (default), english,
1490 %              spanish, french, italian, or any user-supplied language
1491 %              that ships a coppe-lang-<name>.def file.
1492 % * foreign    -- the language used by the foreignabstract environment and
1493 %              by \foreign@... registers. Defaults to english (or to
1494 %              brazilian when main=english) for back-compat.
1495 % * third      -- optional, loaded on demand with \usecoppelanguage{<name>}
1496 %              in the preamble (for citations or quotations in a third
1497 %              language). Has no class option of its own.
1498 % Note: the legacy \if@english boolean (defined and toggled here in
1499 % v3.x for backward compatibility) was removed in v4.0 -- nothing
1500 % inside the class or in the biblatex style files (.bbx/.cbx/.dbx/.ltx)
1501 % tests it. Code that needs to branch on the main language should test
1502 % \ifx\coppe@mainlang\coppe@englishstr instead.
1503 \def\coppe@mainlang{brazilian}
1504 \def\coppe@foreignlang{english}
1505 \DeclareOption{brazilian}{%
1506   \def\coppe@mainlang{brazilian}\def\coppe@foreignlang{english}}
1507 \DeclareOption{english}{%
1508   \def\coppe@mainlang{english}\def\coppe@foreignlang{brazilian}}
1509 \DeclareOption{spanish}{%
1510   \def\coppe@mainlang{spanish}\def\coppe@foreignlang{english}}
1511 \DeclareOption{french}{%
1512   \def\coppe@mainlang{french}\def\coppe@foreignlang{english}}
1513 \DeclareOption{italian}{%
1514   \def\coppe@mainlang{italian}\def\coppe@foreignlang{english}}
1515 \DeclareOption{msc}{%
1516   \newcommand{\@degree}{M.Sc.}
1517   \newcommand{\@degreename}{Mestrado}
1518   \newcommand{\local@degname}{Mestre}
1519   \newcommand{\foreign@degname}{Master}
1520   \newcommand\local@doctype{Disserta{\c c}{\~ a}o}
1521   \newcommand\foreign@doctype{Dissertation}
1522 }
1523 \DeclareOption{dsceexam}{%
1524   \newcommand{\@degree}{D.Sc.}
1525   \newcommand{\@degreename}{Doutorado}
1526   \newcommand{\local@degname}{Doutor}
1527   \newcommand{\foreign@degname}{Doctor}
1528   \setboolean{maledoc}{true}\@coppeexamtrue
1529   \newcommand\local@doctype{Exame de Qualifica{\c c}{\~ a}o}
1530   \newcommand\foreign@doctype{Qualifying Exam}
1531 }
1532 \DeclareOption{msceexam}{%

```

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1533 \newcommand{\@degree}{M.Sc.}
1534 \newcommand{\@degreename}{Mestrado}
1535 \newcommand{\local@degname}{Mestre}
1536 \newcommand{\foreign@degname}{Master}
1537 \setboolean{maledoc}{true}\@coppeexamtrue
1538 \newcommand\local@doctype{Exame de Qualifica{\c c}{\~ a}o}
1539 \newcommand\foreign@doctype{Qualifying Exam}
1540 }
1541 \DeclareOption{dsc}{%
1542 \newcommand{\@degree}{D.Sc.}
1543 \newcommand{\@degreename}{Doutorado}
1544 \newcommand{\local@degname}{Doutor}
1545 \newcommand{\foreign@degname}{Doctor}
1546 \newcommand\local@doctype{Tese}
1547 \newcommand\foreign@doctype{Thesis}
1548 }
1549 \newif\if@coppenumeric\@coppenumericfalse
1550 \DeclareOption{numbers}{\@coppenumerictrue}
1551 % numeraisromanos: show roman numerals on the pre-textual pages. Default: the
1552 % pre-textual pages are counted but not numbered (manual2024 2.7).
1553 \newif\if@copperoman\@copperomanfalse
1554 \DeclareOption{numeraisromanos}{\@copperomantrue}

```

Default line spacing is one-and-a-half; the doublespacing option switches to double spacing. Since `setspace` is loaded at the end of the class (after `hyperref`), the actual `\onehalfspacing`/`\doublespacing` call is deferred: the option only flips a flag, which the late-load block reads.

```

1555 \newif\if@coppedoublespace\@coppedoublespacefalse
1556 \DeclareOption{doublespacing}{\@coppedoublespacetrue}
1557 % Two-sided layout is opt-in since v4.1: the 2026 manual dropped the verso
1558 % margins and the deposit is digital (CEPG 246/2023). Given \texttt{twoside},
1559 % the mirrored 3cm inner / 2cm outer layout is restored after \ProcessOptions.
1560 % With no \fichacatalografica file given, the folha adicional leaves a framed
1561 % blank where the ficha goes. The rascunhoficha option fills that blank with the
1562 % class's own LaTeX-composed ficha instead -- useful while writing, never valid
1563 % for deposit: the official ficha comes from the SiBI generator.
1564 % Annex D of the manual draws one signature rule per board member, with the
1565 % identification underneath. The COPPE folha de aprovacao has listed the names
1566 % compactly, without rules, for years. The compact list stays the default and
1567 % the rules are opt-in, so existing theses do not change shape overnight.
1568 % PDF/A output. Opt-in for now: 2.2(d) of the 2026 manual requires the deposited
1569 % file to be PDF/A, but the constraints (embedded fonts, colour profile, no
1570 % encryption, restricted transparency) can trip up a document that pulls in
1571 % arbitrary images or packages, and breaking existing theses to enforce it would
1572 % be worse than letting the author switch it on when depositing.
1573 \newif\if@coppepdfa \@coppepdfafalse
1574 \DeclareOption{pdfa}{\@coppepdfatrue}
1575 \newif\if@coppeassin \@coppeassinfalse
1576 \DeclareOption{assinaturas}{\@coppeassintrue}
1577 % As paginas de resumo listam os ORIENTADORES abaixo do titulo. O coorientador
1578 % nao aparece ali, e isso e tradicao da COPPE, nao exigencia do manual: a
1579 % 3.1.2.1.4 descreve o resumo e nao lista quem assina a orientacao. Como ha
1580 % Programas que preferem ve-lo, a opcao acrescenta os coorientadores nas tres
1581 % paginas de resumo, no mesmo formato da folha de rosto. Sem a opcao nada muda.

```

```

1582 \newif\if@coppecoadvresumo \@coppecoadvresumofalse
1583 \DeclareOption{coorientador}{\@coppecoadvresumotrue}
1584 % 3.1.2.1.6 says the sumario enumerates the divisions "na mesma ordem e grafia
1585 % em que a materia nele se sucede" and adds "inclusive os elementos
1586 % pos-textuais" -- and then settles the question by pointing at its own sumario
1587 % as the model. That sumario opens at "1 INTRODUCAO": the lists of figures, of
1588 % tables, of abbreviations and of symbols come BEFORE the sumario and are not
1589 % listed in it, which is also the rule of NBR 6027 for pre-textual elements.
1590 % The class was adding all of them, so every thesis carried six entries the
1591 % manual's own example does not have. They come out. Post-textual elements --
1592 % Referencias, Glossario, Apendices, Anexos, Indice -- stay, as required.
1593 %
1594 % Programas that want the old behaviour ask for it by name.
1595 \newif\if@coppelistastoc \@coppelistastocfalse
1596 \DeclareOption{listasnosumario}{\@coppelistastoctrue}
1597 \newif\if@copperascunho \@copperascunhofalse
1598 \DeclareOption{rascunhoficha}{\@copperascunhotrue}
1599 \newif\if@coppetwoside \@coppetwosidefalse
1600 \DeclareOption{twoside}{\@coppetwosidetrue}
1601 \DeclareOption{oneside}{\@coppetwosidefalse}
1602 % --- Fluxos de escrita -----
1603 % TeX has SIXTEEN \write streams and no more. Every list costs one for the
1604 % whole run: the .toc, the .lof, the .lot, and one for each list this class
1605 % adds (.loq for quadros, .lol for programas, .loa for algoritmos), plus the
1606 % .abx and .syx behind the lists of abbreviations and of symbols, the .idx of
1607 % the index, hyperref's .out, biblalex's .bcf and the kernel's .aux and friends.
1608 % A work that asks for all of them sits exactly at the ceiling.
1609 %
1610 % The kernel's failure in that situation -- "No room for a new \write" -- names
1611 % neither the cause nor the way out, and lands on whichever list happens to be
1612 % requested last, which is usually \tableofcontents and has nothing to do with
1613 % it. The class therefore checks before the allocation and says what is
1614 % happening (see \coppe@starttoc below).
1615 %
1616 % There are three ways out and the class chooses none of them for the author:
1617 %
1618 % * compile with LuaLaTeX. It has 128 streams; nothing else has to change,
1619 % and the class supports it.
1620 % * let the class load 'morewrites', which multiplexes the streams so the
1621 % ceiling disappears under pdfLaTeX.
1622 % * drop a list the work does not use. Each one returns a stream.
1623 %
1624 % The DEFAULT is the second one, and it is on. An error telling the author to
1625 % pass an option is advice for someone who already knows: a student two days
1626 % from the deposit reads "No room for a new \write" and concludes the class is
1627 % broken. So under pdfTeX the class loads morewrites by itself.
1628 %
1629 % Three things keep that from becoming a burden:
1630 %
1631 % * it is loaded ONLY IF INSTALLED. Missing, the class carries on without it
1632 % and warns; morewrites does not become a hard dependency of a work that
1633 % would never have reached the ceiling anyway.
1634 % * 'semmorewrites' turns it off, for whoever hits a conflict. morewrites
1635 % patches \newwrite, \openout and \write and buffers what is written until

```

```

1636 %      shipout, so a package that writes a file and reads it back in the SAME run
1637 %      can misbehave. That is the real cost of this default, and it is why the
1638 %      escape exists.
1639 %      * 'morewrites' still exists and now means "I insist": with it, a missing
1640 %      package is an error rather than a warning.
1641 %
1642 % Under LuaTeX none of this applies and nothing is loaded.
1643 \newif\if@coppemorewrites \@coppemorewritetrue
1644 \newif\if@coppemwforced \@coppemwforcedfalse
1645 \DeclareOption{morewrites}{\@coppemorewritetrue\@coppemwforcedtrue}
1646 \DeclareOption{semmorewrites}{\@coppemorewritesfalse}
1647 \ProcessOptions\relax
1648 % morewrites has to come before anything else allocates, so it goes first
1649 % thing after the options are read.
1650 \if@coppemorewrites
1651   \ifdefined\directlua
1652 % Under LuaTeX there is nothing to solve: 128 streams already. The flag is
1653 % cleared too, so the \@starttoc guard further down knows the ceiling is not
1654 % in play here either.
1655   \ClassInfo{coppe}{morewrites dispensado: o LuaTeX ja tem 128 fluxos}%
1656   \@coppemorewritesfalse
1657 \else
1658   \IfFileExists{morewrites.sty}
1659   {\RequirePackage{morewrites}}
1660   {\if@coppemwforced
1661     \ClassError{coppe}{Opcao 'morewrites' pedida, mas morewrites.sty nao
1662       esta instalado}{Instale o pacote morewrites, ou compile com lualatex,
1663       que nao precisa dele.}%
1664     \else
1665       \ClassWarningNoLine{coppe}{0 pacote morewrites nao esta instalado.\MessageBreak
1666         Sem ele o pdfTeX so aguenta 16 fluxos de escrita, e um trabalho\MessageBreak
1667         com todas as listas passa disso. Se a compilacao parar por falta\MessageBreak
1668         de fluxo, instale morewrites ou compile com lualatex}%
1669       \@coppemorewritesfalse
1670     \fi}%
1671 \fi
1672 \fi
1673 % \LoadClass and geometry ran before \ProcessOptions (see above), so the
1674 % two-sided layout can only be applied here. geometry reads left/right as
1675 % inner/outer once twoside is on, so the 3cm/2cm pair set above becomes the
1676 % mirrored gutter with no further arithmetic. \@twosidetrue is what book's own
1677 % twoside option sets, and is what \cleardoublepage and fancyhdr test.
1678 % In one-sided mode fancyhdr treats every page as odd, so the \fancyhead[RO,LE]
1679 % declarations elsewhere already put the folio in the top right corner of every
1680 % folha, as 2.7 requires.
1681 \if@coppetwoside
1682   \@twosidetrue
1683   \geometry{twoside}%
1684 \fi
1685 % Built-in language names (used by the loader and the dispatcher below).
1686 \def\coppe@brazilianstr{brazilian}
1687 \def\coppe@englishstr{english}
1688 % Load Babel with the foreign language first and the main language last
1689 % (Babel treats the LAST option in the list as the main language). With

```

```

1690 % pt-main this produces [english,brazilian]{babel} and with en-main
1691 % [brazilian,english] -- identical to the historical \if@english calls.
1692 \edef\coppe@babelopts{\coppe@foreignlang,\coppe@mainlang}
1693 % Some Babel modules change LaTeX beyond translating captions, and one of
1694 % those changes breaks the norm. With Spanish as the MAIN language,
1695 % babel-spanish redefines \numberline to put a period after the section
1696 % number, so the sumario reads "2.1. SECCION SECUNDARIA" while the section
1697 % heading itself -- which the class formats -- reads "2.1 SECCION SECUNDARIA".
1698 % The 2.6 separates the indicative from the title by a space (NBR 6024) and
1699 % the 3.1.2.1.6 wants the sumario to reproduce the text; the period fails
1700 % both. 'es-nosectiondot' turns off that single adjustment and leaves the rest
1701 % of babel-spanish (quoting, shorthands, roman numerals, captions) alone.
1702 % \coppe@babelextra@<lang> is the general hook: it applies to the MAIN
1703 % language only, which is where these layout modules act, and it is kept out
1704 % of \coppe@babelopts because the language-pack loader below iterates that
1705 % list and would look for coppe-lang-es-nosectiondot.def.
1706 \@namedef{coppe@babelextra@spanish}{,es-nosectiondot}
1707 \let\coppe@babelload\coppe@babelopts
1708 \@ifundefined{coppe@babelextra@\coppe@mainlang}{}{%
1709   \edef\coppe@babelload{\coppe@babelload\@nameuse{coppe@babelextra@\coppe@mainlang}}
1710 \expandafter\RequirePackage\expandafter[\coppe@babelload]{babel}
1711 % --- Multilingual string dispatcher -----
1712 % \copperdefstring{<lang>}{<key>}{<value>} adds one entry to the table for
1713 % language <lang>. The three retrievers below look strings up either at the
1714 % current babel language (\coppestring -- follows \selectlanguage), or pinned
1715 % to the main slot (\coppemainstring) or the foreign slot (\coppeforeignstring).
1716 \providecommand{\copperdefstring}[3]{%
1717   \expandafter\gdef\csname coppe@str@#1@#2\endcsname{#3}}
1718 \providecommand{\coppestring}[1]{\@nameuse{coppe@str\@language @#1}}
1719 \providecommand{\coppemainstring}[1]{\@nameuse{coppe@str\coppe@mainlang @#1}}
1720 \providecommand{\coppeforeignstring}[1]{\@nameuse{coppe@str\coppe@foreignlang @#1}}
1721 % Load an extra language pack at any point in the preamble (third-language
1722 % slot for citations or quotations beyond the main+foreign pair).
1723 \providecommand{\usecoppelanguage}[1]{%
1724   \def\coppe@@cur{#1}%
1725   \ifx\coppe@@cur\coppe@@brazilianstr\else
1726     \ifx\coppe@@cur\coppe@@englishstr\else
1727       \InputIfFileExists{coppe-lang-#1.def}{}
1728       {\ClassWarning{coppe}{Language pack coppe-lang-#1.def not found}}%
1729     \fi\fi}
1730 % \titlein{<lang>}{<text>}: record the title in any language. Will be
1731 % consulted by the cover/folha renderers once they are migrated in step 2.
1732 % In step 1 it is just a no-op store -- the historical \title / \foreigntitle
1733 % still drive every visible site, so behavior for pt/en is unchanged.
1734 \providecommand{\titlein}[2]{%
1735   \expandafter\gdef\csname coppe@title@#1\endcsname{#2}}
1736 % --- Built-in string tables: brazilian + english. The values mirror the
1737 % literals currently scattered through the class; step-1 only populates the
1738 % tables -- no site reads them yet, so the rendered output is unchanged. ----
1739 \copperdefstring{brazilian}{appendix}{\Ap~endice}
1740 \copperdefstring{english}{appendix}{Appendix}
1741 \copperdefstring{brazilian}{annex}{Anexo}
1742 \copperdefstring{english}{annex}{Annex}
1743 \copperdefstring{brazilian}{frame}{Quadro}

```


1744 \copperdefstring{english} {frame} {Frame}
1745 \copperdefstring{brazilian}{listframe} {Lista de Quadros}
1746 \copperdefstring{english} {listframe} {List of Frames}
1747 \copperdefstring{brazilian}{source} {Fonte}
1748 \copperdefstring{english} {source} {Source}
1749 \copperdefstring{brazilian}{program} {Programa}
1750 \copperdefstring{english} {program} {Program}
1751 \copperdefstring{brazilian}{listprogram} {Lista de Programas}
1752 \copperdefstring{english} {listprogram} {List of Programs}
1753 \copperdefstring{brazilian}{references} {Refer\~encias}
1754 \copperdefstring{english} {references} {References}
1755 \copperdefstring{brazilian}{in:} {em}
1756 \copperdefstring{english} {in:} {in}
1757 \copperdefstring{brazilian}{art} {Ilustra\c{c}\~ao}
1758 \copperdefstring{english} {art} {Art}
1759 \copperdefstring{brazilian}{platform} {Plataforma}
1760 \copperdefstring{english} {platform} {Platform}
1761 \copperdefstring{brazilian}{directedby} {Dire\c{c}\~ao}
1762 \copperdefstring{english} {directedby} {Directed by}
1763 \copperdefstring{brazilian}{producedby} {Produ\c{c}\~ao}
1764 \copperdefstring{english} {producedby} {Produced by}
1765 \copperdefstring{brazilian}{listabbreviation}{Lista de Abreviaturas e Siglas}
1766 \copperdefstring{english} {listabbreviation}{List of Abbreviations and Acronyms}
1767 \copperdefstring{brazilian}{listsymbol} {Lista de S{\`i}mbolos}
1768 \copperdefstring{english} {listsymbol} {List of Symbols}
1769 \copperdefstring{brazilian}{glossary} {Gloss{\`a}rio}
1770 \copperdefstring{english} {glossary} {Glossary}
1771 \copperdefstring{brazilian}{advisor} {Orientador}
1772 \copperdefstring{english} {advisor} {Advisor}
1773 \copperdefstring{brazilian}{advisors} {Orientadores}
1774 \copperdefstring{english} {advisors} {Advisors}
1775 \copperdefstring{brazilian}{dept} {Programa}
1776 \copperdefstring{english} {dept} {Department}
1777 \copperdefstring{brazilian}{approvedmale} {Aprovado por}
1778 \copperdefstring{brazilian}{approvedfemale}{Aprovada por}
1779 \copperdefstring{brazilian}{approvedonmale} {Aprovado em}
1780 \copperdefstring{brazilian}{approvedonfemale}{Aprovada em}
1781 \copperdefstring{brazilian}{keywordsname} {Palavras-chave}
1782 \copperdefstring{brazilian}{volumename} {Volume}
1783 \copperdefstring{brazilian}{coadvisor} {Coorientador}
1784 \copperdefstring{brazilian}{coadvisors} {Coorientadores}
1785 \copperdefstring{brazilian}{ofname} {de}
1786 \copperdefstring{english} {approvedmale} {Approved by}
1787 \copperdefstring{english} {approvedfemale}{Approved by}
1788 \copperdefstring{english} {approvedonmale} {Approved on}
1789 \copperdefstring{english} {approvedonfemale}{Approved on}
1790 \copperdefstring{english} {keywordsname} {Keywords}
1791 \copperdefstring{english} {volumename} {Volume}
1792 \copperdefstring{english} {coadvisor} {Co-advisor}
1793 \copperdefstring{english} {coadvisors} {Co-advisors}
1794 \copperdefstring{english} {ofname} {of}
1795 % Institutional names: always Portuguese on the cover/folha (UFRJ template).
1796 \copperdefstring{brazilian}{universityname}{Universidade Federal do Rio de Janeiro}
1797 \copperdefstring{english} {universityname}{Universidade Federal do Rio de Janeiro}

```

1798 \copperdefstring{brazilian}{cityname}      {Rio de Janeiro}
1799 \copperdefstring{english} {cityname}        {Rio de Janeiro}
1800 \copperdefstring{brazilian}{statename}      {RJ}
1801 \copperdefstring{english} {statename}        {RJ}
1802 \copperdefstring{brazilian}{countryname}    {Brasil}
1803 \copperdefstring{english} {countryname}      {Brasil}
1804 % Month names.
1805 \copperdefstring{brazilian}{monthname1} {Janeiro}
1806 \copperdefstring{brazilian}{monthname2} {Fevereiro}
1807 \copperdefstring{brazilian}{monthname3} {Mar{\c c}o}
1808 \copperdefstring{brazilian}{monthname4} {Abril}
1809 \copperdefstring{brazilian}{monthname5} {Maio}
1810 \copperdefstring{brazilian}{monthname6} {Junho}
1811 \copperdefstring{brazilian}{monthname7} {Julho}
1812 \copperdefstring{brazilian}{monthname8} {Agosto}
1813 \copperdefstring{brazilian}{monthname9} {Setembro}
1814 \copperdefstring{brazilian}{monthname10}{Outubro}
1815 \copperdefstring{brazilian}{monthname11}{Novembro}
1816 \copperdefstring{brazilian}{monthname12}{Dezembro}
1817 \copperdefstring{english}{monthname1} {January}
1818 \copperdefstring{english}{monthname2} {February}
1819 \copperdefstring{english}{monthname3} {March}
1820 \copperdefstring{english}{monthname4} {April}
1821 \copperdefstring{english}{monthname5} {May}
1822 \copperdefstring{english}{monthname6} {June}
1823 \copperdefstring{english}{monthname7} {July}
1824 \copperdefstring{english}{monthname8} {August}
1825 \copperdefstring{english}{monthname9} {September}
1826 \copperdefstring{english}{monthname10}{October}
1827 \copperdefstring{english}{monthname11}{November}
1828 \copperdefstring{english}{monthname12}{December}
1829 % algorithm2e package option for this language (empty -> english default).
1830 \copperdefstring{brazilian}{algorithm2eopt}{portuguese}
1831 \copperdefstring{english} {algorithm2eopt}{}
1832 % Babel package name for the foreignabstract \begin{otherlanguage}{...}.
1833 \copperdefstring{brazilian}{babelname}{brazilian}
1834 \copperdefstring{english} {babelname}{english}
1835 % Auto-load language packs for any non-built-in language listed in babelopts.
1836 % Built-ins (brazilian, english) ship their string tables inside coppe.cls
1837 % (defined just above) and need no extra file; others (spanish/french/italian
1838 % /...) ship as coppe-lang-<name>.def alongside the class. The dispatcher
1839 % must already be defined when these run, so the loader sits AFTER the tables.
1840 \@for\coppe@@lang:=\coppe@babelopts\do{%
1841   \edef\coppe@@cur{\coppe@@lang}%
1842   \ifx\coppe@@cur\coppe@@brazilianstr\else
1843     \ifx\coppe@@cur\coppe@@englishstr\else
1844       \InputIfFileExists{coppe-lang-\coppe@@cur.def}
1845       {\ClassInfo{coppe}{Loaded language pack coppe-lang-\coppe@@cur.def}}
1846       {\ClassError{coppe}
1847         {Missing language pack 'coppe-lang-\coppe@@cur.def'.}
1848         {Install it next to coppe.cls, or pick a built-in language.}}}%
1849   \fi\fi}
1850 % Quotation marks: csquotes gives the language-aware \enquote; \citacao is its
1851 % Portuguese alias for short, inline quotations (proposta P3).

```

```

1852 \RequirePackage{csquotes}
1853 \newcommand{\citacao}[1]{\enquote{#1}}
1854 % --- Code listings & algorithms (TODO3 "tratar listagens"): the listings
1855 %     package gives the "Programa" listing (caption on top, ABNT R46) with
1856 %     \listofprogramas; algorithm2e gives algorithms. Names are babel-aware and
1857 %     the language styles below are based on the user's listagens.tex. -----
1858 \RequirePackage{xcolor}
1859 \RequirePackage{listings}
1860 \RequirePackage{fancyvrb}
1861 \DeclareFixedFont{\ttb}{T1}{txtt}{bx}{n}{10}
1862 \DeclareFixedFont{\ttm}{T1}{txtt}{m}{n}{10}
1863 \lstset{captionpos=t,extendedchars=true,basicstyle=\ttfamily,inputencoding=utf8,%
1864   literate=%
1865   {\~}{\~{a}}1 {\~}{\~{a}}1 {\~}{\~{a}}1 {\~}{\~{a}}1 {\~}{\~{e}}1%
1866   {\~}{\~{e}}1 {\~}{\~{e}}1 {\~}{\~{i}}1 {\~}{\~{i}}1 {\~}{\~{o}}1%
1867   {\~}{\~{o}}1 {\~}{\~{o}}1 {\~}{\~{u}}1 {\~}{\~{u}}1 {\~}{\~{c}}1%
1868   {\~}{\~{c}}1 {\~}{\~{A}}1 {\~}{\~{A}}1 {\~}{\~{A}}1 {\~}{\~{E}}1%
1869   {\~}{\~{E}}1 {\~}{\~{I}}1 {\~}{\~{O}}1 {\~}{\~{O}}1 {\~}{\~{O}}1%
1870   {\~}{\~{U}}1%
1871 }
1872 % Babel-aware listing names, set after listings' own babel setup.
1873 \AtBeginDocument{%
1874   \renewcommand{\lstlistingname}{\coppestring{program}}%
1875   \renewcommand{\lstlistlistingname}{\coppestring{listprogram}}%
1876 % \listofprogramas in the coppe list style (heading + ToC + running head).
1877 \newcommand{\listofprogramas}{%
1878   \chapter*{\lstlistlistingname}%
1879   \coppe@listtoc{\lstlistlistingname}%
1880   \@mkboth{\MakeUppercase\lstlistlistingname}{\MakeUppercase\lstlistlistingname}%
1881   \@starttoc{lol}}
1882 % Per-language styles + environments (python, java, xml, html, prolog).
1883 % The red hook that marks a wrapped line is built ONCE, into a box, and the
1884 % styles below insert the box. Writing the colour inline -- postbreak=\mbox{%
1885 % \textcolor{red}{\hookrightarrow}}\space -- worked for years and broke the
1886 % moment the pdfa option was switched on: pdfx puts xcolor into a converting
1887 % mode (\selectcolormodel plus \convertcolorsU/D), and a colour command run
1888 % from inside listings' line-breaking machinery then fails with "Missing
1889 % number, treated as zero" at the first wrapped line. Every listing in the
1890 % document dies, so the class's own example could not be compiled as PDF/A at
1891 % all. A box is immune: it is built here, in ordinary horizontal mode where
1892 % xcolor works normally, and listings only ever copies it.
1893 %
1894 % The \sbox waits for \begin{document} because xcolor's colours are not
1895 % defined while the class is still loading.
1896 \newsavebox\coppe@lstarrow
1897 \AtBeginDocument{\sbox\coppe@lstarrow{\textcolor{red}{\hookrightarrow}}\space}
1898 \newcommand\pythonstyle{\lstset{language=Python,basicstyle=\ttm,numbers=left,%
1899   numberstyle=\small\ttfamily,numbersep=0.5em,commentstyle=\color{purple},%
1900   morekeywords={self,as},keywordstyle=\ttb\color{rgb}{0,0,0.5},%
1901   emph={MyClass,__init__},emphstyle=\ttb\color{rgb}{0.6,0,0},%
1902   stringstyle=\color{rgb}{0,0.5,0},frame=tb,showstringspaces=false,%
1903   extendedchars=true,firstnumber=auto,inputencoding=utf8,breaklines=true,%
1904   escapechar=',postbreak=\usebox\coppe@lstarrow}}
1905 \lstnewenvironment{python}[1][]{\pythonstyle\lstset{#1}}

```

```

1906 \csname\@lst @SetFirstNumber\endcsname}{\csname\@lst @SaveFirstNumber\endcsname}
1907 \newcommand\pythoninline[1]{\pythonstyle\lstinline!#1!}
1908 \newcommand\pythonexternal[2][\pythonstyle\lstinputlisting{#1}{#2}}
1909 \newcommand\xmlstyle{\lstset{language=XML,basicstyle=\ttm,numbers=left,%
1910 numberstyle=\small\ttfamily,numbersep=0.5em,%
1911 morekeywords={xs:schema,xs:element,xs:sequence,xs:complexType},%
1912 keywordstyle=\ttb\color[rgb]{0,0,0.5},%
1913 emph={name,type,attributeFormDefault,elementFormDefault,xmlns:xs},%
1914 emphstyle=\ttb\color[rgb]{0.6,0,0},stringstyle=\color[rgb]{0,0.5,0},frame=tb,%
1915 showstringspaces=false,extendedchars=true,inputencoding=utf8,breaklines=true,%
1916 postbreak=\usebox\coppe@lstarrow}}
1917 \lstnewenvironment{xml}[1][\xmlstyle\lstset{#1}]{%
1918 \newcommand\htmlstyle{\lstset{language=HTML,basicstyle=\ttm,numbers=left,%
1919 numberstyle=\small\ttfamily,numbersep=0.5em,%
1920 keywordstyle=\ttb\color[rgb]{0,0,0.5},emphstyle=\ttb\color[rgb]{0.6,0,0},%
1921 stringstyle=\color[rgb]{0,0.5,0},frame=tb,showstringspaces=false,%
1922 extendedchars=true,inputencoding=utf8,breaklines=true,%
1923 postbreak=\usebox\coppe@lstarrow}}
1924 \lstnewenvironment{html}[1][\htmlstyle\lstset{#1}]{%
1925 \newcommand\prologstyle{\lstset{language=Prolog,basicstyle=\ttm,numbers=left,%
1926 numberstyle=\small\ttfamily,numbersep=0.5em,commentstyle=\color{purple},%
1927 keywordstyle=\ttb\color[rgb]{0,0,0.5},emph={MyClass,__init__},%
1928 emphstyle=\ttb\color[rgb]{0.6,0,0},stringstyle=\color[rgb]{0,0.5,0},frame=tb,%
1929 showstringspaces=false,extendedchars=true,firstnumber=auto,inputencoding=utf8,%
1930 breaklines=true,escapechar=',%
1931 postbreak=\usebox\coppe@lstarrow}}
1932 \lstnewenvironment{prolog}[1][\prologstyle\lstset{#1}%
1933 \csname\@lst @SetFirstNumber\endcsname}{\csname\@lst @SaveFirstNumber\endcsname}
1934 \newcommand\javastyle{\lstset{language=Java,basicstyle=\ttm,numbers=left,%
1935 numberstyle=\small\ttfamily,numbersep=0.5em,commentstyle=\color{purple},%
1936 morekeywords={self,as},keywordstyle=\ttb\color[rgb]{0,0,0.5},%
1937 emphstyle=\ttb\color[rgb]{0.6,0,0},stringstyle=\color[rgb]{0,0.5,0},frame=tb,%
1938 showstringspaces=false,extendedchars=true,firstnumber=auto,inputencoding=utf8,%
1939 breaklines=true,escapechar=',%
1940 postbreak=\usebox\coppe@lstarrow}}
1941 \lstnewenvironment{java}[1][\javastyle\lstset{#1}%
1942 \csname\@lst @SetFirstNumber\endcsname}{\csname\@lst @SaveFirstNumber\endcsname}
1943 \newcommand\javainline[1]{\javastyle\lstinline!#1!}
1944 \newcommand\javaexternal[2][\javastyle\lstinputlisting{#1}{#2}}
1945 % fancyvrb's Verbatim environment is available for program output; the
1946 % \DefineVerbatimEnvironment{gxoutput}/{gxoutputs} aliases that used to live
1947 % here were author-personal (gx prefix) and were removed in v4.0. Use, for
1948 % example,
1949 % \begin{Verbatim}[frame=lines,numbers=left,numbersep=0.5em]
1950 % ...
1951 % \end{Verbatim}
1952 % (or \DefineVerbatimEnvironment{...}{Verbatim}{...} in the manuscript preamble).
1953 \renewcommand{\theFancyVerbLine}{\small\ttfamily\arabic{FancyVerbLine}}
1954 % Algorithms: algorithm2e, language driven by the main-slot \coppemainstring
1955 % lookup (key "algorithm2eopt"). Empty means use algorithm2e's default
1956 % language (English) -- needed because algorithm2e errors on an empty option.
1957 \edef\coppe@@algotlang{\coppemainstring{algorithm2eopt}}
1958 \ifx\coppe@@algotlang\empty
1959 \RequirePackage[linesnumbered,lined,algochapter,ruled]{algorithm2e}

```

```

1960 \else
1961   \expandafter\RequirePackage\expandafter[\coppe@algotlang,linesnumbered,lined,algochapter,]
1962 \fi
1963 % algorithm2e composes its own caption and therefore ignores the class-wide
1964 % \captionsetup{font=footnotesize,labelsep=endash}. Left alone it printed
1965 % "Algoritmo 6.1:Titulo" in bold at body size, while every other illustration
1966 % printed "Figura 6.1 -- Titulo" in footnotesize -- 2.10 asks for the
1967 % designating word, the number, a TRAVESSAO and the title, uniformly.
1968 %
1969 % The separator and the caption fonts are therefore set to match. The 'ruled'
1970 % layout (rules above and below) is kept: 2.10 prescribes where the caption and
1971 % the source go, not whether the float is delimited, and the rules read here
1972 % like the top-and-bottom delimitation a table carries.
1973 \SetAlgoCaptionSeparator{\textendash}
1974 \SetAlCapFnt{\footnotesize}
1975 \SetAlCapNameFnt{\footnotesize}
1976 \SetAlCapSty{textnormal}
1977 \SetAlCapNameSty{textnormal}
1978 \SetKwFor{Para}{para}{faça}{fim do para}
1979 \SetKwFor{Enquanto}{enquanto}{faça}{fim do enquanto}
1980 % setspace + hyperref + biblatex are loaded at the very end of the class
1981 % (see the block just before %</class>), so their hooks see every macro the
1982 % class defines, and the load order setspace -> hyperref -> biblatex is
1983 % respected.

```

`\department` This macro is used to set the author's affiliation. There are twelve options which correspond to all academic units at COPPE/UFRJ. It defines the current and the foreign names of these units.

```

1984 \newcommand\coppe@deptcode{}
1985 \newcommand\meta@deptname{}
1986 \newcommand\department[1]{%
1987 % The programme CODE is kept alongside the resolved name, and each name is
1988 % declared TWICE. \local@deptname keeps the TeX accent escapes with braces
1989 % (Computa{\c c}{\~ a}o) that the cover and the folha de rosto typeset;
1990 % \meta@deptname is the same name in literal UTF-8, and is the one that goes
1991 % into the XMP. pdfx copies whatever it is given straight through, so feeding it
1992 % the escaped form produced "Computa{c}o" in dc:description -- which is why the
1993 % subject line used to fall back to the bare programme code. Both forms are
1994 % written out here rather than derived from one another because there is no
1995 % reliable way to strip TeX accent syntax at the point where the .xmpdata is
1996 % written, and a table of thirteen entries is cheaper than that machinery.
1997 \gdef\coppe@deptcode{#1}%
1998 \ifthenelse{\equal{#1}{PEB}}{
1999   {\global\def\local@deptname{Engenharia Biom{\` e}dica}
2000   \global\def\meta@deptname{Engenharia Biomédica}
2001   \global\def\foreign@deptname{Biomedical Engineering}}{}
2002 \ifthenelse{\equal{#1}{PEC}}{
2003   {\global\def\local@deptname{Engenharia Civil}
2004   \global\def\meta@deptname{Engenharia Civil}
2005   \global\def\foreign@deptname{Civil Engineering}}{}
2006 \ifthenelse{\equal{#1}{PEE}}{
2007   {\global\def\local@deptname{Engenharia El{\` e}trica}
2008   \global\def\meta@deptname{Engenharia Elétrica}
2009   \global\def\foreign@deptname{Electrical Engineering}}{}

```

```

2010 \ifthenelse{\equal{#1}{PEM}}
2011     {\global\def\local@deptname{Engenharia Mec{\~ a}nica}
2012     \global\def\meta@deptname{Engenharia Mecânica}
2013     \global\def\foreign@deptname{Mechanical Engineering}}{}
2014 \ifthenelse{\equal{#1}{PEMM}}
2015     {\global\def\local@deptname{Engenharia Metal{\` u}rgica e de Materiais}
2016     \global\def\meta@deptname{Engenharia Metalúrgica e de Materiais}
2017 \global\def\foreign@deptname{Metallurgical and Materials Engineering}}{}
2018 \ifthenelse{\equal{#1}{PEN}}
2019     {\global\def\local@deptname{Engenharia Nuclear}
2020     \global\def\meta@deptname{Engenharia Nuclear}
2021     \global\def\foreign@deptname{Nuclear Engineering}}{}
2022 \ifthenelse{\equal{#1}{PEN0}}
2023     {\global\def\local@deptname{Engenharia Oce{\~ a}nica}
2024     \global\def\meta@deptname{Engenharia Oceânica}
2025     \global\def\foreign@deptname{Ocean Engineering}}{}
2026 \ifthenelse{\equal{#1}{PPE}}
2027     {\global\def\local@deptname{Planejamento Energ{\` e}tico}
2028     \global\def\meta@deptname{Planejamento Energético}
2029     \global\def\foreign@deptname{Energy Planning}}{}
2030 \ifthenelse{\equal{#1}{PEP}}
2031     {\global\def\local@deptname{Engenharia de Produ{\c c}{\~ a}o}
2032     \global\def\meta@deptname{Engenharia de Produção}
2033     \global\def\foreign@deptname{Production Engineering}}{}
2034 \ifthenelse{\equal{#1}{PEQ}}
2035     {\global\def\local@deptname{Engenharia Qu{\` i}mica}
2036     \global\def\meta@deptname{Engenharia Química}
2037     \global\def\foreign@deptname{Chemical Engineering}}{}
2038 \ifthenelse{\equal{#1}{PESC}}
2039     {\global\def\local@deptname{Engenharia de Sistemas e Computa{\c c}{\~ a}o}
2040     \global\def\meta@deptname{Engenharia de Sistemas e Computação}
2041     \global\def\foreign@deptname{Systems Engineering and Computer Science}}{}
2042 \ifthenelse{\equal{#1}{PET}}
2043     {\global\def\local@deptname{Engenharia de Transportes}
2044     \global\def\meta@deptname{Engenharia de Transportes}
2045     \global\def\foreign@deptname{Transportation Engineering}}{}
2046 \ifthenelse{\equal{#1}{PENT}}
2047     {\global\def\local@deptname{Engenharia de Nanotecnologia}
2048     \global\def\meta@deptname{Engenharia de Nanotecnologia}
2049     \global\def\foreign@deptname{Nanotechnology Engineering}}{}
2050 }

```

`\title` Used to enter the title in Brazilian Portuguese. Mirrors into `\coppe@title@brazilian` so that the per-language title slot used by the cover/folha renderers (`\coppe@selecttitle`) picks the right text for a pt-main thesis.

```

2051 \renewcommand\title[1]{%
2052     \global\def\local@title{#1}%
2053     \expandafter\gdef\csname coppe@title@brazilian\endcsname{#1}%
2054 }

```

`\foreigntitle` Used to enter the foreign title. Also mirrors into `\coppe@title@english` so that the cover renders the right text for an english-main thesis.

```

2055 \newcommand\foreigntitle[1]{%

```

```

2056 \global\def\foreign@title{#1}%
2057 \expandafter\gdef\csname coppe@title@english\endcsname{#1}%
2058 }

```

`\coppe@selecttitle` Expands to the title stored for the current main language. For pt-main this is `\local@title`; for en-main `\foreign@title`; for any other main language the user must call `\titlein{<lang>}{...}` to populate it.

```

2059 \newcommand\coppe@selecttitle{\@nameuse{coppe@title@\coppe@mainlang}}
2060 % Subtitle, mirroring the title machinery slot for slot: \subtitle feeds the
2061 % brazilian slot, \foreignsubtitle the english one, \subtitlein any other.
2062 % 3.1.1 and 3.1.2.1.1(c) require the subtitle on the capa and on the folha de
2063 % rosto, subordinated to the title by a colon.
2064 \newcommand\subtitle[1]{%
2065   \expandafter\gdef\csname coppe@subtitle@brazilian\endcsname{#1}}
2066 \newcommand\foreignsubtitle[1]{%
2067   \expandafter\gdef\csname coppe@subtitle@english\endcsname{#1}}
2068 \newcommand\subtitlein[2]{%
2069   \expandafter\gdef\csname coppe@subtitle@#1\endcsname{#2}}
2070 \newcommand\coppe@selectsubtitle{\@nameuse{coppe@subtitle@\coppe@mainlang}}
2071 % Title as the capa and the folha de rosto print it. Annexes A and B show the
2072 % title in caps and the subtitle in ordinary case after a colon, so the case
2073 % change is applied to the title alone.
2074 \newcommand\coppe@titleblock{%
2075   \MakeUppercase{\coppe@selecttitle}%
2076   \@ifundefined{coppe@subtitle@\coppe@mainlang}{: \coppe@selectsubtitle}}
2077 % Number of volumes (3.1.1 and 3.1.2.1.1(d)): printed only when there is more
2078 % than one, since a single-volume work says nothing about volumes.
2079 % Coorientador (3.1.2.1.1(f)). Same shape as \advisor, including the optional
2080 % institution; the label switches to the plural on the second call, as the
2081 % advisor label already did.
2082 \newcount\@coadvisor\@coadvisor0
2083 \newcommand\coadvisor[5][1]{%
2084   \global\@namedef{CoppeCoadvisorInst:\expandafter\the\@coadvisor}{#1}
2085   \global\@namedef{CoppeCoadvisorTitle:\expandafter\the\@coadvisor}{#2}
2086   \global\@namedef{CoppeCoadvisorName:\expandafter\the\@coadvisor}{#3}
2087   \global\@namedef{CoppeCoadvisorSurname:\expandafter\the\@coadvisor}{#4}
2088   \global\@namedef{CoppeCoadvisorDegree:\expandafter\the\@coadvisor}{#5}
2089   \global\advance\@coadvisor by 1
2090 }
2091 \newcommand\local@coadvisorstring{%
2092   \ifnum\@coadvisor>1
2093     \coppemainstring{coadvisors}%
2094   \else
2095     \coppemainstring{coadvisor}%
2096   \fi}
2097 % Mesma coisa no idioma estrangeiro, para a pagina de abstract.
2098 \newcommand\foreign@coadvisorstring{%
2099   \ifnum\@coadvisor>1
2100     \coppeforeignstring{coadvisors}%
2101   \else
2102     \coppeforeignstring{coadvisor}%
2103   \fi}
2104 \newcommand\coppe@nvolumes{1}
2105 \newcommand\coppe@thisvolume{1}

```

```

2106 \newcommand\volumes[1]{\gdef\coppe@nvolumes{#1}}
2107 \newcommand\volume[1]{\gdef\coppe@thisvolume{#1}}
2108 \newcommand\coppe@volumeline{%
2109   \ifnum\coppe@nvolumes>1
2110     \par\vspace*{4mm}%
2111     \coppemainstring{volumename}\space\coppe@thisvolume\space
2112     \coppemainstring{ofname}\space\coppe@nvolumes
2113 % The closing \par matters: on the capa the next thing is \vspace{\stretch{3}},
2114 % which does not end a paragraph, so without it the city ran on into the volume
2115 % line ("Volume 2 de 3 RIO DE JANEIRO").
2116   \par
2117 \fi}

```

\advisor Defines globally the title, name and academic degree of the advisors.

```

2118 \newcount\@advisor\@advisor0
2119 % The optional first argument is the member's institution, required by
2120 % 3.1.2.1.3(e). Omitting it keeps every pre-4.1 call working unchanged.
2121 \newcommand\advisor[5][]{%
2122   \global\@namedef{CoppeAdvisorInst:\expandafter\the\@advisor}{#1}
2123   \global\@namedef{CoppeAdvisorTitle:\expandafter\the\@advisor}{#2}
2124   \global\@namedef{CoppeAdvisorName:\expandafter\the\@advisor}{#3}
2125   \global\@namedef{CoppeAdvisorSurname:\expandafter\the\@advisor}{#4}
2126   \global\@namedef{CoppeAdvisorDegree:\expandafter\the\@advisor}{#5}
2127   \global\advance\@advisor by 1
2128   \ifnum\@advisor>1
2129     \renewcommand\local@advisorstring{Orientadores}
2130     \renewcommand\foreign@advisorstring{Advisors}
2131   \fi
2132 }

```

\examiner

```

2133 \newcount\@examiner\@examiner0
2134 % Until v4.1 this stored only "#1 #2" and DISCARDED the third argument: the
2135 % titulacao every document passed was silently thrown away and never printed,
2136 % though 3.1.2.1.3(e) requires it. Each part is now kept separately, and the
2137 % optional first argument carries the institution.
2138 % CoppeExaminer: is retained with its old meaning for anything reading it.
2139 \newcommand\examiner[4][]{%
2140   \global\@namedef{CoppeExaminerInst:\expandafter\the\@examiner}{#1}
2141   \global\@namedef{CoppeExaminerTitle:\expandafter\the\@examiner}{#2}
2142   \global\@namedef{CoppeExaminerName:\expandafter\the\@examiner}{#3}
2143   \global\@namedef{CoppeExaminerDegree:\expandafter\the\@examiner}{#4}
2144   \global\@namedef{CoppeExaminer:\expandafter\the\@examiner}{#2\ #3}
2145   \global\advance\@examiner by 1
2146 }

```

\author It was redefined to allow the identification of the author's first names and surname.

```

2147 \renewcommand\author[2]{%
2148   \global\def\@authname{#1}
2149   \global\def\@authsurn{#2}
2150 }

```

\date Sets the *document's* nominal month and year (used by the cover, the folha de rosto, and the catalog). Stores them in private slots (\coppe@docmonth /

\coppe@docyear) rather than mutating TeX's \month / \year primitives, which would clobber \today.

```
2151 \renewcommand\date[2]{%
2152   \def\coppe@docmonth{#1}%
2153   \def\coppe@docyear{#2}}
```

\local@monthname

```
2154 \newcommand\local@monthname{\ifcase\coppe@docmonth\or
2155   Janeiro\or Fevereiro\or Mar{\c c}\o\or Abril\or Maio\or Junho\or
2156   Julho\or Agosto\or Setembro\or Outubro\or Novembro\or Dezembro\fi}
```

\foreign@monthname

```
2157 \newcommand\foreign@monthname{\ifcase\coppe@docmonth\or
2158   January\or February\or March\or April\or May\or June\or
2159   July\or August\or September\or October\or November\or December\fi}
```

\keyword

```
2160 \newcounter{keywords}
2161 \newcommand\keyword[1]{%
2162   \global\@namedef{CoppeKeyword:\expandafter\the\c@keywords}{#1}
2163   \global\addtocounter{keywords}{1}
2164 }
2165 % Keywords for the foreign abstract. Annex F of the manual shows the English
2166 % abstract closing with English keywords, and \keyword alone cannot serve both
2167 % pages. When no \foreignkeyword is given the foreign abstract falls back to
2168 % the \keyword list, which is what a document written before v4.1 expects.
2169 \newcounter{foreignkeywords}
2170 \newcommand\foreignkeyword[1]{%
2171   \global\@namedef{CoppeForeignKeyword:\expandafter\the\c@foreignkeywords}{#1}
2172   \global\addtocounter{foreignkeywords}{1}
2173 }
2174 % Keywords for the optional third abstract (Portuguese, for the banca). Same
2175 % shape and same fallback as \foreignkeyword: without it the Portuguese page
2176 % would close with the main-language keywords under a Portuguese label, which
2177 % is what a Spanish-, French- or Italian-main work would produce.
2178 \newcounter{braziliankeywords}
2179 \newcommand\braziliankeyword[1]{%
2180   \global\@namedef{CoppeBrazilianKeyword:\expandafter\the\c@braziliankeywords}{#1}
2181   \global\addtocounter{braziliankeywords}{1}
2182 }
2183 % 3.1.2.1.4: the keywords close the resumo, preceded by the term, a colon, and
2184 % separated from one another by semicolons.
2185 % #1 label, #2 csname prefix, #3 how many.
2186 \newcommand\coppe@printkeywords[3]{%
2187   \ifnum#3>0
2188     \par\vspace*{7mm}%
2189     \noindent{\singlespacing\textbf{#1:}}\enspace
2190     \count1=0
2191     \@whilenum\count1<#3\do{%
2192       \ifnum\count1>0 ;\space\fi
2193       \csname #2\the\count1\endcsname
2194       \advance\count1 by 1}%
2195     .\par}%
2196   \fi}
```

`\areaconcentracao` Data for the folha adicional (Coleta CAPES) and for the natureza block of the
`\linhapesquisa` folha de rosto and folha de aprovacao. Every one of them defaults to empty;
`\tipoproducao` `\makefolhaadicional` prints an empty rule where a value is missing, so the pro-
`\projetovinculado` gramme can complete the sheet by hand if need be.
`\nomeprojeto` 2197 `\newcommand\coppe@areaconc{}`
`\agenciafomento` 2198 `\newcommand\areaconcentracao[1]{\gdef\coppe@areaconc{#1}}`
2199 `\newcommand\coppe@dataaprov{}`
2200 `\newcommand\dataaprovacao[1]{\gdef\coppe@dataaprov{#1}}`
2201 `\newcommand\coppe@linhapesq{}`
2202 `\newcommand\linhapesquisa[1]{\gdef\coppe@linhapesq{#1}}`
2203 `\newcommand\coppe@tipoproducao{}`
2204 `\newcommand\tipoproducao[1]{\gdef\coppe@tipoproducao{#1}}`
2205 `\newcommand\coppe@projetovinc{}`
2206 `\newcommand\projetovinculado[1]{\gdef\coppe@projetovinc{#1}}`
2207 `\newcommand\coppe@nomeprojeto{}`
2208 `\newcommand\nomeprojeto[1]{\gdef\coppe@nomeprojeto{#1}}`
2209 `\newcounter{coppeagencias}`
2210 `\newcommand\agenciafomento[2]{%`
2211 `\global\@namedef{CoppeAgencia:\expandafter\the\c@coppeagencias}{#1}%`
2212 `\global\@namedef{CoppeAgenciaSigla:\expandafter\the\c@coppeagencias}{#2}%`
2213 `\global\addtocounter{coppeagencias}{1}}`

`\freeconfig` This command allows easy changing of core class parameters.

```

2214 \newcommand\freeconfig[6]{%
2215 \providecommand\@degree{}
2216 \renewcommand\@degree{#1}
2217 \providecommand\@degreename{}
2218 \renewcommand\@degreename{#2}
2219 \providecommand\local@degname{}
2220 \renewcommand\local@degname{#3}
2221 \providecommand\foreign@degname{}
2222 \renewcommand\foreign@degname{#4}
2223 \providecommand\local@doctype{}
2224 \renewcommand\local@doctype{#5}
2225 \providecommand\foreign@doctype{}
2226 \renewcommand\foreign@doctype{#6}%
2227 }
2228 % \end{macrocode}
2229 % \end{macro}
2230 %
2231 % \begin{macro}{\frontmatter}
2232 % The number of pages for both frontmatter and mainmatter printed
2233 % in the cataloging details page is computed by means of simple
2234 % \LaTeX\ labels.
2235 % \begin{macrocode}
2236 \renewcommand\frontmatter{%
2237 \cleardoublepage
2238 \@mainmatterfalse
2239 % The count already started at the folha de rosto (see \maketitle) and simply
2240 % continues here: 2.7 requires every folha from the folha de rosto onwards to
2241 % be counted sequentially, printing arabic numerals only from the first textual
2242 % folha. Resetting the counter here -- as this class did until v4.1 -- made the
2243 % Introducao come out as page 1, which the norm does not allow.
2244 %

```

```

2245 % The numeraisromanos option keeps the legacy behaviour (lowercase roman in the
2246 % pre-textual part, arabic restarting at 1 in the textual part). That departs
2247 % from 2.7, which has the pre-textual folhas counted but NOT numbered, so the
2248 % option warns.
2249 \if@copperoman
2250   \ClassWarningNoLine{coppe}{Option ‘numeraisromanos’ prints roman numerals
2251   on the pre-textual folhas and restarts the arabic count at 1 in the
2252   textual part. Section 2.7 of the SiBI manual (9th rev. ed., 2026) has
2253   those folhas counted but not numbered, and the arabic numbering
2254   continuing the count. Drop the option to comply}%
2255   \pagenumbering{roman}%
2256 \fi
2257 % \thispagestyle vale por UMA pagina. Se a folha de aprovacao transbordar --
2258 % uma banca grande com titulo longo consegue --, a pagina seguinte pegava o
2259 % estilo corrente e saia com folio, na parte pre-textual, onde a 2.7 nao
2260 % admite. O grupo com \pagestyle{empty} cobre quantas paginas forem, e o
2261 % \endgroup devolve o estilo anterior.
2262 \begingroup
2263   \pagestyle{empty}\thispagestyle{empty}
2264   \singlespacing
2265   \makefrontpage
2266   \clearpage
2267 \endgroup
2268 \if@copperoman
2269   \fancypagestyle{plain}{\fancyhf{}\fancyhead[R0,LE]{\coppe@folio}%
2270   \renewcommand{\headrulewidth}{0pt}}%
2271   \pagestyle{coppe}%
2272 \else
2273   \fancypagestyle{plain}{\fancyhf{}\renewcommand{\headrulewidth}{0pt}}%
2274   \pagestyle{empty}%
2275 \fi
2276 }

\mainmatter

2277 \renewcommand\mainmatter{%
2278   \coppe@mainBegin
2279   \cleardoublepage
2280   \@mainmattertrue
2281   \fancypagestyle{plain}{\fancyhf{}\fancyhead[R0,LE]{\coppe@folio}%
2282   \renewcommand{\headrulewidth}{0pt}}%
2283   \pagestyle{coppe}%
2284 % No \pagenumbering{arabic} in the default path: it would reset the counter to
2285 % 1 and make the Introducao folha 1, contrary to 2.7. The counter has been
2286 % arabic and running since the folha de rosto; all that changes here is that
2287 % the folio starts being PRINTED. Only the legacy numeraisromanos path, which
2288 % switched to roman in the front matter, has to switch back -- and there the
2289 % restart at 1 is intrinsic to that (non-conforming) scheme.
2290   \if@copperoman\pagenumbering{arabic}\fi}

\backmatter

2291 \renewcommand\backmatter{%
2292   \if@openright
2293     \cleardoublepage
2294   \else

```

```

2295 \clearpage
2296 \fi}
2297 %

```

\coppetexfinalpage Optional colophon-style closing page intended to face the rear cover. The macro forces a verso (even) page so that, in a two-sided print, the reader sees the colophon on the left and the inside of the rear cover on the right; pushes the text to the bottom of the page with `\vfill`; and types it in Portuguese unconditionally (the colophon is institutional metadata, not body content, so it is rendered in pt-BR regardless of the thesis main language). Three slot macros are user-overridable: `\coppefinalengine` (engine name; auto-detected via `iftex`), `\coppefinalfont` (text body font; defaults to Computer Modern, the L^AT_EX default), and `\coppefinalmanual` (the reference manual title; defaults to the current COPPE/UFRJ norm). Override them before `\coppetexfinalpage` with `\renewcommand` if the document uses a non-default font or follows an earlier edition of the norm.

```

2298 \providecommand{\coppefinalengine}{%
2299 \ifPDFTeX\hologo{pdfLaTeX}\fi
2300 \ifLuaTeX\hologo{LuaLaTeX}\fi
2301 \ifXeTeX\hologo{XeLaTeX}\fi}
2302 % A classe carrega lmodern (as bitmap da Computer Modern saiam em Type 3 e
2303 % inviabilizavam o PDF/A), entao o colofao diz Latin Modern -- quem trocar a
2304 % fonte redefine este comando.
2305 \providecommand{\coppefinalfont}{Latin Modern (\texttt{lmodern})}
2306 \providecommand{\coppefinalmanual}{%
2307 \emph{Norma para a Elabora\c c\~ao Gr\'afica de Teses e Disserta\c c\~oes
2308 da COPPE/UFRJ}}
2309 \newcommand{\coppetexfinalpage}{%
2310 \clearpage
2311 % Force the colophon onto a verso (even) page so it faces the rear cover.
2312 % If the current page is odd (recto), fill it with an empty verso first.
2313 \if@twoside
2314 \ifodd\c@page
2315 \hbox{}\thispagestyle{empty}\newpage
2316 \fi
2317 \fi
2318 \thispagestyle{empty}%
2319 \null\vfill
2320 \begin{otherlanguage}{brazilian}%
2321 \begin{center}
2322 {\footnotesize\setstretch{1.0}%
2323 \textsc{Colof\~ao}\par
2324 \vspace{0.7em}
2325 \begin{minipage}{0.75\textwidth}\centering
2326 Este documento foi composto em \LaTeX{} com \coppefinalengine{}
2327 em \today, na fonte \coppefinalfont. A bibliografia foi processada
2328 por Bib\LaTeX{} e Biber. Foi utilizada a classe \CoppeTeX,
2329 vers\~ao\~ao\coppe@version, dispon\'i vel no reposit\'orio GitHub
2330 \mbox{\url{https://github.com/COPPE-UFRJ/CoppeTeX}}, em
2331 conformidade com a \coppefinalmanual.
2332 \end{minipage}\par}
2333 \end{center}
2334 \end{otherlanguage}%

```

```

2335 \vspace*{2cm}%
2336 \clearpage}

\makecover The ABNT cover (capa, NBR 14724) suggested in the manual's Anexo A. It is
the institution block — the university, the full COPPE name, and the graduate
program taken from \department — followed by the author, the title, and the
city and year, all centred and uppercased. The optional UFRJ and COPPE logos
head the page. The capa is not counted: it precedes the (roman) front matter and
is produced before the titlepage (folha de rosto).

2337 \newcommand\makecover{%
2338 \begin{titlepage}%
2339 % A capa e um modelo institucional, e sai em espaco simples qualquer que seja o
2340 % espacamento do corpo. Sob a opcao doublespacing ela transbordava para uma
2341 % segunda e ate uma terceira folha -- que ainda por cima levavam folio, porque
2342 % o \thispagestyle do titlepage vale por uma pagina so. 0 \pagestyle{empty} e
2343 % local ao ambiente e cobre quantas paginas forem.
2344 \pagestyle{empty}%
2345 \singlespacing
2346 \centering
2347 \makebox[\linewidth]{%
2348 \includegraphics[height=2cm]{ufrj-logo}\hfill
2349 \includegraphics[height=2cm]{coppe-logo}}\par
2350 \vspace{1.5cm}
2351 {\singlespacing
2352 \MakeUppercase{\local@universityname}\par
2353 \nohyphens{\MakeUppercase{Instituto Alberto Luiz Coimbra de \
2354 P{\ ' o}s-Gradua{\c c}{\~ a}o e Pesquisa de Engenharia}}\par
2355 \MakeUppercase{Programa de \local@deptname}\par}
2356 \vspace{\stretch{2}}
2357 \textbf{\MakeUppercase{\textbf{\@authname\ \@authsur}}}\par
2358 \vspace{\stretch{2}}
2359 \nohyphens{\coppe@titleblock}\par
2360 \coppe@volumeline
2361 \vspace{\stretch{3}}
2362 \MakeUppercase{\local@cityname}\par
2363 \coppe@docyear
2364 \end{titlepage}%
2365 }

\maketitle

2366 \renewcommand\maketitle{%
2367 % The capa is neither counted nor numbered (2.7), so it lives in its own
2368 % throwaway numbering scheme; \pagenumbering{arabic} right after it restarts
2369 % the counter, making the FOLHA DE ROSTO page 1 -- which is where 2.7 says the
2370 % count begins. Nothing is printed here: \frontmatter keeps \pagestyle{empty}
2371 % until the textual part starts.
2372 \pagenumbering{alph}
2373 \ltx@ifpackageloaded{hyperref}{\coppe@hypersetup}{}%
2374 \makecover
2375 \pagenumbering{arabic}
2376 \begin{titlepage}
2377 % Mesma razao da capa: modelo institucional, espaco simples, e sem folio ainda
2378 % que transborde.

```

```

2379 \pagestyle{empty}%
2380 \singlespacing
2381 \begin{flushleft}
2382 \vspace*{1.5mm}
2383 \setlength\baselineskip{0pt}
2384 \setlength\parskip{1mm}
2385 \makebox[\linewidth]{%
2386   \includegraphics[height=2cm]{ufrj-logo}\hfill
2387   \includegraphics[height=2cm]{coppe-logo}}
2388 \end{flushleft}
2389 \vspace{1.05cm}
2390 \begin{center}
2391   \nohyphens{\@authname\ \@authsur}\par
2392   \vspace*{3cm}
2393   \nohyphens{\coppe@titleblock}\par
2394   \coppe@volumeline
2395 \end{center}
2396 \vspace*{2.1cm}
2397 \begin{flushright}
2398 % 2.4: on the folha de rosto the natureza/objetivo/instituicao block is set
2399 % SINGLE spaced, and aligned from the middle of the mancha grafica to the right
2400 % margin. 0.5\textwidth tracks the mancha, so the block follows automatically if
2401 % the margins ever change -- the old fixed 8.45cm did not.
2402 \begin{minipage}{0.5\textwidth}
2403 \singlespacing\frontcover@maintext
2404 \end{minipage}\par
2405 \end{flushright}
2406 % Only the natureza block is indented to the right (2.4). The linha de pesquisa
2407 % and the orientadores return to the LEFT MARGIN, as the CAPES model of the
2408 % folha de rosto and Annex B of the manual show. Until v4.1 they sat inside the
2409 % right-hand block, which is what a recent review flagged.
2410 \vspace*{12mm}
2411 {\singlespacing
2412 \ifthenelse{\equal{\coppe@linhapesq}{}}{\}%
2413   \noindent Linha de pesquisa: \coppe@linhapesq\par
2414   \vspace*{5mm}}%
2415 \nohyphens{%
2416 \noindent\begin{tabularx}{\textwidth}[b]{@{}l@{ }}>\raggedright\arraybackslashX@{}}
2417   \local@advisorstring: &
2418   \count1=0
2419   \toks@={}
2420   \@whilenum \count1<\@advisor \do{%
2421     \ifcase\count1 % same as \ifnum0=\count1
2422       \toks@=\expandafter{\csname CoppeAdvisorName:\the\count1%
2423         \expandafter\endcsname\expandafter\space%
2424         \csname CoppeAdvisorSurname:\the\count1\endcsname\\}
2425     \else
2426       \toks@=\expandafter\expandafter\expandafter{%
2427         \expandafter\the\expandafter\toks@%
2428         \expandafter&\expandafter\space%
2429         \csname CoppeAdvisorName:\the\count1\expandafter\endcsname%
2430         \expandafter\space\csname CoppeAdvisorSurname:\the\count1\endcsname\\}
2431     }%
2432 \fi

```

```

2433     \advance\count1 by 1}
2434     \the\toks@
2435 \end{tabularx}}}%
2436 \ifnum\@coadvisor>0
2437     \par\vspace*{2mm}%
2438     \nohyphens{%
2439     \noindent\begin{tabularx}{\textwidth}[b]{\@{ }1@{ }>{\raggedright\arraybackslash}X@{}}
2440         \local@coadvisorstring: &
2441         \count1=0
2442         \toks@={}
2443         \@whilenum \count1<\@coadvisor \do{%
2444             \ifcase\count1
2445                 \toks@=\expandafter{\csname CoppeCoadvisorName:\the\count1%
2446                     \expandafter\endcsname\expandafter\space%
2447                     \csname CoppeCoadvisorSurname:\the\count1\endcsname\\}
2448             \else
2449                 \toks@=\expandafter\expandafter\expandafter{%
2450                     \expandafter\the\expandafter\toks@%
2451                     \expandafter&\expandafter\space%
2452                     \csname CoppeCoadvisorName:\the\count1\expandafter\endcsname%
2453                     \expandafter\space\csname CoppeCoadvisorSurname:\the\count1\endcsname\\}
2454                 }%
2455             \fi
2456             \advance\count1 by 1}
2457         \the\toks@
2458     \end{tabularx}}}%
2459 \fi
2460 \par}
2461 \vspace*{\fill}
2462 % 3.1.2.1.1(g): the folha de rosto carries the ANO DE DEPOSITO alone -- the year
2463 % the revised final version is handed to the Programa. The month was never asked
2464 % for here (the capa, Annex A, also shows only the year), and printing it
2465 % invited confusion with the defence date. \date still takes month and year:
2466 % the month is what the abstract pages and the colophon use.
2467 \begin{center}
2468 \local@cityname\par
2469 \coppe@docyear
2470 \end{center}
2471 \end{titlepage}
2472 % Folha adicional (3.1.2.1.2): a sheet of its own right after the folha de
2473 % rosto, ahead of the approval sheet produced by \frontmatter. Until the 2026
2474 % edition this content lived on the VERSO of the folha de rosto (NBR 14724,
2475 % R57); the verso is gone with the printed copy.
2476 % LaTeX's titlepage environment ends with \setcounter{page}{1} whenever the
2477 % document is one-sided (see \endtitlepage in classes.dtx). That silently ate
2478 % the folha de rosto from the count, putting the folha de aprovacao at 1 instead
2479 % of 2. The count is therefore restated here: the folha de rosto WAS folha 1, so
2480 % whatever follows it is folha 2.
2481 \setcounter{page}{2}%
2482 % The page carrying the ficha catalografica is explicitly excluded from the
2483 % count -- 2.7 of the 2026 manual ("a folha adicional que contem a ficha
2484 % catalografica nao pode ser contada ou numerada"), and the 2025 edition said
2485 % the same of the verso of the folha de rosto. Hence the counter is stepped
2486 % back after it.

```

```

2487 % An exame de qualificacao is not deposited with the library: no ficha
2488 % catalografica, and no Coleta CAPES sheet. Everything else gets both.
2489 \if@coppeexame\else
2490 \makefolhaadicional\clearpage\addtocounter{page}{-1}%
2491 \fi
2492 \global\let\maketitle\relax%
2493 \global\let\and\relax}

2494 % 3.1.2.1.3 lists the elements of the folha de aprovacao in order: (a) the
2495 % AUTHOR, (b) the title in full, (c) the natureza/objetivo/instituicao/area de
2496 % concentracao, (d) the date of approval, (e) the examining board. Until v4.1
2497 % this page opened with the title in caps and the author under it, and set the
2498 % natureza as a wall of uppercase (\frontpage@maintext). Annex D shows neither.
2499 % One board member. \coppe@membroid is the identification line required by
2500 % 3.1.2.1.3(e) -- name, titulacao and institution -- with the last two omitted
2501 % when not given, so a document that never supplied them looks as it did before.
2502 \newcommand\coppe@membroid[4]{%
2503   #1\ #2%
2504   \ifthenelse{\equal{#3}{}}{, #3}%
2505   \ifthenelse{\equal{#4}{}}{, #4}%
2506 % The gap above each signature rule is \coppe@membrogap, set from the size of
2507 % the board just before the list is assembled (see \makefrontpage). A fixed 9mm
2508 % fitted five members and overflowed at seven -- the sheet silently continued
2509 % onto a second page, which also printed a folio in the pre-textual part, where
2510 % 2.7 forbids one. The shrink lives in the length itself -- \newlength makes a
2511 % skip register, so "9mm minus 2mm" survives \setlength and reaches \vspace as
2512 % real glue, and a title long enough to wrap an extra line is absorbed instead
2513 % of costing a page. It has to go through \vspace and not a bare \vskip: the
2514 % board list is emitted inside \nohyphens, i.e. in horizontal mode, where
2515 % \vskip is an error and TeX then typesets the "plus ... minus ..." after it as
2516 % ordinary text -- which is exactly what landed on the page the first time.
2517 \newlength\coppe@membrogap
2518 \setlength\coppe@membrogap{9mm minus 2mm}
2519 \newcount\coppe@bancasize
2520 \newcommand\coppe@membro[4]{%
2521   \if@coppeassin
2522     \vspace{\coppe@membrogap}%
2523     \begin{center}
2524       \rule{95mm}{0.4pt}\par
2525       \nopagebreak{\footnotesize\coppe@membroid{#1}{#2}{#3}{#4}}\par
2526     \end{center}%
2527   \else
2528     \noindent\hspace*{8mm}%
2529     \begin{minipage}{\dimexpr\textwidth-8mm\relax}
2530       \coppe@membroid{#1}{#2}{#3}{#4}
2531     \end{minipage}\par\vspace{2mm}%
2532   \fi}
2533 \newcommand\makefrontpage{%
2534   \begin{center}
2535     \nohyphens{\@authname\ \@authsur}\par
2536     \vspace*{7mm}
2537     \sloppy\nohyphens{\coppe@selecttitle
2538       \@ifundefined{coppe@subtitle@\coppe@mainlang}{:
2539         \coppe@selectsubtitle}}\par

```



```

2540 \end{center}\par
2541 \vspace*{8mm}
2542 % Same block, same indentation and same single spacing as the folha de rosto:
2543 % 2.4 names the two pages together.
2544 \begin{flushright}
2545 \begin{minipage}{0.5\textwidth}
2546 \singlespacing\frontcover@maintext
2547 \end{minipage}
2548 \end{flushright}
2549 \vspace*{12mm}
2550 % (d) date of approval. Annex D prints "Aprovada em:" followed by a blank to be
2551 % filled in; \dataaprovacao supplies it when the date is already known.
2552 \noindent\local@approvedonname:\enspace
2553 \ifthenelse{\equal{\coppe@dataaprova}{}}{\hrulefill}{\coppe@dataaprova}\par
2554 \vspace*{12mm}
2555 % (e) the examining board, with the ORIENTADOR FIRST, as president of the board
2556 % (3.1.2.1.3(e)). Until v4.1 the advisors were printed in a block of their own
2557 % above a second list headed "Aprovada por:", which split the board in two and
2558 % did not mark the advisor as its president. They are now one list.
2559 %
2560 % The list is assembled into \coppe@bancalist with \g@addto@macro instead of
2561 % being built inside a tabularx: each entry now carries four fields (treatment,
2562 % name, titulacao, institution) and the old \toks@/\expandafter accumulation
2563 % could not carry them without becoming unreadable. \protected@edef keeps
2564 % accents and other protected commands in the names intact.
2565 % Size of the board decides the spacing above. The thresholds are measured, not
2566 % guessed: with the 9mm gap a seven-member board pushed three members onto a
2567 % second page, 7mm brought six back onto one sheet, 5mm did the same for seven,
2568 % and eight needed 3.5mm -- eight came up in the adversarial suite, in a
2569 % Spanish thesis whose longer natureza block leaves less room than the
2570 % Portuguese one. Boards of five or fewer keep the original 9mm, so every
2571 % document written before this change renders exactly as it did.
2572 \coppe@bancasize=\@advisor
2573 \advance\coppe@bancasize by \@coadvisor
2574 \advance\coppe@bancasize by \@examiner
2575 \ifnum\coppe@bancasize>7
2576 \setlength\coppe@membrogap{3.5mm minus 1.5mm}%
2577 \else\ifnum\coppe@bancasize>6
2578 \setlength\coppe@membrogap{5mm minus 2mm}%
2579 \else\ifnum\coppe@bancasize>5
2580 \setlength\coppe@membrogap{7mm minus 2mm}%
2581 \fi\fi\fi
2582 \gdef\coppe@bancalist{}%
2583 \count1=0
2584 \@whilenum\count1<\@advisor\do{%
2585 \protected@edef\coppe@tmp{\noexpand\coppe@membro
2586 {\csname CoppeAdvisorTitle:\the\count1\endcsname}%
2587 {\csname CoppeAdvisorName:\the\count1\endcsname}\space
2588 {\csname CoppeAdvisorSurname:\the\count1\endcsname}%
2589 {\csname CoppeAdvisorDegree:\the\count1\endcsname}%
2590 {\csname CoppeAdvisorInst:\the\count1\endcsname}}%
2591 \expandafter\g@addto@macro\expandafter\coppe@bancalist
2592 \expandafter{\coppe@tmp}%
2593 \advance\count1 by 1}%

```

```

2594 % Coorientadores come right after the orientadores and before the rest of the
2595 % board, keeping the order 3.1.2.1.3(e) prescribes.
2596 \count1=0
2597 \@whilenum\count1<\@coadvisor\do{%
2598   \protected@edef\coppe@tmp{\noexpand\coppe@membro
2599     {\csname CoppeCoadvisorTitle:\the\count1\endcsname}%
2600     {\csname CoppeCoadvisorName:\the\count1\endcsname}\space
2601     \csname CoppeCoadvisorSurname:\the\count1\endcsname}%
2602     {\csname CoppeCoadvisorDegree:\the\count1\endcsname}%
2603     {\csname CoppeCoadvisorInst:\the\count1\endcsname}}%
2604   \expandafter\g@addto@macro\expandafter\coppe@bancalist
2605   \expandafter{\coppe@tmp}%
2606   \advance\count1 by 1}%
2607 \count1=0
2608 \@whilenum\count1<\@examiner\do{%
2609   \protected@edef\coppe@tmp{\noexpand\coppe@membro
2610     {\csname CoppeExaminerTitle:\the\count1\endcsname}%
2611     {\csname CoppeExaminerName:\the\count1\endcsname}%
2612     {\csname CoppeExaminerDegree:\the\count1\endcsname}%
2613     {\csname CoppeExaminerInst:\the\count1\endcsname}}%
2614   \expandafter\g@addto@macro\expandafter\coppe@bancalist
2615   \expandafter{\coppe@tmp}%
2616   \advance\count1 by 1}%
2617 \noindent\local@approvedname:\par
2618 \nohyphens{\singlespacing\coppe@bancalist}\par
2619 % No city/date block at the foot: 3.1.2.1.3 ends the folha de aprovacao at the
2620 % examining board, and Annex D shows nothing below the signature lines. The date
2621 % that belongs on this page is the date of APPROVAL, printed above.
2622 % \frontpage@bottomtext stays defined for anyone calling it directly.
2623 \vspace*{\fill}}

2624 \newcommand\coppe@hypersetup{%
2625 \begingroup
2626 % changes to \toks@ and \count@ are kept local;
2627 % it's not necessary for them, but it is usually the case
2628 % for \count1, because the first ten counters are written
2629 % to the DVI file, thus you got lucky because of PDF output
2630 \toks@={}% in this special case not necessary
2631 \count@=0 %
2632 \@whilenum\count@<\value{keywords}\do{%
2633   % * a keyword separator is not necessary,
2634   %   if there is just one keyword
2635   % * \csname CoppeKeyword:\the\count@\endcsname must be expanded
2636   %   at least once, to get rid of the loop depended \count@
2637   \ifcase\count@ % same as \ifnum0=\count@
2638     \toks@=\expandafter{\csname CoppeKeyword:\the\count@\endcsname}%
2639   \else
2640     \toks@=\expandafter\expandafter\expandafter{%
2641       \expandafter\the\expandafter\toks@
2642       \expandafter;\expandafter\space
2643       \csname CoppeKeyword:\the\count@\endcsname
2644     }%
2645   \fi
2646   \advance\count@ by 1 %

```

```

2647 }%
2648 \edef\x{\endgroup
2649   \noexpand\hypersetup{%
2650     pdfkeywords={\the\toks0}%
2651   }%
2652 }%
2653 \x
2654 \hypersetup{%
2655   pdfauthor={\@authname\ \@authsur},
2656   pdftitle={\local@title},
2657   pdfsubject={\local@doctype\ de \@degreename\ em \local@deptname\ da COPPE/UFRJ},
2658   pdfcreator={LaTeX with CoppeTeX toolkit},
2659   breaklinks={true},
2660   raiselinks={true},
2661   pageanchor={true},
2662 }}

```

`\makecatalog` When the document has illustrations, it is required to insert “: il.,” between the number of pages of the textual part and the page dimension. We have created a label to flag the existence of lists of figures. It is checked to be undefined using the plain T_EX command `\isundefined` (cf. the T_EX FAQ).

```

2663 \newcommand\makecatalog{%
2664   \vspace*{\fill}
2665   \begin{center}
2666     \setlength{\fboxsep}{5mm}
2667     \framebox[120mm][c]{\makebox[5mm][c]{}%
2668       \begin{minipage}[c]{105mm}
2669         \setlength{\parindent}{5mm}
2670         \noindent\sloppy\nohyphens\@authsur,
2671         \nohyphens\@authname\par
2672         \nohyphens{%
2673           \coppe@selecttitle/\@authname\ \@authsur. -- \local@cityname:
2674           UFRJ/COPPE, \coppe@docyear.}\par
2675           \pageref{front:pageno},
2676           \pageref{LastPage}
2677           p.\@ifundefined{r@cat:lofflag}{\pageref{cat:lofflag}}{$29,7$cm.\par
2678           % There is an issue here. When the last entry must be split between lines,
2679           % the spacing between it and the next paragraph becomes smaller.
2680           % Should we manually introduce a fixed space? But how could we know that
2681           % a name was split? Is this happening yet?
2682           \nohyphens{%
2683             \begin{tabularx}{100mm}[b]{@{}l@{ }>{\raggedright\arraybackslash}X@{}}
2684               \local@advisorstring: &
2685               \count1=0
2686               \toks@={}
2687               \@whilenum \count1<\@advisor \do{%
2688                 \ifcase\count1 % same as \ifnum0=\count1
2689                   \toks@=\expandafter{\csname CoppeAdvisorName:\the\count1%
2690                     \expandafter\endcsname\expandafter\space%
2691                     \csname CoppeAdvisorSurname:\the\count1\endcsname\\}
2692                 \else
2693                   \toks@=\expandafter\expandafter\expandafter{%
2694                     \expandafter\the\expandafter\toks@
2695                     \expandafter&\expandafter\space

```

```

2696         \csname CoppeAdvisorName:\the\count1\expandafter\endcsname%
2697         \expandafter\space\csname CoppeAdvisorSurname:\the\count1\endcsname\
2698     }%
2699     \fi
2700     \advance\count1 by 1}
2701     \the\toks@
2702 \end{tabularx}}\par
2703 \nohyphens{\local@doctype\ ({\MakeLowercase\@degreename}) --
2704 UFRJ/COPPE/Programa de \local@deptname, \coppe@docyear.}\par
2705 Refer{\^ e}ncias Bibliogr{\' a}ficas: p. \pageref{bib:begin} -- \pageref{bib:end}.\par
2706 \count1=0
2707 \count2=1
2708 \nohyphens{\@whilenum \count1<\value{keywords} \do {%
2709     \number\count2. \csname CoppeKeyword:\the\count1 \endcsname.
2710     \advance\count1 by 1
2711     \advance\count2 by 1}
2712 I. \csname CoppeAdvisorSurname:0\endcsname,%
2713 \ \csname CoppeAdvisorName:0\endcsname%
2714 \ifthenelse{\@advisor>1}{\ \emph{et~al.}}{}}.
2715 II. \local@universityname, COPPE, Programa de \local@deptname.
2716 III. T{\' i}tulo.}
2717 \end{minipage}}
2718 \end{center}
2719 \vspace*{\fill}}

```

\fichacatalografica The folha adicional required from August 2026 (SiBI manual, 9th rev. ed.,
\makefolhaaditional 3.1.2.1.2 and Annex H): the CAPES collection fields plus the ficha cata-
lografica, sitting immediately after the folha de rosto and counted by no-
body. \fichacatalografica takes the PDF produced by the SiBI generator at
<<http://fichacatalografica.sibi.ufrj.br/>>, optionally with a page number.

The sheet is institutional, so it is always in Portuguese whatever the main lan-
guage of the work – and deliberately without \selectlanguage, since the brazilian
babel language need not be loaded in, say, a Spanish-main document.

```

2720 \newcommand\coppe@fichafile{}
2721 \newcommand\coppe@fichapage{1}
2722 \newcommand\fichacatalografica[2][1]{%
2723     \gdef\coppe@fichapage{#1}\gdef\coppe@fichafile{#2}}
2724 \newcommand\coppe@marcatipo[1]{%
2725     \ifthenelse{\equal{\coppe@tipoproducao}{#1}}{$\times$}{\phantom{$\times$}}}
2726 \newcommand\coppe@marcavinc[1]{%
2727     \ifthenelse{\equal{\coppe@projetoavinc}{#1}}{$\times$}{\phantom{$\times$}}}
2728 \newcommand\coppe@campo[1]{%
2729     \ifthenelse{\equal{#1}{}}{\hrulefill}{#1\enspace\hrulefill}}
2730 \newcommand\coppe@checatipo{%
2731     \ifthenelse{\equal{\coppe@tipoproducao}{}}{}}{%
2732         \ifthenelse{\equal{\coppe@tipoproducao}{bibliografica}}\OR
2733             \equal{\coppe@tipoproducao}{artistica}\OR
2734             \equal{\coppe@tipoproducao}{tecnologica}\OR
2735             \equal{\coppe@tipoproducao}{tecnica}}{}}{%
2736     \ClassWarning{coppe}{Unknown \string\coppe@tipoproducao{ } value
2737         '\coppe@tipoproducao'. Annex H of the SiBI manual allows only
2738         bibliografica, artistica, tecnologica or tecnica; no box will be
2739         ticked}}}}

```

```

2740 \newcommand\makefolhaadicional{%
2741 \clearpage
2742 \thispagestyle{empty}%
2743 \coppe@checativo
2744 {\singlespacing
2745 \begin{center}
2746 {\large\bfseries Informa\c c\~oes Coleta CAPES}
2747 \end{center}
2748 \vspace{7mm}
2749 \noindent Tipo de produ\c c\~ao intelectual:\par
2750 \vspace{1mm}
2751 \hspace*{10mm}(\coppe@marcatipo{bibliografica})\enspace Bibliogr\'afica\par
2752 \hspace*{10mm}(\coppe@marcatipo{artistica})\enspace Art'\i{stica}\par
2753 \hspace*{10mm}(\coppe@marcatipo{tecnologica})\enspace Tecnol\'ogica\par
2754 \hspace*{10mm}(\coppe@marcatipo{tecnica})\enspace T\'ecnica\par
2755 \vspace{5mm}
2756 \noindent Projeto de pesquisa vinculado \'a produ\c c\~ao intelectual?\par
2757 \vspace{1mm}
2758 \hspace*{10mm}(\coppe@marcavinc{sim})\enspace Sim\par
2759 \hspace*{10mm}(\coppe@marcavinc{nao})\enspace N\~ao\par
2760 \vspace{5mm}
2761 \noindent Nome do projeto de pesquisa:\par
2762 \vspace{1mm}
2763 \noindent\coppe@campo{\coppe@nomeprojeto}\par
2764 \vspace{5mm}
2765 \noindent \'Area de concentra\c c\~ao da produ\c c\~ao intelectual:\par
2766 \vspace{1mm}
2767 \noindent\coppe@campo{\coppe@areaconc}\par
2768 \vspace{5mm}
2769 \noindent Ag\~encia(s) de fomento financiadora(s) da produ\c c\~ao
2770 intelectual:\par
2771 \vspace{1mm}
2772 \ifthenelse{\value{coppeagencias}=0}{%
2773 \noindent\hrulefill\par}{%
2774 \count1=0
2775 \@whilenum\count1<\value{coppeagencias}\do{%
2776 \noindent\hspace*{10mm}\csname CoppeAgencia:\the\count1\endcsname
2777 \ (\csname CoppeAgenciaSigla:\the\count1\endcsname)\par
2778 \advance\count1 by 1}}%
2779 \vspace{9mm}
2780 \begin{center}
2781 {\large\bfseries Ficha catalogr\'afica}\par
2782 \vspace{4mm}
2783 \ifthenelse{\equal{\coppe@fichafile}{}}{%
2784 \if@copperascunho
2785 \makecatalog
2786 \else
2787 \framebox[125mm][c]{\parbox[c][62mm][c]{115mm}{\centering\footnotesize
2788 Espa\c co reservado \'a ficha catalogr\'afica.\.[1ex]
2789 Gere-a em \texttt{fichacatalografica.sibi.ufrj.br} e inclua o
2790 arquivo com \texttt{\string\fichacatalografica\{arquivo.pdf\}},
2791 ou solicite-a \'a biblioteca do seu Programa.}}%
2792 \fi}{%
2793 \includegraphics[page=\coppe@fichapage,width=125mm]{\coppe@fichafile}}%

```

```

2794 \end{center}
2795 \par}}

```

`\dedication`

```

2796 \newcommand\dedication[1]{
2797   \gdef\@dedic{#1}
2798   \cleardoublepage
2799   \vspace*{\fill}
2800   \begin{flushright}
2801     \begin{minipage}{60mm}
2802       \raggedleft \it \normalsize \@dedic
2803     \end{minipage}
2804   \end{flushright}}

```

`\coppe@coadvisorrows` The coadvisor block of the abstract pages, in the shape the folha de rosto already uses. It prints nothing at all unless the `coorientador` class option is given AND the document actually declared a `\coadvisor`, so the default output is byte for byte what it was. The argument is the label macro, which differs per page: the main-language pages pass `\local@coadvisorstring` and the foreign one passes `\foreign@coadvisorstring`.

```

2805 \newcommand\coppe@coadvisorrows[1]{%
2806   \if@coppecoadvresumo
2807     \ifnum\@coadvisor>0
2808       \par\vspace*{2mm}%
2809       \noindent\begin{tabularx}{\textwidth}[b]{@{}l@{ }>{\raggedright\arraybackslash}X@{}}
2810         #1: &
2811         \count1=0
2812         \toks@={}
2813         \@whilenum \count1<\@coadvisor \do{%
2814           \ifcase\count1
2815             \toks@=\expandafter{\csname CoppeCoadvisorName:\the\count1%
2816               \expandafter\endcsname\expandafter\space%
2817               \csname CoppeCoadvisorSurname:\the\count1\endcsname\\}
2818           \else
2819             \toks@=\expandafter\expandafter\expandafter{%
2820               \expandafter\the\expandafter\toks@%
2821               \expandafter&\expandafter\space%
2822               \csname CoppeCoadvisorName:\the\count1\expandafter\endcsname%
2823               \expandafter\space\csname CoppeCoadvisorSurname:\the\count1\endcsname\\}
2824             }%
2825           \fi
2826           \advance\count1 by 1}
2827         \the\toks@
2828       \end{tabularx}%
2829     \fi
2830   \fi}

```

`abstract (env.)` This is a specialization of the abstract in the article standard class.

```

2831 \newenvironment{abstract}{%
2832   \cleardoublepage
2833   \thispagestyle{plain}
2834   \abstract@toptext\par
2835   \vspace*{8.6mm}

```

```

2836 \begin{center}
2837   \sloppy\nohyphens{\MakeUppercase\local@title}\par
2838   \vspace*{13.2mm}
2839   \@authname\ \@authsur \par
2840   \vspace*{7mm}
2841   \local@monthname/\coppe@docyear
2842 \end{center}\par
2843 \vspace*{1.5cm}
2844 \noindent%
2845 \begin{tabularx}{\textwidth}[b]{@{}l@{ }>{\raggedright\arraybackslash}X@{}}
2846   \local@advisorstring: &
2847   \count1=0
2848   \toks@={}
2849   \@whilenum \count1<\@advisor \do{%
2850     \ifcase\count1 % same as \ifnum0=\count1
2851       \toks@=\expandafter{\csname CoppeAdvisorName:\the\count1%
2852         \expandafter\endcsname\expandafter\space%
2853         \csname CoppeAdvisorSurname:\the\count1\endcsname\\}
2854     \else
2855       \toks@=\expandafter\expandafter\expandafter{%
2856         \expandafter\the\expandafter\toks@
2857         \expandafter&\expandafter\space
2858         \csname CoppeAdvisorName:\the\count1\expandafter\endcsname%
2859         \expandafter\space\csname CoppeAdvisorSurname:\the\count1\endcsname\\}
2860     }%
2861   \fi
2862   \advance\count1 by 1}
2863   \the\toks@
2864 \end{tabularx}%
2865 \coppe@coadvisorrows{\local@coadvisorstring}\par
2866 \vspace*{2mm}
2867 \noindent\local@deptstring: \local@deptname\par
2868 \vspace*{7mm}}{%
2869 \coppe@printkeywords{\coppemainstring{keywordsname}}%
2870 {CoppeKeyword:}{\value{keywords}}}%
2871 \vspace*{\fill}}

```

foreignabstract (*env.*)

```

2872 \newenvironment{foreignabstract}{%
2873   \clearoublepage
2874   \thispagestyle{plain}
2875   \begin{otherlanguage}{\coppe@foreignlang}
2876   \foreignabstract@toptext\par
2877   \vspace*{8.6mm}
2878   \begin{center}
2879     \sloppy\nohyphens{\MakeUppercase\foreign@title}\par
2880     \vspace*{13.2mm}
2881     \@authname\ \@authsur \par
2882     \vspace*{7mm}
2883     \foreign@monthname/\coppe@docyear
2884   \end{center}\par
2885   \vspace*{1.5cm}
2886   \noindent%
2887   \begin{tabularx}{\textwidth}[b]{@{}l@{ }>{\raggedright\arraybackslash}X@{}}

```

```

2888 \foreign@advisorstring: &
2889 \count1=0
2890 \toks@={}
2891 \@whilenum \count1<\@advisor \do{%
2892 \ifcase\count1 % same as \ifnum0=\count1
2893 \toks@=\expandafter{\csname CoppeAdvisorName:\the\count1%
2894 \expandafter\endcsname\expandafter\space%
2895 \csname CoppeAdvisorSurname:\the\count1\endcsname\\}
2896 \else
2897 \toks@=\expandafter\expandafter\expandafter{%
2898 \expandafter\the\expandafter\toks@
2899 \expandafter&\expandafter\space
2900 \csname CoppeAdvisorName:\the\count1\expandafter\endcsname%
2901 \expandafter\space\csname CoppeAdvisorSurname:\the\count1\endcsname\\
2902 }%
2903 \fi
2904 \advance\count1 by 1}
2905 \the\toks@
2906 \end{tabularx}%
2907 \coppe@coadvisorrows{\foreign@coadvisorstring}\par
2908 \vspace*{2mm}
2909 \noindent\foreign@deptstring: \foreign@deptname\par
2910 \vspace*{7mm}}{%
2911 \ifnum\value{foreignkeywords}>0
2912 \coppe@printkeywords{\coppeforeignstring{keywordsname}}%
2913 {CoppeForeignKeyword:}{\value{foreignkeywords}}%
2914 \else
2915 \coppe@printkeywords{\coppeforeignstring{keywordsname}}%
2916 {CoppeKeyword:}{\value{keywords}}%
2917 \fi
2918 \end{otherlanguage}
2919 \vspace*{\fill}
2920 \global\let\@author\@empty
2921 \global\let\@date\@empty
2922 \global\let\foreign@title\@empty
2923 \global\let\foreign@title\relax
2924 \global\let\local@title\@empty
2925 \global\let\local@title\relax
2926 \global\let\author\relax
2927 \global\let\author\relax
2928 \global\let\date\relax}

```

brazilianabstract (*env.*) Optional third abstract for non-pt main languages: a Brazilian Portuguese abstract printed in addition to **abstract** (main language) and **foreignabstract** (English by default). Wraps its body in `\begin{otherlanguage}{brazilian}` so the babel typography (hyphenation, quotation marks) is correct regardless of the current main language. Uses the same `\local@...` Portuguese strings the **abstract** env uses today. Meaningful only when main is neither **brazilian** nor **english**; harmless otherwise (the document just won't include it).

```

2929 \newenvironment{brazilianabstract}{%
2930 \cleardoublepage
2931 \thispagestyle{plain}
2932 \begin{otherlanguage}{brazilian}

```



```

2933 \abstract@top text\par
2934 \vspace*{8.6mm}
2935 \begin{center}
2936 \sloppy\nohyphens{\MakeUppercase\local@title}\par
2937 \vspace*{13.2mm}
2938 \@authname\ \@authsur\ \par
2939 \vspace*{7mm}
2940 \local@monthname/\coppe@docyear
2941 \end{center}\par
2942 \vspace*{1.5cm}
2943 \noindent%
2944 \begin{tabularx}{\textwidth}[b]{@{}l@{ }>{\raggedright\arraybackslash}X@{}}
2945 \local@advisorstring: &
2946 \count1=0
2947 \toks@={}
2948 \@whilenum \count1<\@advisor \do{%
2949 \ifcase\count1
2950 \toks@=\expandafter{\csname CoppeAdvisorName:\the\count1%
2951 \expandafter\endcsname\expandafter\space%
2952 \csname CoppeAdvisorSurname:\the\count1\endcsname\\}
2953 \else
2954 \toks@=\expandafter\expandafter\expandafter{%
2955 \expandafter\the\expandafter\toks@
2956 \expandafter&\expandafter\space
2957 \csname CoppeAdvisorName:\the\count1\expandafter\endcsname%
2958 \expandafter\space\csname CoppeAdvisorSurname:\the\count1\endcsname\\}
2959 }%
2960 \fi
2961 \advance\count1 by 1}
2962 \the\toks@
2963 \end{tabularx}%
2964 \coppe@coadvisorrows{\local@coadvisorstring}\par
2965 \vspace*{2mm}
2966 \noindent\local@deptstring: \local@deptname\par
2967 \vspace*{7mm}}{%
2968 % \coppestring takes ONE argument and resolves against the current language;
2969 % this runs before \end{otherlanguage}, so \language is already brazilian.
2970 \ifnum\value{braziliankeywords}>0
2971 \coppe@printkeywords{\coppestring{keywordsname}}%
2972 {CoppeBrazilianKeyword:}{\value{braziliankeywords}}%
2973 \else
2974 \coppe@printkeywords{\coppestring{keywordsname}}%
2975 {CoppeKeyword:}{\value{keywords}}%
2976 \fi
2977 \end{otherlanguage}
2978 \vspace*{\fill}}

```

\listoffigures

```

2979 \renewcommand\listoffigures{%
2980 \coppe@hasLof
2981 \if@twocolumn
2982 \@restonecoltrue\onecolumn
2983 \else
2984 \@restonecolfalse

```

```

2985 \fi
2986 \chapter*{\listfigurename}%
2987 \coppe@listtoc{\listfigurename}%
2988 \mkboth{\MakeUppercase\listfigurename}%
2989 {\MakeUppercase\listfigurename}%
2990 \@starttoc{lof}%
2991 \if@restonecol\twocolumn\fi
2992 }

```

\listoftables

```

2993 \renewcommand\listoftables{%
2994 \if@twocolumn
2995 \@restonecoltrue\onecolumn
2996 \else
2997 \@restonecolfalse
2998 \fi
2999 \chapter*{\listtablename}%
3000 \coppe@listtoc{\listtablename}%
3001 \mkboth{%
3002 \MakeUppercase\listtablename}%
3003 {\MakeUppercase\listtablename}%
3004 \@starttoc{lot}%
3005 \if@restonecol\twocolumn\fi
3006 }

```

\printlosymbols

```

3007 \newcommand\printlosymbols{%
3008 \renewcommand\glossaryname{\listsymbolname}%
3009 \begingroup\@coppeglosspretrue
3010 \@input@{\jobname.los}\endgroup}

```

\makelosymbols

```

3011 \def\makelosymbols{%
3012 \newwrite\@losfile
3013 \immediate\openout\@losfile=\jobname.syx
3014 \newcommand\symb1[3][\@bsphack\begingroup
3015 \ifstrempy{##1}{\def\@tempsymb1{##2=}}{\def\@tempsymb1{##1=}}%
3016 \@sanitize%
3017 \@wrlos{\@tempsymb1}{##2}{##3}\typeout%
3018 {Writing index of symbols file \jobname.syx}%
3019 \let\makelosymbols\@empty%
3020 }%
3021 \@onlypreamble\makelosymbols

3022 \AtBeginDocument{%
3023 \@ifpackageloaded{hyperref}{%
3024 \newcommand\@wrlos[3]{%
3025 \protected@write\@losfile{%
3026 {\string\indexentry{#1[#2] #3|hyperpage}{\thepage}}%
3027 \endgroup%
3028 \@esphack}}{%
3029 \newcommand\@wrlos[3]{%
3030 \protected@write\@losfile{%
3031 {\string\indexentry{#1[#2] #3}{\thepage}}%

```

```

3032 \endgroup%
3033 \@esphack}}}%

```

\printloabbreviations

```

3034 \newcommand\printloabbreviations{%
3035 \renewcommand\glossaryname{\listabbreviationname}%
3036 \begingroup\@coppeglosspretrue
3037 \@input@{\jobname.lab}\endgroup}

```

\makeloabbreviations

```

3038 \def\makeloabbreviations{%
3039 \newwrite\@labfile
3040 \immediate\openout\@labfile=\jobname.abx
3041 \newcommand\abbrev[3][\@bsphack\begingroup
3042 \ifstrempy{##1}{\def\@tempsymb1{##2=}}{\def\@tempsymb1{##1=}}
3043 \@sanitize
3044 \@wrlab{\@tempsymb1}{##2}{##3}}\typeout
3045 {Writing index of abbreviations file \jobname.abx}%
3046 \let\makeloabbreviations\@empty
3047 }
3048 \@onlypreamble\makeloabbreviations

3049 \AtBeginDocument{%
3050 \ifpackageloaded{hyperref}{%
3051 \newcommand\@wrlab[3]{%
3052 \protected@write\@labfile{%
3053 {\string\indexentry{#1[#2] #3|hyperpage}}{\thepage}}%
3054 \endgroup%
3055 \@esphack}}{%
3056 \newcommand\@wrlab[3]{%
3057 \protected@write\@labfile{%
3058 {\string\indexentry{#1[#2] #3}}{\thepage}}%
3059 \endgroup%
3060 \@esphack}}}%

```

\newsigla Acronyms (siglas), manual2024 2.8 / R43. \newsigla{key}{SIGLA}{full form}
\sigla declares one; \sigla{key} prints “full form (SIGLA)” on first use and registers it in the list of abbreviations and acronyms (through \abbrev, so \makeloabbreviations must be active), then just “SIGLA” on later uses. \sigla*{key} forces the full form. The first-use flag is set globally so it survives groups (floats, footnotes).

```

3061 \newcommand\newsigla[3]{%
3062 \@namedef{coppe@sigla@s@#1}{#2}%
3063 \@namedef{coppe@sigla@l@#1}{#3}%
3064 \newcommand\coppe@sigla@reg[1]{%
3065 \@ifundefined{abbrev}{%
3066 {\abbrev{\@nameuse{coppe@sigla@s@#1}}{\@nameuse{coppe@sigla@l@#1}}}}
3067 \newcommand\coppe@sigla@full[1]{%
3068 \@nameuse{coppe@sigla@l@#1}\space(\@nameuse{coppe@sigla@s@#1})}
3069 \newcommand\coppe@sigla@undef[1]{%
3070 \PackageWarning{coppe}{Sigla '#1' nao declarada com \string\newsigla}%
3071 \textbf{??#1??}}
3072 \newcommand\coppe@sigla@first[1]{%
3073 \global\@namedef{coppe@sigla@done@#1}{\coppe@sigla@reg{#1}\coppe@sigla@full{#1}}
3074 \newcommand\coppe@sigla@norm[1]{%

```

```

3075 \ifundefined{coppe@sigla@s@#1}{\coppe@sigla@undef{#1}}%
3076 {\ifundefined{coppe@sigla@done@#1}{\coppe@sigla@first{#1}}%
3077 {\@nameuse{coppe@sigla@s@#1}}}%
3078 \newcommand\coppe@sigla@star[1]{%
3079 \ifundefined{coppe@sigla@s@#1}{\coppe@sigla@undef{#1}}%
3080 {\global\@namedef{coppe@sigla@done@#1}{\coppe@sigla@full{#1}}}%
3081 \newcommand\sigla{\ifstar\coppe@sigla@star\coppe@sigla@norm}

3082 %%% \AtBeginDocument{%
3083 %%% \ifpackageloaded{hyperref}{%
3084 %%% \def\@wrlab#1#2{%
3085 %%% \protected@write\@labfile{%
3086 %%% {\string\indexentry{[#1] #2|hyperpage}{\thepage}}%
3087 %%% \endgroup
3088 %%% \@esphack}}{%
3089 %%% \def\@wrlab#1#2{%
3090 %%% \protected@write\@labfile{%
3091 %%% {\string\indexentry{[#1] #2}{\arabic{page}}}%
3092 %%% \endgroup
3093 %%% \@esphack}}}

3094 % Catalog-label writers. Two language-related traps must be avoided here:
3095 % 1. The label NAMES contain ':' and '.', which babel makes active in
3096 % some languages (e.g. Brazilian's "shorthand" set). If the write
3097 % argument is tokenised while one of those is active, pdfTeX aborts
3098 % with "<inserted text> }\endwrite" because the shorthand expansion
3099 % injects unbalanced braces. \detokenize emits its argument as
3100 % catcode-12 characters regardless of the current activation, so the
3101 % label NAMES survive whatever language is current at writer time.
3102 % 2. Spanish babel redefines \@roman (and therefore \roman{...}) to
3103 % produce small-caps roman numerals via \es@scroman, whose expansion
3104 % contains an \uppercase group that is unbalanced inside \write. The
3105 % same crash signature appears. The fix is to bypass LaTeX's
3106 % \roman/\arabic counter wrappers entirely and use the raw e-TeX
3107 % primitives \romannumeral and \number directly on the \c@page count
3108 % register --- both are fully expandable, language-agnostic, and
3109 % always produce plain ASCII digits or lowercase roman letters.
3110 % \Roman{page} is left in place because babel-spanish only redefines
3111 % \@Roman under the (non-default) ucroman option.
3112 \AtBeginDocument{%
3113 \ltx@ifpackageloaded{hyperref}{%
3114 \def\coppe@bibEnd{%
3115 \immediate\write\@auxout{%
3116 \string\newlabel{\detokenize{bib:end}}%
3117 {\number\c@page}{\detokenize{page.}\number\c@page}}}%
3118 \def\coppe@bibBegin{%
3119 \immediate\write\@auxout{%
3120 \string\newlabel{\detokenize{bib:begin}}%
3121 {\number\c@page}{\detokenize{page.}\number\c@page}}}%
3122 \def\coppe@mainBegin{%
3123 \immediate\write\@auxout{%
3124 \string\newlabel{\detokenize{front:pageno}}%
3125 {\Roman{page}}{\detokenize{page.}\romannumeral\c@page}}}%
3126 \def\coppe@hasLof{%
3127 \immediate\write\@auxout{%

```

```

3128     \string\newlabel{\detokenize{cat:lofflag}}%
3129     {{{\detokenize{:~il.;}}}{\detokenize{page.}\romannumeral\c@page}{}}}%
3130 }{%
3131     \def\coppe@bibEnd{%
3132     \immediate\write\@auxout{%
3133     \string\newlabel{\detokenize{bib:end}}{ {\number\c@page} } }%
3134     \def\coppe@bibBegin{%
3135     \immediate\write\@auxout{%
3136     \string\newlabel{\detokenize{bib:begin}}{ {\number\c@page} } }%
3137     \def\coppe@mainBegin{%
3138     \immediate\write\@auxout{%
3139     \string\newlabel{\detokenize{front:pageno}}{ {\Roman{page}} } }%
3140     \def\coppe@hasLof{%
3141     \immediate\write\@auxout{%
3142     \string\newlabel{\detokenize{cat:lofflag}}{ {\detokenize{:~il.;}} } }%
3143     }%
3144 }
3145 % The reference list is now produced by biblatex's \printbibliography (see the
3146 % bibliography setup after babel, above). The former thebibliography
3147 % redefinition and its \bibindent are no longer used and have been removed; the
3148 % bib:begin/bib:end catalog labels are now written by \AtBeginBibliography /
3149 % \AtEndBibliography via \coppe@bibBegin and \coppe@bibEnd, defined just above.

```

longquote (*env.*)

```

3150 \newlength{\recuolongquote}
3151 \setlength{\recuolongquote}{4cm}
3152
3153 \newenvironment*{longquote}[1][default]{%
3154 \begin{list}{}%
3155 \setlength{\leftmargin}{\recuolongquote}%
3156 \setlength{\rightmargin}{0pt}%
3157 \setlength{\itemindent}{0pt}%
3158 \setlength{\listparindent}{0pt}%
3159 \setlength{\parsep}{0pt}%
3160 \setlength{\topsep}{\baselineskip}%
3161 }%
3162 \item\relax
3163 \footnotesize
3164 \singlespacing
3165 \ifthenelse{\not\equal{#1}{default}}%
3166 {\itshape\selectlanguage{#1}}%
3167 {}%
3168 \ignorespaces
3169 }{%
3170 \end{list}%
3171 \ignorespacesafterend
3172 }
3173 \newenvironment{theglossary}{%
3174 \if@twocolumn%
3175 \@restonecoltrue\onecolumn%
3176 \else%
3177 \@restonecolfalse%
3178 \fi%

```

```

3179 \@mkboth{\MakeUppercase\glossaryname}%
3180 {\MakeUppercase\glossaryname}%
3181 \chapter*{\glossaryname}%
3182 \if@coppeglosspre
3183   \coppe@listtoc{\glossaryname}%
3184 \else
3185   \addcontentsline{toc}{chapter}{\glossaryname}%
3186 \fi
3187 \list{}
3188 {\setlength{\listparindent}{0in}%
3189  \setlength{\labelwidth}{1.0in}%
3190  \setlength{\leftmargin}{1.5in}%
3191  \setlength{\labelsep}{0.5in}%
3192  \setlength{\itemindent}{0in}}%
3193 \sloppy}%
3194 {\if@restonecol\twocolumn\fi%
3195 \endlist}
3196 %
3197 \renewenvironment{theindex}{%
3198   \if@twocolumn
3199     \@restonecolfalse
3200   \else
3201     \@restonecoltrue
3202   \fi
3203   \twocolumn[\@makeschapterhead{\indexname}]%
3204   \@mkboth{\MakeUppercase\indexname}%
3205   {\MakeUppercase\indexname}%
3206   \thispagestyle{plain}\parindent\z@
3207   \addcontentsline{toc}{chapter}{\indexname}
3208   \parskip\z@ \@plus .3\p@\relax
3209   \columnseprule \z@
3210   \columnsep 35\p@
3211   \let\item\@idxitem}
3212   {\if@restonecol\onecolumn\else\clearpage\fi}
3213 \newcommand\listabbreviationname{\coppemainstring{listabbreviation}}
3214 \newcommand\listsymbolname{\coppemainstring{listsymbol}}
3215 \newcommand\glossaryname{\coppemainstring{glossary}}
3216 %
3217 \newcommand\local@advisorstring{Orientador}
3218 \newcommand\foreign@advisorstring{Advisor}
3219 % Fed by the string dispatcher instead of a hard-coded Portuguese literal.
3220 % Expansion is lazy (it happens inside \makefrontpage), so \coppe@mainlang is
3221 % already set by the time this is used.
3222 \ifthenelse{\boolean{maledoc}}{%
3223   \newcommand\local@approvedname{\coppemainstring{approvedmale}}%
3224 }{%
3225   \newcommand\local@approvedname{\coppemainstring{approvedfemale}}%
3226 }
3227 \ifthenelse{\boolean{maledoc}}{%
3228   \newcommand\local@approvedonname{\coppemainstring{approvedonmale}}%
3229 }{%
3230   \newcommand\local@approvedonname{\coppemainstring{approvedonfemale}}%
3231 }
3232 \newcommand\local@universityname{Universidade Federal do Rio de Janeiro}

```

```

3233 \newcommand\local@deptstring{Programa}
3234 \newcommand\foreign@deptstring{Department}
3235 \newcommand\local@cityname{Rio de Janeiro}
3236 \newcommand\local@statename{RJ}
3237 \newcommand\local@countryname{Brasil}
3238 %
3239 % 3.1.2.1.1(e) of the 2026 manual requires the natureza, the objetivo, the
3240 % name of the institution AND the area de concentracao in this block. The area
3241 % was missing; it is appended as its own sentence when \areaconcentracao was
3242 % given, and simply omitted otherwise.
3243 \newcommand\frontcover@maintext{
3244   \sloppy\nohyphens{\local@doctype\ de \@degreename\
3245   \ifthenelse{\boolean{maledoc}}{apresentado}{apresentada}
3246   ao Programa de P{\' o}s-gradua{\c c}{\~ a}o em \local@deptname,
3247   COPPE, da \local@universityname, como parte dos requisitos
3248   necess{\' a}rios {\' a} obten{\c c}{\~ a}o do t{\' i}tulo de
3249   \local@degname\ em \local@deptname.}%
3250   \ifthenelse{\equal{\coppe@areaconc}{}}{ }{%
3251     \par\vspace{1ex}%
3252     \sloppy\nohyphens{\' Area de concentra{\c c}{\~ a}o: \coppe@areaconc.}}%
3253 }
3254 %
3255 \newcommand\frontpage@maintext{
3256   \noindent {\MakeUppercase\local@doctype}
3257   \ifthenelse{\boolean{maledoc}}{SUBMETIDO}{SUBMETIDA}
3258   \sloppy\nohyphens{AO CORPO DOCENTE DO INSTITUTO ALBERTO LUIZ COIMBRA
3259   DE P{\' O}S-GRADUA{\c C}{\~ A}O E PESQUISA DE ENGENHARIA DA
3260   UNIVERSIDADE FEDERAL DO RIO DE JANEIRO COMO PARTE DOS REQUISITOS
3261   NECESS{\' A}RIOS PARA A OBTEN{\c C}{\~ A}O DO GRAU DE
3262   {\MakeUppercase\local@degname} EM CI{\^ E}NCIAS EM
3263   {\MakeUppercase\local@deptname.\par}}%
3264 }
3265 %
3266 \newcommand\frontpage@bottomtext{%
3267   \begin{center}
3268   {\MakeUppercase{\local@cityname, \local@statename\ -- \local@countryname}}\par
3269   {\MakeUppercase\local@monthname\ DE \coppe@docyear}
3270   \end{center}%
3271 }
3272 %
3273 \newcommand\abstract@toptext{%
3274   \noindent Resumo \ifthenelse{\boolean{maledoc}}{do}{da}
3275   \local@doctype\ \ifthenelse{\boolean{maledoc}}{apresentado}{apresentada}
3276   \sloppy\nohyphens{{\\' a} COPPE/UFRJ como parte dos requisitos
3277   necess{\' a}rios para a obten{\c c}{\~ a}o do grau de
3278   \local@degname\ em Ci{\^ e}ncias (\@degree)}
3279 }
3280 \newcommand\foreignabstract@toptext{%
3281   \noindent \sloppy\nohyphens{Abstract of \foreign@doctype\ presented to
3282   COPPE/UFRJ as a partial fulfillment of the requirements for the
3283   degree of \foreign@degname\ of Science (\@degree)}
3284 }
3285 %

```

Late-load packages: setspace, hyperref, biblatex

These three packages *must* be the last things the class loads, and in this exact order:

- **setspace** – supplies `\singlespacing`, `\onehalfspacing` and friends. It is loaded here so the line-spacing state it carries around is settled before **hyperref** patches all the cross-reference and TOC plumbing.
- **hyperref** – patches an enormous number of LaTeX internals (`\section`, `\label`, `\ref`, `\bibcite`, list-of-* commands, ...). Loading it as late as possible is the standard rule of thumb so it sees the final version of those macros, including the ones the **coppe** class redefines (e.g. `\@chapter`, `\listofquados`).
- **biblatex** – must come after **hyperref** so its citation links and back-references hook into **hyperref** correctly; otherwise the `\cite` commands print as plain (non-clickable) text.

```
3286 \RequirePackage{setspace}
3287 % Apply the default line spacing now that setspace is available. The flag was
3288 % set above by \DeclareOption{doublespacing}; absent the option, one-and-a-half.
3289 \if@coppedoublespace\doublespacing\else\onehalfspacing\fi
3290 % pdfx MUST precede hyperref -- it loads hyperref itself, with the options
3291 % PDF/A needs, and cannot fix up an already-loaded one. Level a-2b rather than
3292 % a-1b (the "ISO 19005-1" of Annex I): a-1b forbids transparency, which
3293 % tcolorbox and several graphics packages produce, so a-1b would fail on
3294 % documents that are otherwise perfectly fine. a-2b is ISO 19005-2, accepted by
3295 % Pantheon and by the SiBI's own instructions, which also list PDF/A-2b.
3296 \if@coppepdfa
3297   \RequirePackage[a-2b]{pdfx}
3298 % pdfTeX stamps /PTEX.Fullbanner into the document information dictionary.
3299 % It is not one of the keys ISO 19005-2 6.6.2.3.2 maps to an XMP property, and
3300 % no extension schema declares it, so it is a DocInfo entry with no XMP
3301 % counterpart -- a finding a PDF/A checker has every reason to raise, and one
3302 % that costs nothing to remove. -1 sets every suppression bit.
3303 %
3304 % The two engines spell this differently, and not just in the name. pdfTeX has
3305 % \pdfsuppressptexinfo, whose bits only ever govern PTEX.* entries, so -1 (all
3306 % of them) is safe. LuaTeX moved the pdfTeX integer parameters behind
3307 % \pdfvariable in 0.85 AND renamed this one to suppressoptionalinfo, whose bits
3308 % reach further: besides the four PTEX.* entries (1, 2, 4, 8) it can also
3309 % suppress Creator, the dates, Producer, Trapped and -- the dangerous one --
3310 % the trailer /ID, which PDF/A requires. So -1 is exactly what must NOT be
3311 % written there; 15 is the four PTEX.* bits and nothing else. Checked on a real
3312 % LuaTeX run: /ID survives.
3313 %
3314 % Getting the name wrong is not a quiet failure. \pdfvariable with an unknown
3315 % variable name makes LuaTeX typeset the name, and the document stops with
3316 % "Missing \begin{document}" -- which is how the first attempt at this was
3317 % caught.
3318 \ifdefined\pdfsuppressptexinfo
3319   \pdfsuppressptexinfo=-1
3320 \else
3321   \ifdefined\pdfvariable
```



```

3322     \pdfvariable suppressoptionalinfo=15
3323     \fi
3324 \fi
3325 % pdfx takes the XMP metadata from \jobname.xmpdata, which it reads when it is
3326 % LOADED -- before the preamble has run, so before \title, \author and
3327 % \keyword exist. The file is therefore WRITTEN by the class and picked up on
3328 % the NEXT run. The build already runs pdflatex three times for cross-
3329 % references, so the deposited PDF always carries current metadata; only a
3330 % first-ever compilation of a brand-new file emits pdfx's "no \jobname.xmpdata"
3331 % warning and lacks dc:title.
3332 %
3333 % The write happens at the END of the document, not at \begin{document}. This
3334 % class has always documented -- and every example follows it -- that \title,
3335 % \author, \department and \keyword go AFTER \begin{document}, next to
3336 % \maketitle. A write at \begin{document} therefore ran before any of them
3337 % existed and died with "Undefined control sequence" on \@authname, so the
3338 % pdfa option was unusable in exactly the style the manual teaches. It only
3339 % ever worked for a document that set its metadata in the preamble. At the end
3340 % everything is set whichever style the author chose, and \immediate still puts
3341 % the file on disk in time for the next run.
3342 %
3343 % The hook is enddocument/afteraux, and the stream is \@mainaux rather than one
3344 % of our own. TeX has sixteen output streams and example.tex was already using
3345 % all of them (toc, lof, lot, loa, lol, loq, los, idx, glo, aux, bcf, ...), so
3346 % \newwrite here failed with "No room for a new \write" on precisely the
3347 % document that most needs validating. enddocument/afteraux fires after LaTeX
3348 % has closed the .aux, which frees that stream for the one \immediate\openout
3349 % we need and costs the document nothing. Kernels older than 2020-10-01 have no
3350 % such hook; there the class falls back to \AtEndDocument with a stream of its
3351 % own, which is the pre-existing behaviour.
3352 %
3353 % Every field is guarded, and the guards are not cosmetic. \coppe@selecttitle
3354 % is \@nameuse{coppe@title@<lang>}, and \csname on a name that does not exist
3355 % yet MAKES it \relax: unexpandable, so \write puts the control sequence NAME
3356 % into the file instead of the title. The .xmpdata then carried the literal
3357 % \coppe@title@brazilian, pdfx read it back on the following run, and the
3358 % document stopped compiling at all -- a single bad run bricked the file until
3359 % someone deleted the .xmpdata by hand. A field that is not set is now simply
3360 % omitted, which pdfx handles, instead of poisoning the next run.
3361 %
3362 % The writes must be \immediate. \protected@write defers to the shipout, and
3363 % the stream is closed here and now, so deferred writes would land nowhere --
3364 % which is exactly what happened first time round: a 0-byte .xmpdata.
3365 % \let\protect\string inside the group gives the protection that
3366 % \protected@write would have given: accents and other robust commands are
3367 % written literally instead of being expanded. pdfx reads the .xmpdata as
3368 % LaTeX, so \a arrives there intact.
3369 \ifdefined\AddToHook
3370   \let\coppe@xmpout\@mainaux
3371   \def\coppe@xmphook{\AddToHook{enddocument/afteraux}}%
3372 \else
3373   \newwrite\coppe@xmpout
3374   \let\coppe@xmphook\AtEndDocument
3375 \fi

```

```

3376 \coppe@xmphook{%
3377   \gdef\coppe@xmpkeys{%
3378     \count@=0
3379     \@whilenum\count@<\value{keywords}\do{%
3380       \ifnum\count@>0
3381         \g@addto@macro\coppe@xmpkeys{\string\sep\space}%
3382       \fi
3383       \protected@edef\coppe@tmp{\csname CoppeKeyword:\the\count@\endcsname}%
3384       \expandafter\g@addto@macro\expandafter\coppe@xmpkeys
3385       \expandafter{\coppe@tmp}%
3386       \advance\count@ by 1}%
3387 \immediate\openout\coppe@xmpout=\jobname.xmpdata
3388 \begingroup
3389   \let\protect\string
3390   \@ifundefined{coppe@title@\coppe@mainlang}{\{%
3391     \immediate\write\coppe@xmpout{\string\Title{\coppe@selecttitle}}}%
3392   \@ifundefined{@authname}{\{%
3393     \immediate\write\coppe@xmpout{%
3394       \string\Author{\@authname\space\@authsur}}}%
3395   \immediate\write\coppe@xmpout{\string\Keywords{\coppe@xmpkeys}}%
3396   \ifthenelse{equal{\coppe@deptcode}{}}{\{%
3397     \immediate\write\coppe@xmpout{\string\Subject{\local@doctype\space de
3398       \@degreename. Programa de \meta@deptname\space(\coppe@deptcode),
3399       COPPE/UFRJ}}}%
3400   \immediate\write\coppe@xmpout{\string\Publisher{\local@universityname}}}%
3401   \@ifundefined{coppe@docyear}{\{%
3402     \immediate\write\coppe@xmpout{\string\Date{\coppe@docyear}}}%
3403 \endgroup
3404 \immediate\closeout\coppe@xmpout}%
3405 \fi
3406 \RequirePackage{hyperref}
3407 % Bibliography engine: the full coppe BibLaTeX/biber ABNT style, replacing the
3408 % former natbib+bibtex setup. The numbers class option selects the numeric
3409 % citation style; otherwise the author-date default is used. natbib=true keeps
3410 % \citet/\citep available for users migrating from natbib.
3411 \if@coppenumeric
3412   \RequirePackage[backend=biber,database=coppe,%
3413     bibstyle=coppe-numeric,citestyle=coppe-numeric,natbib=true]{bibtex}
3414 \else
3415   \RequirePackage[backend=biber,database=coppe,%
3416     bibstyle=coppe,citestyle=coppe,natbib=true]{bibtex}
3417 \fi
3418 % Reference-list heading "Referencias" (pt) / "References" (en) -- manual2024
3419 % 3.1.4.1 requires the post-textual list to be titled "Referencias", not
3420 % "Bibliografia" (babel's default \bibname). The title is rendered through
3421 % \coppestring so it follows the current main language in BOTH the chapter
3422 % heading and the .toc entry (bibtex's \bibstring proved unreliable across
3423 % the .toc write, and babel's \bibname caption override was equally fragile).
3424 \defbibheading{bibliography}{%
3425   \chapter*{\coppestring{references}}%
3426   \addcontentsline{toc}{chapter}{\coppestring{references}}
3427 % The catalog page range (bib:begin/bib:end) was written by the old
3428 % thebibliography redefinition, which bibtex's \printbibliography does not
3429 % use. bibtex offers \AtBeginBibliography but no end counterpart, so the

```

```

3430 % begin label uses that hook and the end label is emitted by wrapping
3431 % \printbibliography to write it just after the reference list.
3432 \AtBeginBibliography{\coppe@bibBegin}
3433 \let\coppe@orig@printbibliography\printbibliography
3434 \renewcommand\printbibliography[1][]{\%
3435   \coppe@orig@printbibliography[#1]\%
3436   \coppe@bibEnd}
3437 \</class>

3438 \<*glossary>
3439 actual '='
3440 quote '!'
3441 level '>'
3442 %%% delim_0 " , p. "
3443 delim_0 "\\dotfill "
3444 lethead_flag 0
3445 headings_flag 0
3446 preamble
3447 "\n\begin{theglossary}\n \\makeatletter"
3448 postamble
3449 "\n \\end{theglossary}\n"
3450 \</glossary>

3451 \<*class>
3452 \newcommand{\annex}{
3453   \renewcommand{\appendixname}{\coppestring{annex}}\%
3454   \appendix
3455   \@coppeannextrue
3456 % \appendix has already re-issued the centred chapter format, but it baked in
3457 % the \appendixname of that moment. Re-issue it now that the name is "Anexo".
3458   \coppe@appendixheadings
3459 }
3460 \</class>

```

BibLaTeX style files

Generated from this .dtx by coppe.ins (one docstrip module each).

```

3461 \<*dbx>
3462 \ProvidesFile{coppe.dbx}[2026/09/09 v4.1 CoppeTeX ABNT datamodel]
3463 % Custom entry types beyond biblatex's standard datamodel (games, TV, maps).
3464 % map = cartographic document (NBR 6023 4.2.12); biblatex has no native @map.
3465 % The remaining ABNT "exotic" types (legislation, jurisdiction, legal, patent,
3466 % standard, music, audio, image, artwork, video, dataset, letter, periodical,
3467 % performance) are already in biblatex's standard datamodel.
3468 \DeclareDatamodelEntrytypes{game,videogame,boardgame,tvshow,map}
3469 % Custom fields.
3470 \DeclareDatamodelFields[type=field,datatype=literal]{course} % "(... em <course>)"
3471 \DeclareDatamodelFields[type=field,datatype=literal]{coppedegree} % degree word
3472 \DeclareDatamodelFields[type=field,datatype=literal]{platform} % videogame platform
3473 \DeclareDatamodelFields[type=field,datatype=literal]{scale} % map scale (escala)
3474 \DeclareDatamodelFields[type=list,datatype=literal]{artist} % game/illustration art
3475 \DeclareDatamodelFields[type=list,datatype=literal]{director} % movie/tvshow/video
3476 \DeclareDatamodelFields[type=list,datatype=literal]{producer}
3477 \DeclareDatamodelFields[type=field,datatype=literal]{caderno} % newspaper section

```

```

3478 \DeclareDatamodelFields[type=field,datatype=literal]{relator}      % jurisprudence rapporteur
3479 \DeclareDatamodelEntryfields{course,coppedegree}
3480 \DeclareDatamodelEntryfields[game,videogame,boardgame,software,movie,tvshow,video]%
3481   {artist,platform,director,producer,version}
3482 \DeclareDatamodelEntryfields[map]{scale}
3483 \DeclareDatamodelEntryfields[article]{caderno}
3484 \DeclareDatamodelEntryfields[jurisdiction]{relator}
3485 </dbx>
3486 <*bbx>
3487 \ProvidesFile{coppe.bbx}[2026/09/09 v4.1 CoppeTeX ABNT bibliography style]
3488 % The reference list is single-spaced (ABNT 2.4) via \singlespacing in
3489 % \AtBeginBibliography. coppe.cls already loads setspace, but the style must work
3490 % standalone too (e.g. \usepackage[bibstyle=coppe]{biblatex} in a plain article),
3491 % so load setspace here as well (idempotent).
3492 \RequirePackage{setspace}
3493 % ABNT reference list keeps the date at the end (imprenta), same for author-date
3494 % and numeric modes, so we derive from the standard style, not authoryear.
3495 \RequireBibliographyStyle{standard}
3496
3497 % Note: standard style never dashes repeated authors, which already matches the
3498 % post-2020 ABNT rule (repeat the entry in full), so no 'dashed' option is needed.
3499 \ExecuteBibliographyOptions{%
3500   maxbibnames=99,      % ABNT 4.3.1.3: list all authors by default
3501   giveninits=false,
3502   uniquename=false,
3503   uniquelist=false,
3504   labeldateparts=true, % needed for author-date extradata (1999a/1999b)
3505   sorting=nyt,
3506 }
3507
3508 % Language strings (Disponível em / Acesso em / In, etc.)
3509 \DeclareLanguageMapping{brazilian}{brazilian-coppe}
3510 \DeclareLanguageMapping{english}{english-coppe}
3511
3512 % --- Names: all reversed "FAMILY, Given", separated by ";" in the list ---
3513 \DeclareNameAlias{author}{family-given}
3514 \DeclareNameAlias{editor}{family-given}
3515 \DeclareNameAlias{translator}{family-given}
3516 \DeclareNameAlias{default}{family-given}
3517 \DeclareDelimFormat[bib,biblist]{multinamedelim}{\addsemicolon\space}
3518 \DeclareDelimFormat[bib,biblist]{finalnamedelim}{\addsemicolon\space}
3519
3520 % ABNT (NBR 6023 8.1.1.4): a collection entered under the name of its
3521 % organizer prints the kind of responsibility in parentheses after the
3522 % name, abbreviated, lowercase and in the singular even for several
3523 % organizers -- e.g. "SOUSA, A. M. de; OLIVEIRA, L. da C. (org.)". The
3524 % standard style prints ", Org." (comma, no parentheses) when the editor
3525 % takes the author slot, so override editor+others to wrap the role string
3526 % in "(...)". The host position ("In: NAME (org.)") already parentheseses
3527 % the role via coppe:bookeditor below. The role word itself is the
3528 % lowercase, singular "org." (and "ed.", "cur.", ...) set in the language
3529 % packs (brazilian-coppe.lbx et al.).
3530 \renewbibmacro*{editor+others}{%
3531   \ifboolexpr{ test \ifuseeditor and not test {\ifnameundef{editor}} }

```

```

3532     {\printnames{editor}%
3533      \setunit*{\addspace}%
3534      \printtext{\mkbibparens{\usebibmacro{editor+othersstrg}}}%
3535      \clearname{editor}}
3536     {}
3537 % Same parenthesised role for drivers that take the editor through the plain
3538 % editor macro instead of editor+others (e.g. @manual).
3539 \renewbibmacro*{editor}{%
3540   \ifboolexpr{ test \ifuseeditor and not test {\ifnameundef{editor}} }
3541     {\printnames{editor}%
3542      \setunit*{\addspace}%
3543      \printtext{\mkbibparens{\usebibmacro{editorstrg}}}%
3544      \clearname{editor}}
3545     {}
3546
3547 % Uppercase the surname (family + prefix) only inside the bibliography list;
3548 % citations keep normal caps (post-2020 rule).
3549 % ABNT date (NBR 6023 §4.3.5.5.2): "dia mês. ano" -- without the brazilian "de"
3550 % between the parts (e.g. "4 fev. 1992", "abr. 1996"). #1=year, #2=month, #3=day
3551 % field names; the month stays abbreviated via \mkbibmonth.
3552 \protected\def\coppe@abntdate#1#2#3{%
3553   \iffieldundef{#3}{\stripzeros{\thefield{#3}}\nobreakspace}%
3554   \iffieldundef{#2}{\mkbibmonth{\thefield{#2}}\iffieldundef{#1}{\space}}%
3555   \iffieldundef{#1}{\stripzeros{\thefield{#1}}}}
3556 \AtBeginBibliography{%
3557   \singlespacing % entries single-spaced internally (2.4)
3558   \setlength{\bibitemsep}{\baselineskip}% one blank line between entries (2.4)
3559   \renewcommand*{\mkbibnamefamily}[1]{\MakeUppercase{#1}}%
3560   \renewcommand*{\mkbibnameprefix}[1]{\MakeUppercase{#1}}%
3561   \let\mkbibdatelong\coppe@abntdate % ABNT date format (drop the "de")
3562   \let\mkbibdateshort\coppe@abntdate
3563 }
3564
3565 % --- Titles: bold for whole-document types; plain for parts; entry-by-title
3566 % (no author/editor) -> not bold, first word in CAPS (ABNT 4.2). ---
3567 % \coppeUCtitle uppercases the first SIGNIFICANT word of the title (ABNT 4.2).
3568 % If the title starts with an article (o/a/os/as/um/uma/uns/umas, the/an), the
3569 % article AND the next word are uppercased; otherwise just the first word. The
3570 % delimited macros split off words; \coppeUCkeep strips the trailing-space
3571 % sentinel so no spurious space is left before the following punctuation.
3572 \protected\def\coppeUCtitle#1{\coppeUCtitleA#1 \coppeUCstop}
3573 \protected\def\coppeUCtitleA#1 #2\coppeUCstop{%
3574   \coppeUCifart{#1}
3575   {\MakeUppercase{#1}\ifblank{#2}{\space\coppeUCtitleB#2\coppeUCstop}}%
3576   {\MakeUppercase{#1}\ifblank{#2}{\space\coppeUCkeep#2\coppeUCstop}}%
3577 \protected\def\coppeUCtitleB#1 #2\coppeUCstop{%
3578   \MakeUppercase{#1}\ifblank{#2}{\space\coppeUCkeep#2\coppeUCstop}}
3579 \protected\def\coppeUCkeep#1 \coppeUCstop{#1}
3580 % \coppeUCifart{word}{if-article}{if-not}: case-insensitive membership test.
3581 \protected\def\coppeUCifart#1{%
3582   \begingroup
3583   \lowercase{\def\coppeUC@w{#1}}%
3584   \edef\coppeUC@w{\coppeUC@w}%
3585   \expandafter\in@\expandafter{\coppeUC@w}{a,o,as,os,um,uma,uns,umas,the,an,}%

```

```

3586 \ifin@aftergroup\@firstoftwo\else\aftergroup\@secondoftwo\fi
3587 \endgroup}
3588 \DeclareFieldFormat{title}{%
3589 \ifboolexpr{ test {\ifnameundef{labelname}} }
3590 {#1}%
3591 {\mkbibbold{#1}}}
3592 % First word of the title in CAPS for entry-by-title (ABNT 4.2). The titlecase
3593 % format receives the RAW title text (the title format above only sees the
3594 % field-printing commands), so \coppeUCtitle can split off and uppercase the
3595 % first word here.
3596 \DeclareFieldFormat{titlecase}{%
3597 \ifboolexpr{ test {\ifnameundef{labelname}} }
3598 {\coppeUCtitle{#1}}%
3599 {#1}}
3600 \DeclareFieldFormat[article,incollection,inbook,inproceedings,suppperiodical,review]{title}{%
3601 \DeclareFieldFormat[thesis]{title}{\mkbibbold{#1}}
3602 % biblatex's standard style quotes patent (and unpublished) titles; ABNT bolds
3603 % the title of a whole document, so override those back to bold.
3604 \DeclareFieldFormat[patent]{title}{\mkbibbold{#1}}
3605 % Legal documents (NBR 6023 4.2.6): the law's identification/ementa is NOT
3606 % highlighted; the entry leads with the jurisdiction (entity author, CAPS).
3607 \DeclareFieldFormat[legislation,jurisdiction,legal]{title}{#1}
3608 \DeclareFieldFormat[journaltitle]{\mkbibbold{#1}}
3609 \DeclareFieldFormat{booktitle}{\mkbibbold{#1}}
3610 \DeclareFieldFormat{maintitle}{\mkbibbold{#1}}
3611
3612 % --- Pages: ABNT uses "p." (also for ranges); theses use "f." for pagetotal ---
3613 \DeclareFieldFormat{pagetotal}{#1\addnbspace p\adddot}
3614 \DeclareFieldFormat[thesis]{pagetotal}{#1\addnbspace f\adddot}
3615 \DeclareFieldFormat{pages}{p\adddot\addnbspace#1}
3616
3617 % --- Periodical volume / number prefixes ---
3618 \DeclareFieldFormat[article]{volume}{v\adddot\addnbspace#1}
3619 \DeclareFieldFormat[article]{number}{n\adddot\addnbspace#1}
3620
3621 % --- "In:" before host title (incollection/inproceedings/inbook) ---
3622 \renewbibmacro*{in:}{\printtext{\bibstring{in}\addcolon\space}}
3623
3624 % --- Online: "Disponível em: <url>." + "Acesso em: <date>." via lbx strings ---
3625 \DeclareFieldFormat{url}{\bibstring{availablefrom}\addcolon\space\url{#1}}
3626
3627 % --- ABNT article driver: no "In:"; Journal, Local, v., n., p., date, DOI/URL.
3628 % Newspapers (entrysubtype=newspaper, set by the @artigodejornal source map)
3629 % differ: the newspaper name (journaltitle) and city, then either
3630 % "<date>. <caderno>, p. <pages>" (when a caderno/section is given), or
3631 % "p. <pages>, <date>" (page before date, when none),
3632 % per NBR 6023 §4.2.3.6. ---
3633 \DeclareBibliographyDriver{article}{%
3634 \usebibmacro{bibindex}%
3635 \usebibmacro{begentry}%
3636 \usebibmacro{author/translator+others}%
3637 \setunit{\labelnamepunct}\newblock
3638 \usebibmacro{title}%
3639 \newunit\newblock

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```

3640 \printfield{journaltitle}%
3641 \setunit{\addcomma\space}\printlist{location}%
3642 \iffieldequalstr{entrysubtype}{newspaper}
3643   {\iffieldundef{caderno}
3644     {\setunit{\addcomma\space}\printfield{pages}%
3645       \setunit{\addcomma\space}\printdate}%
3646     {\setunit{\addcomma\space}\printdate%
3647       \setunit{\adddot\space}\printfield{caderno}%
3648       \setunit{\addcomma\space}\printfield{pages}}}%
3649 \setunit{\addcomma\space}\printfield{volume}%
3650 \setunit{\addcomma\space}\printfield{number}%
3651 \setunit{\addcomma\space}\printfield{pages}%
3652 \setunit{\addcomma\space}\printdate}%
3653 \newunit\newblock
3654 \printfield{doi}%
3655 \newunit\newblock
3656 \usebibmacro{url+urldate}%
3657 \usebibmacro{finentry}}
3658
3659 % --- incollection: ABNT puts the host organizer before the booktitle, as
3660 % "AUTOR. Parte. In: ORGANIZADOR (Org.). Booktitle. Local: Editora, ano,
3661 % p. X-Y." The organizer role string is "Org./Orgs." (see the .lbx). ---
3662 \newbibmacro*{coppe:bookeditor}{%
3663   \ifnameundef{editor}
3664     {}
3665     {\printnames{editor}%
3666       \setunit*{\addspace}\printtext{\mkbibparens{\usebibmacro{editorstrg}}}%
3667       \clearname{editor}%
3668       \setunit{\adddot\space}}
3669 \newbibmacro*{coppe:partdriver}{%
3670   \usebibmacro{bibindex}\usebibmacro{begentry}%
3671   \usebibmacro{author/translator+others}%
3672   \setunit{\labelnamepunct}\newblock
3673   \usebibmacro{title}%
3674   \newunit\newblock
3675   \usebibmacro{in:}%
3676   \usebibmacro{coppe:bookeditor}%
3677   \printfield{booktitle}%
3678   \newunit\newblock
3679   \printfield{edition}%
3680   \newunit\newblock
3681   \printlist{location}%
3682   \setunit*{\addcolon\space}\printlist{publisher}%
3683   \setunit*{\addcomma\space}\printdate%
3684   \newunit\newblock
3685   \printfield{pages}%
3686   \setunit{\addcomma\space}\printfield{chapter}%
3687   \newunit\newblock
3688   \usebibmacro{url+urldate}%
3689   \usebibmacro{finentry}}
3690 \DeclareBibliographyDriver{incollection}{\usebibmacro{coppe:partdriver}}
3691 \DeclareBibliographyDriver{inbook}{\usebibmacro{coppe:partdriver}}
3692
3693 % --- type field: resolve a known bibliography string (babel-aware), else literal ---

```

```

3694 \DeclareFieldFormat{type}{\ifbibstring{#1}{\bibstring{#1}}{#1}}
3695
3696 % --- Thesis: "<type> (em <course>)" using the custom course field (ABNT). ---
3697 \DeclareFieldFormat{course}{\mkbibparens{\coppestring{in:}\addspace#1}}
3698 \DeclareBibliographyDriver{thesis}{%
3699   \usebibmacro{bibindex}\usebibmacro{begentry}%
3700   \usebibmacro{author}%
3701   \setunit{\labelnamepunct}\newblock
3702   \usebibmacro{title}%
3703   \newunit\newblock
3704   \printfield{type}\setunit*\addspace\printfield{course}%
3705   \newunit\newblock
3706   \printlist{institution}%
3707   \setunit*\addcomma\space\printlist{location}%
3708   \setunit*\addcomma\space\printdate%
3709   \newunit\newblock
3710   \printfield{pagetotal}%
3711   \newunit\newblock
3712   \printfield{note}%
3713   \newunit\newblock
3714   \usebibmacro{url+urldate}%
3715   \usebibmacro{finentry}}
3716 % --- Games / audiovisual: render the custom fields (artist, platform,
3717 %       director, producer, version) the generic misc driver ignores. ---
3718 \newbibmacro*{coppe:media}{%
3719   \iflistundef{artist}{\setunit{\addperiod\space}%
3720     \printtext{\coppestring{art}\addcolon\space}\printlist{artist}}%
3721   \iffieldundef{platform}{\setunit{\addperiod\space}%
3722     \printtext{\coppestring{platform}\addcolon\space}\printfield{platform}}%
3723   \iflistundef{director}{\setunit{\addperiod\space}%
3724     \printtext{\coppestring{directedby}\addcolon\space}\printlist{director}}%
3725   \iflistundef{producer}{\setunit{\addperiod\space}%
3726     \printtext{\coppestring{producedby}\addcolon\space}\printlist{producer}}%
3727 \newbibmacro*{coppe:mediadriver}{%
3728   \usebibmacro{bibindex}\usebibmacro{begentry}%
3729   \usebibmacro{author/translator+others}%
3730   \setunit{\labelnamepunct}\newblock
3731   \usebibmacro{title}%
3732   \newunit\newblock
3733   \usebibmacro{coppe:media}%
3734   \newunit\newblock
3735   \printlist{publisher}%
3736   \setunit*\addcomma\space\printlist{location}%
3737   \setunit*\addcomma\space\printdate%
3738   \newunit\newblock
3739   \printfield{version}%
3740   \newunit\newblock
3741   \printfield{note}%
3742   \newunit\newblock
3743   \usebibmacro{url+urldate}%
3744   \usebibmacro{finentry}}
3745 \DeclareBibliographyDriver{game}{\usebibmacro{coppe:mediadriver}}
3746 \DeclareBibliographyDriver{videogame}{\usebibmacro{coppe:mediadriver}}
3747 \DeclareBibliographyDriver{boardgame}{\usebibmacro{coppe:mediadriver}}

```



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3748 \DeclareBibliographyDriver{movie}{\usebibmacro{coppe:mediadriver}}
3749 \DeclareBibliographyDriver{tvshow}{\usebibmacro{coppe:mediadriver}}
3750 % Cartographic document (map, NBR 6023 4.2.12): like a book (author/title/
3751 % imprenta) plus the mandatory scale ("Escala 1:N").
3752 \DeclareFieldFormat{scale}{\bibstring{coppescale}\addnbspace#1}
3753 \DeclareBibliographyDriver{map}{%
3754   \usebibmacro{bibindex}\usebibmacro{begentry}%
3755   \usebibmacro{author/translator+others}%
3756   \setunit{\labelnamepunct}\newblock
3757   \usebibmacro{title}%
3758   \newunit\newblock
3759   \printlist{location}%
3760   \setunit*{\addcolon\space}\printlist{publisher}%
3761   \setunit*{\addcomma\space}\printdate%
3762   \newunit\newblock
3763   \printfield{scale}%
3764   \newunit\newblock
3765   \usebibmacro{url+urldate}%
3766   \usebibmacro{finentry}}
3767 % Patent (NBR 6023 4.2.5): inventor. \textbf{title}. Depositante: holder. type
3768 % number. date. The holder (depositante) is printed in normal case, not the
3769 % ALL-CAPS family treatment the bibliography gives the leading author.
3770 \newbibmacro*{coppe:depositor}{%
3771   \ifnameundef{holder}
3772   {}
3773   {\printtext{\bibstring{depositor}\addcolon\space}%
3774     \begingroup
3775       \def\mkbibnamefamily##1{##1}%
3776       \def\mkbibnameprefix##1{##1}%
3777       \printnames{holder}%
3778     \endgroup}}
3779 \DeclareBibliographyDriver{patent}{%
3780   \usebibmacro{bibindex}%
3781   \usebibmacro{begentry}%
3782   \usebibmacro{author/translator+others}%
3783   \setunit{\labelnamepunct}\newblock
3784   \usebibmacro{title}%
3785   \newunit\newblock
3786   \usebibmacro{coppe:depositor}%
3787   \newunit\newblock
3788   \printfield{type}%
3789   \setunit*{\addspace}\printfield{number}%
3790   \newunit\newblock
3791   \printdate%
3792   \newunit\newblock
3793   \usebibmacro{url+urldate}%
3794   \usebibmacro{finentry}}
3795 % Exotic ABNT types that biblatex's standard style leaves without a driver:
3796 % give each a baseline (book-like for published docs: norma/partitura; misc for
3797 % the rest) so they print. Per-type ABNT refinements can follow later.
3798 \DeclareBibliographyDriver{standard}{\usedriver{}{book}}
3799 \DeclareBibliographyDriver{music}{\usedriver{}{book}}
3800 % Legal documents (NBR 6023 4.2.6-4.2.8): jurisdiction/entity (CAPS author),
3801 % the law id + ementa (plain, not bold), then location and date.

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```

3802 \newbibmacro*{coppe:legaldriver}{%
3803   \usebibmacro{bibindex}\usebibmacro{begentry}%
3804   \usebibmacro{author/translator+others}%
3805   \setunit{\labelnamepunct}\newblock
3806   \usebibmacro{title}%
3807   \newunit\newblock
3808   % Jurisprudência: "Relator: <nome>." after the decision title (NBR 6023).
3809   \iffielddundef{relator}{\}\printtext{Relator\addcolon\space}\printfield{relator}\newunit\ne
3810   \printlist{location}%
3811   \setunit{\addcomma\space}\printdate%
3812   \newunit\newblock
3813   \usebibmacro{url+urldate}%
3814   \usebibmacro{finentry}}
3815 \DeclareBibliographyDriver{legislation}{\usebibmacro{coppe:legaldriver}}
3816 \DeclareBibliographyDriver{jurisdiction}{\usebibmacro{coppe:legaldriver}}
3817 \DeclareBibliographyDriver{legal}{\usebibmacro{coppe:legaldriver}}
3818 \DeclareBibliographyDriver{audio}{\usedriver{}{misc}}
3819 \DeclareBibliographyDriver{video}{\usedriver{}{misc}}
3820 \DeclareBibliographyDriver{software}{\usedriver{}{misc}}
3821 \DeclareBibliographyDriver{image}{\usedriver{}{misc}}
3822 \DeclareBibliographyDriver{artwork}{\usedriver{}{misc}}
3823 \DeclareBibliographyDriver{performance}{\usedriver{}{misc}}
3824 \DeclareBibliographyDriver{dataset}{\usedriver{}{misc}}
3825
3826 % --- biber source map: Portuguese synonyms (entry types + fields) + thesis family ---
3827 \DeclareSourcemap{%
3828   \maps[datatype=bibtex]{%
3829     % entry-type PT synonyms -> canonical English
3830     \map{step[typesource=livro,typetarget=book]}
3831     \map{step[typesource=artigo,typetarget=article]}
3832     \map{step[typesource=artigodejornal,typetarget=article,final]\step[fieldset=entrysubtyp
3833     \map{step[typesource=partedelivro,typetarget=incollection]}
3834     \map{step[typesource=emlivro,typetarget=inbook]}
3835     \map{step[typesource=verbete,typetarget=inreference]}
3836     \map{step[typesource=anais,typetarget=proceedings]}
3837     \map{step[typesource=trabalhodeevento,typetarget=inproceedings]}
3838     \map{step[typesource=periodico,typetarget=periodical]}
3839     \map{step[typesource=relatorio,typetarget=report]}
3840     \map{step[typesource=relatoriotecnico,typetarget=report]}
3841     \map{step[typesource>manual,typetarget>manual]}
3842     \map{step[typesource=correspondencia,typetarget=letter]}
3843     \map{step[typesource=patente,typetarget=patent]}
3844     \map{step[typesource=diverso,typetarget=misc]}
3845     \map{step[typesource=jogo,typetarget=game]}
3846     \map{step[typesource=jogoeletronico,typetarget=videogame]}
3847     \map{step[typesource=jogodetabuleiro,typetarget=boardgame]}
3848     \map{step[typesource=filme,typetarget=movie]}
3849     \map{step[typesource=serietv,typetarget=tvshow]}
3850     \map{step[typesource=programadetelevisao,typetarget=tvshow]}
3851     % Exotic ABNT types (NBR 6023 4.2.5-4.2.14); targets are biblatex-native
3852     % except map (declared in coppe.dbx).
3853     \map{step[typesource=legislacao,typetarget=legislation]}
3854     \map{step[typesource=atonomativo,typetarget=legislation]}
3855     \map{step[typesource=jurisprudencia,typetarget=jurisdiction]}

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3856 \map{\step[typesource=documentojuridico,typetarget=legal]}
3857 \map{\step[typesource=documentocivil,typetarget=legal]}
3858 \map{\step[typesource=norma,typetarget=standard]}
3859 \map{\step[typesource=partitura,typetarget=music]}
3860 \map{\step[typesource=iconografico,typetarget=image]}
3861 \map{\step[typesource=obradearte,typetarget=artwork]}
3862 \map{\step[typesource=tridimensional,typetarget=artwork]}
3863 \map{\step[typesource=cartografico,typetarget=map]}
3864 \map{\step[typesource=mapa,typetarget=map]}
3865 \map{\step[typesource=audiovisual,typetarget=video]}
3866 \map{\step[typesource=conjuntodedados,typetarget=dataset]}
3867 \map{\step[typesource=fasciculo,typetarget=supperiodical]}
3868 \map{\step[typesource=trabalhoapresentado,typetarget=unpublished]}
3869 % thesis family: retype to thesis and preset the (babel-aware) type word
3870 \map{\step[typesource=tese,typetarget=thesis]}
3871 \map{\step[typesource=dissertacao,typetarget=thesis]}
3872 \map{\step[typesource=masterdissertation,typetarget=thesis,final]\step[fieldset=type,fiel
3873 \map{\step[typesource=mscdissertation,typetarget=thesis,final]\step[fieldset=type,fiel
3874 \map{\step[typesource=dissertacaomestrado,typetarget=thesis,final]\step[fieldset=type,fi
3875 \map{\step[typesource=dscthesis,typetarget=thesis,final]\step[fieldset=type,fieldvalue=d
3876 \map{\step[typesource=tesedoutorado,typetarget=thesis,final]\step[fieldset=type,fieldval
3877 \map{\step[typesource=phdthesis,typetarget=thesis,final]\step[fieldset=type,fieldvalue=p
3878 \map{\step[typesource=tesephd,typetarget=thesis,final]\step[fieldset=type,fieldvalue=phd
3879 \map{\step[typesource=tcc,typetarget=thesis,final]\step[fieldset=type,fieldvalue=tcc]}
3880 % field PT synonyms (unaccented) -> canonical English
3881 \map{\step[fieldsource=periodico,fieldtarget=journaltitle]}
3882 \map{\step[fieldsource=revista,fieldtarget=journaltitle]}
3883 \map{\step[fieldsource=jornal,fieldtarget=journaltitle]}
3884 \map{\step[fieldsource=titulo,fieldtarget=title]}
3885 \map{\step[fieldsource=subtitulo,fieldtarget=subtitle]}
3886 \map{\step[fieldsource=autor,fieldtarget=author]}
3887 \map{\step[fieldsource=editora,fieldtarget=publisher]}
3888 \map{\step[fieldsource=local,fieldtarget=location]}
3889 \map{\step[fieldsource=ano,fieldtarget=year]}
3890 \map{\step[fieldsource=data,fieldtarget=date]}
3891 \map{\step[fieldsource=edicao,fieldtarget=edition]}
3892 \map{\step[fieldsource=paginas,fieldtarget=pages]}
3893 \map{\step[fieldsource=numero,fieldtarget=number]}
3894 \map{\step[fieldsource=tradutor,fieldtarget=translator]}
3895 \map{\step[fieldsource=organizador,fieldtarget=editor]}
3896 \map{\step[fieldsource=instituicao,fieldtarget=institution]}
3897 \map{\step[fieldsource=tipo,fieldtarget=type]}
3898 \map{\step[fieldsource=nota,fieldtarget=note]}
3899 \map{\step[fieldsource=dataacesso,fieldtarget=urldate]}
3900 \map{\step[fieldsource=curso,fieldtarget=course]}
3901 \map{\step[fieldsource=grau,fieldtarget=coppedegree]}
3902 \map{\step[fieldsource=artista,fieldtarget=artist]}
3903 \map{\step[fieldsource=plataforma,fieldtarget=platform]}
3904 \map{\step[fieldsource=diretor,fieldtarget=director]}
3905 \map{\step[fieldsource=produtor,fieldtarget=producer]}
3906 \map{\step[fieldsource=versao,fieldtarget=version]}
3907 \map{\step[fieldsource=escala,fieldtarget=scale]}
3908 }%
3909 }

```

```

3910
3911 % --- ABNT list layout: flush left, single spacing, blank line between entries ---
3912 \setlength{\bibhang}{0pt}
3913 \setlength{\bibitemsep}{\baselineskip}
3914
3915 </bbx>
3916 <*cbx>
3917 \ProvidesFile{coppe.cbx}[2026/09/09 v4.1 CoppeTeX ABNT author-date citations]
3918 % Author-date is the CoppeTeX 4.0 default. authoryear-comp gives compressed
3919 % author-year cites with normal-caps names (post-2020 rule) out of the box.
3920 \RequireCitationStyle{authoryear-comp}
3921
3922 % --- NBR 10520:2023 author separator -----
3923 % Parenthetical citations separate authors with ";" -> (Silva; Souza, 2020).
3924 % Narrative citations (\textcite, natbib \citet) keep comma + "e/and" ->
3925 % Silva, Souza e Oliveira (2020). The reference list keeps its own ";" via
3926 % coppe.bbx ([bib,biblist]); these contextless formats cover the cite side.
3927 \DeclareDelimFormat{multinamedelim}{\addsemicolon\space}
3928 \DeclareDelimFormat{finalnamedelim}{\addsemicolon\space}
3929 \DeclareDelimFormat[textcite]{multinamedelim}{\addcomma\space}
3930 \DeclareDelimFormat[textcite]{finalnamedelim}{\addspace\bibstring{and}\space}
3931
3932 % --- apud: citation of a citation (ABNT NBR 6023 §4.1.2.2 / NBR 10520) -----
3933 % ABNT lists only the work actually CONSULTED. So only the consulted work (#4, a
3934 % .bib key) is cited and enters the reference list; the ORIGINAL work (#2 author,
3935 % #3 year) was not consulted and is given as plain text -- it is never a cite key,
3936 % so it is never added to the bibliography.
3937 % \citetapud[<page>]{<orig author>}{<orig year>}{<consulted key>}
3938 % -> Orig Author (year) apud Consulted (yr[, page]) [narrative]
3939 % \citepapud[<page>]{<orig author>}{<orig year>}{<consulted key>}
3940 % -> (Orig Author, year apud CONSULTED, yr[, page]) [parenthetical]
3941 % #1 = optional postnote (page/loc) on the consulted work; "apud" is latin -> italic.
3942 \DeclareRobustCommand{\citetapud}[4][]{%
3943 #2\space(#3)\space\mkbibemph{apud}\space\textcite[#1]{#4}}
3944 \DeclareRobustCommand{\citepapud}[4][]{%
3945 \mkbibparens{#2\addcomma\space#3\space\mkbibemph{apud}\space
3946 \citeauthor{#4}\addcomma\space\citeyear{#4}}%
3947 \ifblank{#1}{\addcomma\space\mkpageprefix[pagination]{#1}}{}}
3948 </cbx>
3949 <*lbxbr>
3950 \ProvidesFile{brazilian-coppe.lbx}[2026/09/09 v4.1 CoppeTeX brazilian strings]
3951 \InheritBibliographyExtras{brazilian}
3952 \NewBibliographyString{availablefrom,mscdiss,dscthesi,tcc,depositor,coppescale}
3953 \DeclareBibliographyStrings{%
3954 inherit = {brazilian},
3955 availablefrom = {{Dispon'\i vel em}{Dispon'\i vel em}},
3956 urlseen = {{Acesso em}{Acesso em}},
3957 in = {{In}{In}},
3958 editor = {{organizador}{org\adddot}},
3959 editors = {{organizadores}{org\adddot}},
3960 depositor = {{Depositante}{Depositante}},
3961 coppescale = {{Escala}{Escala}},
3962 mscdiss = {{Disserta\c{c}\~{a}o de Mestrado}{Disserta\c{c}\~{a}o de Mestrado}},
3963 dscthesi = {{Tese de Doutorado}{Tese de Doutorado}},

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3964 phdthesis      = {{Tese de Doutorado}{Tese de Doutorado}},
3965 tcc              = {{Trabalho de Conclus\~{a}o de Curso}{Trabalho de Conclus\~{a}o de Curso}}
3966 }
3967 \</lbr>
3968 \<lbren>
3969 \ProvidesFile{english-coppe.lbx}[2026/09/09 v4.1 CoppeTeX english strings]
3970 \InheritBibliographyExtras{english}
3971 \NewBibliographyString{availablefrom,mscdiss,dscthesi,tcc,depositor,coppescale}
3972 \DeclareBibliographyStrings{%
3973   inherit          = {english},
3974   availablefrom    = {{Available from}{Available from}},
3975   urlseen          = {{Access on}{Access on}},
3976   in               = {{In}{In}},
3977   editor           = {{editor}{ed\addot}},
3978   editors          = {{editors}{ed\addot}},
3979   depositor        = {{Depositor}{Depositor}},
3980   coppescale       = {{Scale}{Scale}},
3981   mscdiss          = {{M\addot Sc\addot\ Dissertation}{M\addot Sc\addot\ Diss\addot}},
3982   dscthesi         = {{D\addot Sc\addot\ Thesis}{D\addot Sc\addot\ Thesis}},
3983   phdthesis        = {{Ph\addot D\addot\ Thesis}{Ph\addot D\addot\ Thesis}},
3984   tcc              = {{Undergraduate Thesis}{Undergraduate Thesis}},
3985 }
3986 \</lbren>
3987 \<lbxes>
3988 \ProvidesFile{spanish-coppe.lbx}[2026/09/09 v4.1 CoppeTeX spanish strings]
3989 \InheritBibliographyExtras{spanish}
3990 \NewBibliographyString{availablefrom,mscdiss,dscthesi,tcc,depositor,coppescale}
3991 \DeclareBibliographyStrings{%
3992   inherit          = {spanish},
3993   availablefrom    = {{Disponible en}{Disponible en}},
3994   urlseen          = {{Consultado el}{Consultado el}},
3995   in               = {{En}{En}},
3996   editor           = {{editor}{ed\addot}},
3997   editors          = {{editores}{ed\addot}},
3998   depositor        = {{Solicitante}{Solicitante}},
3999   coppescale       = {{Escala}{Escala}},
4000   mscdiss          = {{Tesis de Maestr'\{i\}a}{Tesis de Maestr'\{i\}a}},
4001   dscthesi         = {{Tesis Doctoral}{Tesis Doctoral}},
4002   phdthesis        = {{Tesis Doctoral}{Tesis Doctoral}},
4003   tcc              = {{Trabajo de Fin de Carrera}{Trabajo de Fin de Carrera}},
4004 }
4005 \</lbxes>
4006 \<lbxfr>
4007 \ProvidesFile{french-coppe.lbx}[2026/09/09 v4.1 CoppeTeX french strings]
4008 \InheritBibliographyExtras{french}
4009 \NewBibliographyString{availablefrom,mscdiss,dscthesi,tcc,depositor,coppescale}
4010 \DeclareBibliographyStrings{%
4011   inherit          = {french},
4012   availablefrom    = {{Disponible \{a\}}{Disponible \{a\}}},
4013   urlseen          = {{Consult\{e\} le}{Consult\{e\} le}},
4014   in               = {{Dans}{Dans}},
4015   editor           = {{\{editeur\}}{\{ed\addot\}}},
4016   editors          = {{\{editeurs\}}{\{ed\addot\}}},
4017   depositor        = {{D\{eposant\}}{D\{eposant\}}},

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4018 coppescale = {{\Echelle}{\Echelle}},
4019 mscdiss = {{M\emoire de Master}{M\emoire de Master}},
4020 dscthesi = {{Th\ese de Doctorat}{Th\ese de Doctorat}},
4021 phdthesis = {{Th\ese de Doctorat}{Th\ese de Doctorat}},
4022 tcc = {{M\emoire de fin d'\etudes}{M\emoire de fin d'\etudes}},
4023 }
4024 \lbrack
4025 \lbrack
4026 \ProvidesFile{italian-coppe.lbx}[2026/09/09 v4.1 CoppeTeX italian strings]
4027 \InheritBibliographyExtras{italian}
4028 \NewBibliographyString{availablefrom,mscdiss,dscthesi,tcc,depositor,coppescale}
4029 \DeclareBibliographyStrings{%
4030 inherit = {italian},
4031 availablefrom = {{Disponibile da}{Disponibile da}},
4032 urlseen = {{Consultato il}{Consultato il}},
4033 in = {{In}{In}},
4034 editor = {{curatore}{cur\adddot}},
4035 editors = {{curatori}{cur\adddot}},
4036 depositor = {{Depositante}{Depositante}},
4037 coppescale = {{Scala}{Scala}},
4038 mscdiss = {{Tesi di Laurea Magistrale}{Tesi di Laurea Magistrale}},
4039 dscthesi = {{Tesi di Dottorato}{Tesi di Dottorato}},
4040 phdthesis = {{Tesi di Dottorato}{Tesi di Dottorato}},
4041 tcc = {{Tesi di Laurea Triennale}{Tesi di Laurea Triennale}},
4042 }
4043 \lbrack
4044 \lbrack
4045 %% coppe-lang-spanish.def -- CoppeTeX 4.x Spanish class-string pack.
4046 %% Loaded by coppe.cls when \documentclass[spanish]{coppe} (or any other
4047 %% spanish slot) is active. Populates the dispatcher table for the
4048 %% spanish language and registers the biblalex language mapping.
4049 %% Institutional names (universityname/cityname/statename/countryname)
4050 %% are deliberately kept Portuguese: the COPPE/UFRJ cover and folha de
4051 %% rosto are a Brazilian institutional template that does not translate.
4052 \ProvidesFile{coppe-lang-spanish.def}[2026/09/09 v4.1 CoppeTeX spanish class strings]
4053 \copperdefstring{spanish}{appendix} {Ap\endice}
4054 \copperdefstring{spanish}{annex} {Anexo}
4055 \copperdefstring{spanish}{frame} {Cuadro}
4056 \copperdefstring{spanish}{listframe} {Lista de Cuadros}
4057 \copperdefstring{spanish}{source} {Fuente}
4058 \copperdefstring{spanish}{program} {Programa}
4059 \copperdefstring{spanish}{listprogram} {Lista de Programas}
4060 \copperdefstring{spanish}{references} {Referencias}
4061 \copperdefstring{spanish}{in:} {en}
4062 \copperdefstring{spanish}{art} {Arte}
4063 \copperdefstring{spanish}{platform} {Plataforma}
4064 \copperdefstring{spanish}{directedby} {Direcci\on}
4065 \copperdefstring{spanish}{producedby} {Producci\on}
4066 \copperdefstring{spanish}{listabbreviation} {Lista de Abreviaturas y Siglas}
4067 \copperdefstring{spanish}{listsymbol} {Lista de S\imbolos}
4068 \copperdefstring{spanish}{glossary} {Glosario}
4069 \copperdefstring{spanish}{advisor} {Director}
4070 \copperdefstring{spanish}{advisors} {Directores}
4071 \copperdefstring{spanish}{dept} {Programa}

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4072 \copperdefstring{spanish}{approvedmale}    {Aprobado por}
4073 \copperdefstring{spanish}{approvedfemale}  {Aprobada por}
4074 \copperdefstring{spanish}{approvedonmale}   {Aprobado el}
4075 \copperdefstring{spanish}{approvedonfemale}{Aprobada el}
4076 \copperdefstring{spanish}{keywordsname}    {Palabras clave}
4077 \copperdefstring{spanish}{volumename}       {Volumen}
4078 \copperdefstring{spanish}{coadvisor}        {Codirector}
4079 \copperdefstring{spanish}{coadvisors}       {Codirectores}
4080 \copperdefstring{spanish}{ofname}           {de}
4081 \copperdefstring{spanish}{universityname}    {Universidade Federal do Rio de Janeiro}
4082 \copperdefstring{spanish}{cityname}         {Rio de Janeiro}
4083 \copperdefstring{spanish}{statename}        {RJ}
4084 \copperdefstring{spanish}{countryname}       {Brasil}
4085 \copperdefstring{spanish}{monthname1}       {enero}
4086 \copperdefstring{spanish}{monthname2}       {febrero}
4087 \copperdefstring{spanish}{monthname3}       {marzo}
4088 \copperdefstring{spanish}{monthname4}       {abril}
4089 \copperdefstring{spanish}{monthname5}       {mayo}
4090 \copperdefstring{spanish}{monthname6}       {junio}
4091 \copperdefstring{spanish}{monthname7}       {julio}
4092 \copperdefstring{spanish}{monthname8}       {agosto}
4093 \copperdefstring{spanish}{monthname9}       {septiembre}
4094 \copperdefstring{spanish}{monthname10}      {octubre}
4095 \copperdefstring{spanish}{monthname11}      {noviembre}
4096 \copperdefstring{spanish}{monthname12}     {diciembre}
4097 \copperdefstring{spanish}{algorithm2eopt}{spanish}
4098 \copperdefstring{spanish}{babelname}        {spanish}
4099 %% biblatex is loaded by coppe.cls AFTER this .def file is read; defer the
4100 %% language-mapping registration until \begin{document} so biblatex's
4101 %% \DeclareLanguageMapping is available.
4102 \AtBeginDocument{\DeclareLanguageMapping{spanish}{spanish-coppe}}
4103 \

```

```

4126 \copperdefstring{french}{advisors}      {Directeurs}
4127 \copperdefstring{french}{dept}          {Programme}
4128 \copperdefstring{french}{approvedmale}   {Approuv\'e par}
4129 \copperdefstring{french}{approvedfemale} {Approuv\'ee par}
4130 \copperdefstring{french}{approvedonmale} {Approuv\'e le}
4131 \copperdefstring{french}{approvedonfemale}{Approuv\'ee le}
4132 \copperdefstring{french}{keywordsname} {Mots-cl\'es}
4133 \copperdefstring{french}{volumename} {Volume}
4134 \copperdefstring{french}{coadvisor} {Co-directeur}
4135 \copperdefstring{french}{coadvisors} {Co-directeurs}
4136 \copperdefstring{french}{ofname} {de}
4137 \copperdefstring{french}{universityname}{Universidade Federal do Rio de Janeiro}
4138 \copperdefstring{french}{cityname} {Rio de Janeiro}
4139 \copperdefstring{french}{statename} {RJ}
4140 \copperdefstring{french}{countryname} {Brasil}
4141 \copperdefstring{french}{monthname1} {janvier}
4142 \copperdefstring{french}{monthname2} {f\'evrier}
4143 \copperdefstring{french}{monthname3} {mars}
4144 \copperdefstring{french}{monthname4} {avril}
4145 \copperdefstring{french}{monthname5} {mai}
4146 \copperdefstring{french}{monthname6} {juin}
4147 \copperdefstring{french}{monthname7} {juillet}
4148 \copperdefstring{french}{monthname8} {ao\^ut}
4149 \copperdefstring{french}{monthname9} {septembre}
4150 \copperdefstring{french}{monthname10}{octobre}
4151 \copperdefstring{french}{monthname11}{novembre}
4152 \copperdefstring{french}{monthname12}{d\'ecembre}
4153 \copperdefstring{french}{algorithm2eopt}{french}
4154 \copperdefstring{french}{babelname} {french}
4155 \AtBeginDocument{\DeclareLanguageMapping{french}{french-coppe}}
4156 \end{langfr}
4157 \end{*langit}
4158 %% coppe-lang-italian.def -- CoppeTeX 4.x Italian class-string pack.
4159 %% See coppe-lang-spanish.def for the structure. The algorithm2e
4160 %% language option for Italian is spelled "italiano" -- that's the
4161 %% spelling algorithm2e ships, even though babel calls the language
4162 %% "italian".
4163 \ProvidesFile{coppe-lang-italian.def}[2026/09/09 v4.1 CoppeTeX italian class strings]
4164 \copperdefstring{italian}{appendix}      {Appendice}
4165 \copperdefstring{italian}{annex}         {Allegato}
4166 \copperdefstring{italian}{frame}         {Riquadro}
4167 \copperdefstring{italian}{listframe}     {Elenco dei riquadri}
4168 \copperdefstring{italian}{source}        {Fonte}
4169 \copperdefstring{italian}{program}       {Programma}
4170 \copperdefstring{italian}{listprogram}   {Elenco dei programmi}
4171 \copperdefstring{italian}{references}    {Riferimenti}
4172 \copperdefstring{italian}{in:}          {in}
4173 \copperdefstring{italian}{art}           {Illustrazione}
4174 \copperdefstring{italian}{platform}     {Piattaforma}
4175 \copperdefstring{italian}{directedby}    {Regia di}
4176 \copperdefstring{italian}{producedby}    {Prodotto da}
4177 \copperdefstring{italian}{listabbreviation}{Elenco di abbreviazioni e sigle}
4178 \copperdefstring{italian}{listsymbol}    {Elenco dei simboli}
4179 \copperdefstring{italian}{glossary}      {Glossario}

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4180 \copperdefstring{italian}{advisor}      {Relatore}
4181 \copperdefstring{italian}{advisors}     {Relatori}
4182 \copperdefstring{italian}{dept}         {Programma}
4183 \copperdefstring{italian}{approvedmale}  {Approvato da}
4184 \copperdefstring{italian}{approvedfemale}{Approvata da}
4185 \copperdefstring{italian}{approvedonmale}{Approvato il}
4186 \copperdefstring{italian}{approvedonfemale}{Approvata il}
4187 \copperdefstring{italian}{keywordsname} {Parole chiave}
4188 \copperdefstring{italian}{volumename} {Volume}
4189 \copperdefstring{italian}{coadvisor}    {Correlatore}
4190 \copperdefstring{italian}{coadvisors}   {Correlatori}
4191 \copperdefstring{italian}{ofname}       {di}
4192 \copperdefstring{italian}{universityname}{Universidade Federal do Rio de Janeiro}
4193 \copperdefstring{italian}{cityname}     {Rio de Janeiro}
4194 \copperdefstring{italian}{statename}    {RJ}
4195 \copperdefstring{italian}{countryname} {Brasil}
4196 \copperdefstring{italian}{monthname1}  {gennaio}
4197 \copperdefstring{italian}{monthname2}  {febbraio}
4198 \copperdefstring{italian}{monthname3}  {marzo}
4199 \copperdefstring{italian}{monthname4}  {aprile}
4200 \copperdefstring{italian}{monthname5}  {maggio}
4201 \copperdefstring{italian}{monthname6}  {giugno}
4202 \copperdefstring{italian}{monthname7}  {luglio}
4203 \copperdefstring{italian}{monthname8}  {agosto}
4204 \copperdefstring{italian}{monthname9}  {settembre}
4205 \copperdefstring{italian}{monthname10}{ottobre}
4206 \copperdefstring{italian}{monthname11}{novembre}
4207 \copperdefstring{italian}{monthname12}{dicembre}
4208 \copperdefstring{italian}{algorithm2eopt}{italiano}
4209 \copperdefstring{italian}{babelname}   {italian}
4210 \AtBeginDocument{\DeclareLanguageMapping{italian}{italian-coppe}}
4211 \</langit>
4212 \<numbbx>
4213 \ProvidesFile{coppe-numeric.bbx}[2026/09/09 v4.1 CoppeTeX ABNT numeric bibliography]
4214 % Numeric reference list for the ABNT numeric system (NBR 10520). It reuses the
4215 % full coppe ABNT style (drivers, CAPS surnames, bold titles, "In:", p./f.,
4216 % the PT->EN source map and the language mappings) and only adds the [n] list
4217 % label and the numeric sorting on top.
4218 \RequireBibliographyStyle{coppe}
4219
4220 % ABNT numeric system lists references in the order they are first cited
4221 % (NBR 10520), i.e. unsorted. Override with sorting=nty for an alphabetical
4222 % numbered list. 'labelnumber' generates the [n] field.
4223 \ExecuteBibliographyOptions{labelnumber,sorting=none}
4224
4225 % Bracketed labels, both in the list and for any shorthand entries.
4226 \DeclareFieldFormat{labelnumberwidth}{\mkbibbrackets{#1}}
4227 \DeclareFieldFormat{shorthandwidth}{\mkbibbrackets{#1}}
4228
4229 % Reference list driven by the [n] label (copied from biblatex's numeric.bbx),
4230 % keeping the ABNT blank line between entries via \bibitemsep set in coppe.bbx.
4231 \defbibenvironment{bibliography}
4232 {list
4233   {\printtext[labelnumberwidth]{%

```

```

4234         \printfield{labelprefix}%
4235         \printfield{labelnumber}}}%
4236     {\setlength{\labelwidth}{\labelnumberwidth}%
4237      \setlength{\leftmargin}{\labelwidth}%
4238      \setlength{\labelsep}{\biblabelsep}%
4239      \addtolength{\leftmargin}{\labelsep}%
4240      \setlength{\itemsep}{\bibitemsep}%
4241      \setlength{\parsep}{\bibparsep}}}%
4242     \renewcommand*{\makelabel}[1]{\hss#1}}
4243 {\endlist}
4244 {\item}
4245
4246 </numbbx>
4247 <*numcbx>
4248 \ProvidesFile{coppe-numeric.cbx}[2026/09/09 v4.1 CoppeTeX ABNT numeric citations]
4249 % Numeric citation system (ABNT NBR 10520): in-text references are bracketed
4250 % numbers [1], compressed and sorted within a single cite group. This is the
4251 % opt-in alternative to the author-date default (coppe.cbx); it pairs with
4252 % bibstyle=coppe-numeric. With the biblatex natbib option, \citep -> [1] and
4253 % \citet -> Author [1].
4254 \RequireCitationStyle{numeric-comp}
4255 </numcbx>

```

10.2 The bibliographic databases

The two BibLaTeX/biber databases shipped with the class — `example.bib` (the sample database cited by the example monograph) and `coppe.bib` (the references of this manual) — are generated from this source too, so the whole distribution still has a single editable origin. They live inside an `\iffalse` block below: the documentation driver never typesets them, and only `coppe.ins` extracts them, through the `examplebib` and `coppebib` docstrip targets. Edit the entries here, never the generated `.bib` files.

10.3 The generated companion files

Everything the distribution ships is extracted from this file: the class, the BibLaTeX styles, the language packs, the databases, the `.ist`, the demonstration documents and the `latexmkrc`. One `pdflatex coppe.ins` rebuilds all of it and there is no second place to edit. Each one sits in an `\iffalse` block: the documentation driver never typesets them, and only `coppe.ins` extracts them.

The line is drawn at what is DISTRIBUTED. What exists only to prove the class works stays outside, as ordinary files under version control:

`tests/` the regression suite. Nobody installing CoppeTeX needs it, and generating it from here would put proof and product in the same box.

`adversativa/` twelve documents — four kinds of work by three languages — each firing everything the class offers at once.

`tools/*.ps1` the build harness itself.

`NORMA_COPPE_2026.tex`, `futuremanual2026.tex` documents about the norm that live in this repository with their own life cycle.

Both suites are run by one command, `tools\prova.ps1`, and that is what has to come out clean before a version is tagged.

10.3.1 example_pt.tex -- pt-main demo

`example_pt.tex` – CoppeTeX 4.0 multilingual demo, main language: brazilian. Short demonstration document (cover + folha + abstracts + 1 chapter) used by the CPGP submission package. The full pt-main example with bibliography and lists is the historical `example.tex`.

10.3.2 example_en.tex -- en-main demo

`example_en.tex` – CoppeTeX 4.0 multilingual demo, main language: english.

10.3.3 example_es.tex -- es-main demo

`example_es.tex` – CoppeTeX 4.0 multilingual demo, main language: spanish. Requires the language pack files `coppe-lang-spanish.def` and `spanish-coppe.lbx` to be installed alongside `coppe.cls`.

10.3.4 example_fr.tex -- fr-main demo

`example_fr.tex` – CoppeTeX 4.0 multilingual demo, main language: french. Requires the language pack files `coppe-lang-french.def` and `french-coppe.lbx` to be installed alongside `coppe.cls`.

10.3.5 example_it.tex -- it-main demo

`example_it.tex` – CoppeTeX 4.0 multilingual demo, main language: italian. Requires the language pack files `coppe-lang-italian.def` and `italian-coppe.lbx` to be installed alongside `coppe.cls`.

10.3.6 example_pdfa.tex -- the full example as PDF/A

`example_pdfa.tex` – o `example.tex` completo, compilado com a opção ‘pdfa’.

Existe para dar ao veraPDF o documento mais exigente que a classe produz: o exemplo inteiro, com bibliografia, listas, figuras, tabelas, algoritmos e as caixas do `tcolorbox` – ou seja, transparência, que é justamente o que o a-1b proibiria e o a-2b admite. Validar só um documento curto não diria nada sobre o que vai de fato para o depósito.

`\PassOptionsToClass` tem de vir ANTES do `\documentclass` que o `example.tex` traz dentro de si; é o único jeito de acrescentar uma opção de classe a um arquivo gerado pelo docstrip sem duplicá-lo aqui e deixar as duas cópias divergirem com o tempo.

10.3.7 covers_5languages.tex -- the five covers side by side

`covers_5languages.tex` – One-page visual demonstration of the CoppeTeX 4.0 multilingual model: the five built-in main languages side by side, all sharing the same Portuguese institutional template, differing only in the title (and the body language, not visible on the cover).

This file uses the article class deliberately – it is a meta-document about the coppe class, not a thesis. It expects the five cover PNGs `cover_pt.png`, `cover_en.png`, `cover_es.png`, `cover_fr.png`, `cover_it.png` to live in the same directory (generated by `pdftoppm` from the `example_<lang>.pdf` demos).

10.3.8 latexmkrc -- latexmk/Overleaf build configuration

latexmkrc – CoppeTeX build configuration (works locally and on Overleaf).

Run with: `latexmk example` (or `latexmk futuremanual2026`, your thesis...) `latexmk` reads this file automatically when it is in the build directory. On Overleaf, place this file at the project root; Overleaf runs `latexmk`.

Engine: pdfLaTeX by default (fast). For LuaLaTeX (recommended, full Unicode) set `$pdf_mode = 4` below, or use the `"% !TeX program = lualatex"` magic comment / the Overleaf "Menu > Compiler" setting.

Acknowledgments

Thanks to all COPPET_EX users who have reported their experience with this class. We also thank to professor Fernando Lizarralde and Heiko Oberdiek for their helpful comments. The authors would like to thank the National Council for Scientific and Technological Development (CNPq) of Brazil.

Index

Numbers written in *italic* refer to the page where the corresponding entry is described; numbers underlined refer to the code line of the definition; numbers in *roman* refer to the code lines where the entry is used.

Symbols		
\%	 856	\@coppecoadvresumofalse 1582
\,	 822, 831	\@coppecoadvresumotrue 1583
\;	 756, 758, 760	\@coppedoublespacefalse 1555
\@M	 1290	\@coppedoublespacetrue 1556
\@Roman	 3111	\@coppeexamefalse 1483
\@advisor	 2118,	\@coppeexametrue .. 1528, 1537
	2122, 2123,	\@coppeglossprefalse 1458
	2124, 2125,	\@coppeglosspretrue 3009, 3036
	2126, 2127,	\@coppelistastocfalse 1595
	2128, 2420,	\@coppelistastotrue 1596
	2572, 2584,	\@coppemorewritesfalse .. 1646, 1656, 1669
	2687, 2714,	\@coppemorewritetrue 1643, 1645
	2849, 2891, 2948	\@coppemwforcedfalse 1644
\@afterheading	... 1180	\@coppemwforcedtrue 1645
\@afterindent	 1104	\@coppenumericfalse 1549
\@afterindentfalse	1105	\@coppenumerictrue 1550
\@authname	... 2148,	\@coppepdfafalse . 1573
	2357, 2391,	\@coppepdfatrue .. 1574
	2535, 2655,	\@copperascunhofalse 1597
	2671, 2673,	\@copperascunhotrue 1598
	2839, 2881,	\@copperomanfalse 1553
	2938, 3337, 3394	\@copperomantrue . 1554
\@author	 2920	\@coppetwosidefalse 1599, 1601
\@authsurn	2149, 2357,	\@coppetwosidettrue 1600
	2391, 2535,	\@date 2921
	2655, 2670,	\@dedic 2797, 2802
	2673, 2839,	\@degree 1516,
	2881, 2938, 3394	1524, 1533,
\@chapapp	 1157	1542, 2215,
\@chapter	 1153	2216, 3278, 3283
\@coadvisor		
	... 2082, 2084,	\@degreename .. 1517,
	2085, 2086,	1525, 1534,
	2087, 2088,	1543, 2217,
	2089, 2092,	2218, 2657,
	2099, 2436,	2703, 3244, 3398
	2443, 2573,	\@dotsep 1296, 1300, 1302
	2597, 2807, 2813	\@dottedtocline .. 1232
\@coppeannexfalse	1146	\@examiner 2133, 2140,
\@coppeannextrue	. 3455	2141, 2142,
\@coppeappendixfalse	 1120	2143, 2144,
\@coppeappendixtrue	 1141	2145, 2574, 2608
\@coppeassinfalse	1575	\@firstoftwo 3586
\@coppeassintrue	. 1576	\@footnotetext
		 1382, 1383
		\@for 1840
		\@highpenalty 1312, 1314
		\@ifstar 3081
		\@labfile 3039,
		3040, 3052,
		3057, 3085, 3090
		\@losfile 3012,
		3013, 3025, 3030
		\@lst .. 1906, 1933, 1942
		\@mainaux .. 3343, 3370
		\@makechapterhead .
		 1177, 1179
		\@nameuse 1709,
		1718, 1719,
		1720, 2059,
		2070, 3066,
		3068, 3077, 3354
		\@pnumwidth 1298
		\@roman 3102
		\@secondoftwo 3586
		\@tempdima 1257, 1258,
		1264, 1265,
		1266, 1267,
		1275, 1276, 1277
		\@tempsymb1 ... 3015,
		3017, 3042, 3044
		\@tocrmarg 1289
		\@topnewpage 1177
		\@twosidettrue 1676, 1682
		\@width 1384
		\@wrlab . 3044, 3051,
		3056, 3084, 3089
		\@wrls 3017, 3024, 3029
		\{ 794, 2790

373, 391, 529,	\coppe@@cur	2363, 2469,
534, 536, 565,	. . . 1724, 1725,	2674, 2704,
574, 586, 614,	1726, 1841,	2841, 2883,
712, 723, 752, 831	1842, 1843,	2940, 3269, 3402
\captionsetup 1332, 1964	1844, 1845, 1847	\coppe@fichafile . .
\chaptermark 1173	\coppe@@englishstr .	 2720,
\citacao . . . 1850, 1853	 1502,	2723, 2783, 2793
\cite . 583, 595, 597,	1687, 1726, 1843	\coppe@fichapage . .
598, 599, 600,	\coppe@@ext . 1436, 1437	. . 2721, 2723, 2793
601, 603, 604,	\coppe@@lang 1840, 1841	\coppe@folio . . 1459,
605, 607, 623,	\coppe@@toc . 1436, 1437	1460, 2269, 2281
625, 626, 627,	\coppe@abntdate . . .	\coppe@foreignlang .
628, 629, 631,	. . 3552, 3561, 3562	. . . 1504, 1506,
632, 633, 635, 643	\coppe@appendixheadings	1508, 1510,
\citeauthor 3946	. . 1133, 1141, 3458	1512, 1514,
\citep 214, 217,	\coppe@areaconc . . .	1692, 1720, 2875
583, 595, 597,	. . . 2197, 2198,	\coppe@glosspre . . 1454
598, 599, 600,	2767, 3250, 3252	\coppe@hasLof
602, 603, 604,	\coppe@babelextra@ 1702	. . 2980, 3126, 3140
606, 607, 643,	\coppe@babelload . .	\coppe@hypersetup .
647, 852, 3410, 4252	. . 1707, 1709, 1710	 2373, 2624
\citepapud	\coppe@babelopts . .	\coppe@linhapesq . .
221, 651, 3939, 3944	 1692,	 2201,
\citet 217, 583,	1704, 1707, 1840	2202, 2412, 2413
623, 625, 626,	\coppe@bancalist . .	\coppe@listtoc
627, 628, 630,	 2560,	. . . 1357, 1377,
631, 632, 634,	2582, 2591,	1452, 1879,
635, 643, 852,	2604, 2614, 2618	2987, 3000, 3183
3410, 3924, 4253	\coppe@bancasize . .	\coppe@lstarrow . . .
\citetapud	. . . 2519, 2572,	. . . 1896, 1897,
221, 651, 3937, 3942	2573, 2574,	1904, 1916,
\citetitle 127	2575, 2577, 2579	1923, 1931, 1940
\citeyear 3946	\coppe@bibBegin 3118,	\coppe@mainBegin . .
\ClassInfo . 1655, 1845	3134, 3149, 3432	. . 2278, 3122, 3137
\ClassWarningNoLine	\coppe@bibEnd . 3114,	\coppe@mainlang . . .
. 1007,	3131, 3149, 3436	. . . 1502, 1503,
1047, 1665, 2250	\coppe@campo	1506, 1508,
\clearname	. . 2728, 2763, 2767	1510, 1512,
. . 3535, 3544, 3667	\coppe@checatipo . .	1514, 1692,
\closeout 3404	 2730, 2743	1708, 1709,
\coadvisor 2083	\coppe@checkwrites .	1719, 2059,
\color 1899,	 1406, 1434	2070, 2076,
1900, 1901,	\coppe@coadvisorrows	2538, 3220, 3390
1902, 1912,	 2805,	\coppe@marcatipo . .
1914, 1920,	2865, 2907, 2964	. . . 2724, 2751,
1921, 1926,	\coppe@dataaprova . .	2752, 2753, 2754
1927, 1928,	. . 2199, 2200, 2553	\coppe@marcavinc . .
1935, 1936, 1937	\coppe@date . . . 964, 967	. . 2726, 2758, 2759
\convertcolorsU . . 1887	\coppe@deptcode 1984,	\coppe@membro . 2520,
\coppe@@algotlang . .	1997, 3396, 3398	2585, 2598, 2609
. . 1957, 1958, 1961	\coppe@docmonth 972,	\coppe@membrogap . .
\coppe@@brazilianstr	. . 2152, 2154, 2157	. . . 2506, 2517,
. . 1686, 1725, 1842	\coppe@docyear	2518, 2522,
	 973, 2153,	2576, 2578, 2580

\coppe@membroid 2499,	\coppe@tocapp 1147, 1162	\copperdefstring ..
2502, 2525, 2530	\coppe@tocline 1286,	 1712,
\coppe@nomeprojeto .	1313, 1316,	1716, 1739,
.. 2207, 2208, 2763	1318, 1320, 1322	1740, 1741,
\coppe@nvolumes 2104,	\coppe@tocnumberline	1742, 1743,
2106, 2109, 2112	 1240,	1744, 1745,
\coppe@orig@appendix	1256, 1430, 1437	1746, 1747,
..... 1139, 1141	\coppe@tocnummax ..	1748, 1749,
\coppe@orig@footnotetext	 1246,	1750, 1751,
..... 1382, 1383	1255, 1258,	1752, 1753,
\coppe@orig@printbibliography	1270, 1273, 1275	1754, 1755,
..... 3433, 3435	\coppe@tocnumsaved .	1756, 1757,
\coppe@orig@starttoc	 1264, 1272	1758, 1759,
..... 1429, 1438	\coppe@tocnumsep ..	1760, 1761,
\coppe@printkeywords	 1247,	1762, 1763,
..... 2186,	1249, 1265, 1276	1764, 1765,
2869, 2912,	\coppe@tocnumwidth .	1766, 1767,
2915, 2971, 2974	... 1245, 1254,	1768, 1769,
\coppe@projetovinc .	1259, 1266,	1770, 1771,
.. 2205, 2206, 2727	1267, 1277, 1293	1772, 1773,
\coppe@selectsubtitle	\coppe@version	1774, 1775,
.. 2070, 2076, 2539	... 964, 966, 2329	1776, 1777,
\coppe@selecttitle .	\coppe@volumeline .	1778, 1779,
... 2059, 2537,	.. 2108, 2360, 2394	1780, 1781,
2673, 3353, 3391	\coppe@wrapstarttoc	1782, 1783,
\coppe@sigla@first .	 1428, 1440	1784, 1785,
..... 3072, 3076	\coppe@xmphook	1786, 1787,
\coppe@sigla@full .	.. 3371, 3374, 3376	1788, 1789,
.. 3067, 3073, 3080	\coppe@xmpkeys 3377,	1790, 1791,
\coppe@sigla@norm .	3381, 3384, 3395	1792, 1793,
..... 3074, 3081	\coppe@xmpout . 3370,	1794, 1796,
\coppe@sigla@reg ..	3373, 3387,	1797, 1798,
..... 3064, 3073	3391, 3393,	1799, 1800,
\coppe@sigla@star .	3395, 3397,	1801, 1802,
..... 3078, 3081	3400, 3402, 3404	1803, 1805,
\coppe@sigla@undef .	\coppefinalengine .	1806, 1807,
.. 3069, 3075, 3079	 2298, 2326	1808, 1809,
\coppe@starttoc .. 1614	\coppefinalfont ...	1810, 1811,
\coppe@thisvolume .	 2305, 2327	1812, 1813,
.. 2105, 2107, 2111	\coppefinalmanual .	1814, 1815,
\coppe@tipoproducao	 2306, 2331	1816, 1817,
... 2203, 2204,	\coppeforeignstring	1818, 1819,
2725, 2731,	 1715,	1820, 1821,
2732, 2733,	1720, 2100,	1822, 1823,
2734, 2735, 2737	2102, 2912, 2915	1824, 1825,
\coppe@title@brazilian	\coppemainstring ..	1826, 1827,
..... 3357	 1715,	1828, 1830,
\coppe@titleblock .	1719, 1954,	1831, 1833,
.. 2074, 2359, 2393	1957, 2093,	1834, 4053,
\coppe@tmp ... 2585,	2095, 2111,	4054, 4055,
2592, 2598,	2112, 2869,	4056, 4057,
2605, 2609,	3213, 3214,	4058, 4059,
2615, 3383, 3385	3215, 3223,	4060, 4061,
\coppe@tocanx 1150, 1160	3225, 3228, 3230	4062, 4063,

4064,	4065,	4191,	4192,	\DeclareBibliographyDriver
4066,	4067,	4193,	4194,	... 3633, 3690,
4068,	4069,	4195,	4196,	3691, 3698,
4070,	4071,	4197,	4198,	3745, 3746,
4072,	4073,	4199,	4200,	3747, 3748,
4074,	4075,	4201,	4202,	3749, 3753,
4076,	4077,	4203,	4204,	3779, 3798,
4078,	4079,	4205,	4206,	3799, 3815,
4080,	4081,	4207, 4208, 4209		3816, 3817,
4082,	4083,	\coppestring		3818, 3819,
4084,	4085,	... 1148, 1151,		3820, 3821,
4086,	4087,	1347, 1348,		3822, 3823, 3824
4088,	4089,	1364, 1714,	\DeclareBibliographyStrings	
4090,	4091,	1718, 1874,	... 3953, 3972,	
4092,	4093,	1875, 2968,	3991, 4010, 4029	
4094,	4095,	2971, 2974,	\DeclareDatamodelEntryfields	
4096,	4097,	3421, 3425,	... 3479, 3480,	
4098,	4109,	3426, 3453,	3482, 3483, 3484	
4110,	4111,	3697, 3720,	\DeclareDatamodelEntrytypes	
4112,	4113,	3722, 3724, 3726	... 3468	
4114,	4115,	\CoppeTeX 165,	\DeclareDatamodelFields	
4116,	4117,	944, 990, 1472, 2328	... 3470, 3471,	
4118,	4119,	\coppetexfinalpage .	3472, 3473,	
4120,	4121,	... 950, 965, 2298	3474, 3475,	
4122,	4123,	\coppeUC@w	3476, 3477, 3478	
4124,	4125,	.. 3583, 3584, 3585	\DeclareDelimFormat	
4126,	4127,	\coppeUCifart	... 3517,	
4128,	4129,	.. 3574, 3580, 3581	3518, 3927,	
4130,	4131,	\coppeUCkeep .. 3570,	3928, 3929, 3930	
4132,	4133,	3576, 3578, 3579	\DeclareFieldFormat	
4134,	4135,	\coppeUCstop	... 3588,	
4136,	4137,	... 3572, 3573,	3596, 3600,	
4138,	4139,	3575, 3576,	3601, 3604,	
4140,	4141,	3577, 3578, 3579	3607, 3608,	
4142,	4143,	\coppeUCtitle . 3567,	3609, 3610,	
4144,	4145,	3572, 3594, 3598	3613, 3614,	
4146,	4147,	\coppeUCtitleA	3615, 3618,	
4148,	4149,	... 3572, 3573	3619, 3625,	
4150,	4151,	\coppeUCtitleB	3694, 3697,	
4152,	4153,	... 3575, 3577	3752, 4226, 4227	
4154,	4164,	\cpfonte 229, 1362, 1368	\DeclareFixedFont .	
4165,	4166,	\cpsource	... 1861, 1862	
4167,	4168,	229, 1362, 1365,	\DeclareLanguageMapping	
4169,	4170,	1368, 1369, 1370	... 3509,	
4171,	4172,	\cpsourcenam 1364, 1367	3510, 4101,	
4173,	4174,		4102, 4155, 4210	
4175,	4176,		\DeclareMathOperator	
4177,	4178,		.. 12, 13, 804, 805	
4179,	4180,	D	\DeclareNameAlias .	
4181,	4182,	\Dados 754	... 3513,	
4183,	4184,	\dataaprovacao	3514, 3515, 3516	
4185,	4186,	... 2200, 2551	\DeclareRobustCommand	
4187,	4188,	\Date 3402	... 3942, 3944	
4189,	4190,	\date .. 45, 968, 970,	... 3827	
		2151, 2465, 2928	\DeclareSourcemap	

\dedication ... 69, <u>2796</u>	\Enquanto 749	2830, 2861,
\defbibenvironment 4231	\enquote 203, 1850, 1853	2903, 2917,
\defbibheading ... 3424	\enspace 2189, 2552,	2960, 2976,
\DefineVerbatimEnvironment	2729, 2751,	2985, 2991,
..... 1946, 1952	2752, 2753,	2998, 3005,
\department	2754, 2758, 2759	3178, 3186,
... 44, <u>1984</u> , 3335	environments:	3194, 3202,
\det 779	abstract <u>2831</u>	3212, 3289,
\detokenize	brazilianabstract	3323, 3324,
... 3099, 3116,	 <u>2929</u>	3375, 3382,
3117, 3120,	foreignabstract <u>2872</u>	3405, 3417, 3586
3121, 3124,	longquote <u>3150</u>	\fichacatalografica
3125, 3128,	\es@scroman 3103	... 64, <u>1560</u> , <u>2720</u>
3129, 3133,	\examiner 39,	\filcenter . 1135, 1185
3136, 3139, 3142	40, 41, 42, 43, <u>2133</u>	\floatname . 1350, 1375
\diag .. 13, 805, 810, 811	\ExecuteBibliographyOptions	\floatstyle . 1341, 1374
\dimexpr 2529	 3499, 4223	\fonte 229, 255, 285,
\directlua . 1407, 1651		310, 344, 363,
\document 1260	F	383, 519, 538,
\dots 870	\familydefault .. 24, 25	567, 610, 638,
\draw 563	\fancyhead ... 1460,	714, 725, 1360, 1370
	1678, 2269, 2281	\foo 867
E	\fancyhf 1460,	\footnote
\else 1038, 1123, 1161,	2269, 2273, 2281	. 174, 180, 835, 852
1164, 1167,	\fancypagestyle 1460,	\footnotemark 835
1170, 1178,	2269, 2273, 2281	\footnoterule 1384
1287, 1407,	\fi 1009, 1056, 1123,	\footnotetext 835
1408, 1467,	1163, 1166,	\foreign@ 1493
1657, 1664,	1169, 1172,	\foreign@advisorstring
1725, 1726,	1181, 1258,	.. 2130, 2888, 3218
1842, 1843,	1268, 1274,	\foreign@coadvisorstring
1960, 2094,	1279, 1280,	 2098, 2907
2101, 2272,	1299, 1420,	\foreign@degname ..
2294, 2425,	1421, 1422,	... 1519, 1527,
2448, 2489,	1437, 1453,	1536, 1545,
2527, 2577,	1469, 1470,	2221, 2222, 3283
2579, 2639,	1471, 1670,	\foreign@deptname .
2692, 2786,	1671, 1672,	 2001,
2818, 2854,	1684, 1729,	2005, 2009,
2896, 2914,	1849, 1962,	2013, 2017,
2953, 2973,	2096, 2103,	2021, 2025,
2983, 2996,	2117, 2131,	2029, 2033,
3176, 3184,	2156, 2159,	2037, 2041,
3200, 3212,	2192, 2196,	2045, 2049, 2909
3289, 3320,	2256, 2275,	\foreign@deptstring
3372, 3414, 3586	2290, 2296,	 2909, 3234
\emergencystretch .	2299, 2300,	\foreign@doctype ..
..... 1110, 1113	2301, 2316,	... 1521, 1530,
\endfirsthead 393	2317, 2432,	1539, 1547,
\endfoot 401	2455, 2459,	2225, 2226, 3281
\endhead 398	2491, 2532,	\foreign@monthname .
\endlastfoot 406	2581, 2645,	 <u>2157</u> , 2883
\endtitlepage 2477	2699, 2792,	\foreign@title 2056,
\endwrite 3098	2825, 2829,	2879, 2922, 2923

<code>foreignabstract</code>	<code>\hss</code> 1298, 4242	<code>\iflistundef</code>
<code>(env.)</code> 2872	<code>\htmlstyle</code> . . 1918, 1924	.. 3719, 3723, 3725
<code>\foreignabstract@toptext</code>		<code>\ifLuaTeX</code> 2300
.. 2876, 3280		<code>\ifnameundef</code> . . 3531,
<code>\foreignkeyword</code> . . .	I	3540, 3589,
.. 2167, 2170, 2175	<code>\i</code> 2329, 2752, 3955, 4000	3597, 3663, 3771
<code>\foreignsubtitle</code> . .	<code>\if@coppeannex</code>	<code>\ifodd</code> 1467, 2314
.. 2061, 2066	 1146, 1159	<code>\ifPDFTeX</code>
<code>\foreigntitle</code>	<code>\if@coppeappendix</code> . .	.. 1005, 1036, 2299
.. 30, 1732, 2055	.. 1120, 1123, 1158	<code>\ifstrempty</code> . . 3015, 3042
<code>\freeconfig</code> 2214	<code>\if@coppeassin</code>	<code>\ifuseeditor</code> 3531, 3540
<code>\frontcover@maintext</code>	 1575, 2521	<code>\ifXeTeX</code> 2301
.. 2403, 2546, 3243	<code>\if@coppecoadvresumo</code>	<code>\ignorespaces</code> 3168
<code>\frontmatter</code>	 1582, 2806	<code>\ignorespacesafterend</code>
68, 1393, 2231,	<code>\if@coppedoublespace</code>	 3171
2236, 2370, 2473	 1555, 3289	<code>\in@</code> 3585
<code>\frontpage@bottomtext</code>	<code>\if@coppeexame</code>	<code>\index</code> 848
.. 2622, 3266	 1483, 2489	<code>\InheritBibliographyExtras</code>
<code>\frontpage@maintext</code>	<code>\if@coppeglosspre</code> . .	 3951, 3970,
.. 2498, 3255	 1458, 3182	3989, 4008, 4027
	<code>\if@coppelistastoc</code> . .	<code>\InputIfFileExists</code> .
G	 1453, 1595	 1727, 1844
<code>\g@addto@macro</code> 2560,	<code>\if@coppemorewrites</code>	<code>\int</code> 822
2591, 2604,	.. 1408, 1643, 1650	<code>\interlinepenalty</code> 1290
2614, 3381, 3384	<code>\if@coppemwforced</code> . .	<code>\itemsep</code> 4240
<code>\ge</code> 795	 1644, 1660	<code>\itshape</code> 1199, 1208,
<code>\geometry</code> 1683	<code>\if@coppenumeric</code> . .	1320, 1322, 3166
<code>\glossaryname</code> . 3008,	 1549, 3411	
3035, 3179,	<code>\if@coppepdfa</code> 1573, 3296	J
3180, 3181,	<code>\if@copperascunho</code> . .	<code>\javaexternal</code> 1944
3183, 3185, 3215	 1597, 2784	<code>\javainline</code> . . 748, 1943
	<code>\if@copperoman</code> 1553,	<code>\javastyle</code> . . . 1934,
H	2249, 2268, 2290	1941, 1943, 1944
<code>\hat</code> 809	<code>\if@coppetwoside</code> . .	
<code>\hb@xt@</code> 1259, 1298	 1599, 1681	K
<code>\hbar</code> 8, 765, 1050	<code>\if@english</code> . . 1498, 1691	<code>\keyword</code> 47, 48,
<code>\headheight</code> . . 1082, 1442	<code>\if@filesw</code> 1271	49, 2160, 3327, 3335
<code>\headrulewidth</code> 1461,	<code>\if@mainmatter</code> . . . 1155	<code>\Keywords</code> 3395
2270, 2273, 2282	<code>\if@openright</code> 2292	
<code>\hfil</code> 1259	<code>\if@twoside</code> . . 1466, 2313	
<code>\hfill</code> 531,	<code>\ifbibstring</code> 3694	L
1297, 2348, 2386	<code>\ifblank</code> 3575,	<code>\l@chapter</code> 1311
<code>\hologo</code> . . . 180, 552,	3576, 3578, 3947	<code>\l@figure</code> 1354
579, 581, 765,	<code>\ifboolexpr</code> . . . 3531,	<code>\l@paragraph</code> 1321
816, 857, 989,	3540, 3589, 3597	<code>\l@quadro</code> . . 1353, 1354
991, 992, 993,	<code>\ifdefined</code> 1407, 1651,	<code>\l@section</code> 1315
994, 995, 996,	3318, 3321, 3369	<code>\l@subsection</code> 1317
997, 998, 999,	<code>\ifdim</code> 1258,	<code>\l@subsubsection</code> . 1319
2299, 2300, 2301	1266, 1270, 1277	<code>\labelnamepunct</code> . . .
<code>\hookrightarrow</code> . . .	<code>\iffieldequalstr</code> . 3642	.. . 3637, 3672,
.. 1885, 1897	<code>\iffieldundef</code> . 3553,	3701, 3730,
<code>\href</code> 135, 137, 844	3554, 3555,	3756, 3783, 3805
<code>\hrule</code> 1384	3643, 3721, 3809	<code>\labelnumberwidth</code> 4236
<code>\hskip</code> 1282, 1294	<code>\IfFileExists</code> 1658	<code>\lambda</code> 810, 811
	<code>\ifin@</code> 3586	<code>\language</code> 1718, 2969

2044, 2048, 3398	1549, 1553,	\parbox 2787
\midrule 303,	1555, 1573,	\parfillskip 1289
338, 357, 378, 624	1575, 1582,	\parsep 3159, 4241
\MiKTeX 999	1595, 1597,	\pdfLaTeX 995
\mkbibbold . . . 3591,	1599, 1643, 1644	\pdfsuppressptexinfo
3601, 3604,	\newlength . . . 1245,	. . . 3305, 3318, 3319
3608, 3609, 3610	1246, 1247,	\pdfTeX 996
\mkbibbrackets	2510, 2517, 3150	\pdfvariable . . 3307,
. 4226, 4227	\newlist 1388	3314, 3321, 3322
\mkbibdatelong . . . 3561	\newsavebox 1896	\penalty 1314
\mkbibdateshort . . 3562	\newsigla	\phantom . . . 2725, 2727
\mkbibemph . . 3943, 3945	901, 902, 904, 3061	\printbibliography .
\mkbibmonth . . 3551, 3554	\newunit 3639,	 655,
\mkbibnamefamily . .	3653, 3655,	919, 3145, 3428,
. 3559, 3775	3674, 3678,	3431, 3433, 3434
\mkbibnameprefix . .	3680, 3684,	\printdate 3645, 3646,
. 3560, 3776	3687, 3703,	3652, 3683,
\mkbibparens	3705, 3709,	3708, 3737,
. . . 3534, 3543,	3711, 3713,	3761, 3791, 3811
3666, 3697, 3945	3732, 3734,	\printfield . . . 3640,
\mkern 1296	3738, 3740,	3644, 3647,
\mkpageprefix 3947	3742, 3758,	3648, 3649,
\multicolumn	3762, 3764,	3650, 3651,
. 395, 400, 404	3785, 3787,	3654, 3677,
	3790, 3792,	3679, 3685,
	3807, 3809, 3812	3686, 3704,
N	\nobreak 1294, 1295, 1297	3710, 3712,
\NewBibliographyString	\nobreakspace	3722, 3739,
. . . 3952, 3971,	. . . 1149, 1152, 3553	3741, 3763,
3990, 4009, 4028	\node . . 561, 562, 702, 705	3788, 3789,
\newbibmacro . . 3662,	\nomeprojeto . . 57, 2197	3809, 4234, 4235
3669, 3718,	\nopagebreak 2525	\printindex 848
3727, 3770, 3802	\normalfont	\printlist 3641, 3681,
\newblock 3637, 3639,	. . . 1122, 1135,	3682, 3706,
3653, 3655,	1185, 1193,	3707, 3720,
3672, 3674,	1196, 1199,	3724, 3726,
3678, 3680,	1208, 1316,	3735, 3736,
3684, 3687,	1318, 1320, 1322	3759, 3760, 3810
3701, 3703,	\not 3165	\printloabbreviations
3705, 3709,	\null 1294, 2319	 117, 3034
3711, 3713,	\numberline	\printlosymbols . . .
3730, 3732,	. . . 1165, 1282,	 118, 3007
3734, 3738,	1430, 1437, 1695	\printnames . . . 3532,
3740, 3742,		3541, 3665, 3777
3756, 3758,	O	\printtext 3534, 3543,
3762, 3764,	\OR . . . 2732, 2733, 2734	3622, 3666,
3783, 3785,	\or 2154, 2155, 2156,	3720, 3722,
3787, 3790,	2157, 2158, 2159	3724, 3726,
3792, 3805,		3773, 3809, 4233
3807, 3809, 3812		\projetovinculado .
\newcoppefloat	P	 56, 2197
. . . 231, 1371, 1373	\PackageWarning . . 3070	\prologstyle 1925, 1932
\newfloat . . . 1349, 1374	\Para 749, 757	\protected . . . 3552,
\newif . . . 1120, 1146,	\paragraph . . . 1204,	3572, 3573,
1458, 1483,	1207, 1209, 1210	

3577, 3579, 3581	\RMSE 12, 804, 809	\singlespacing 1367, 1383,
\protected@edef 2563, 2585,	\romannumeral 3107, 3125, 3129	2189, 2264,
2598, 2609, 3383	\rule 2524	2345, 2351,
\providecommand 1347,		2380, 2403,
1348, 1351,	S	2411, 2546,
1716, 1718,	\sbox 1894, 1897	2618, 2744,
1719, 1720,	\section 157, 168,	3164, 3488, 3557
1723, 1734,	176, 200, 234,	\smallskipamount . 1366
2215, 2217,	292, 313, 521,	\source 229, 231, 267,
2219, 2221,	544, 554, 572,	574, 739, 1324,
2223, 2225,	641, 657, 728,	1360, 1369, 1372
2298, 2305, 2306	767, 774, 782,	\step 3830, 3831, 3832,
\ProvidesFile 3462, 3487,	791, 800, 814,	3833, 3834,
3917, 3950,	829, 833, 837,	3835, 3836,
3969, 3988,	841, 846, 850,	3837, 3838,
4007, 4026,	856, 866, 869,	3839, 3840,
4052, 4108,	872, 875, 881, 1192	3841, 3842,
4163, 4213, 4248	\selectcolormodel 1887	3843, 3844,
\Publisher 3400	\selectlanguage 1714, 3166	3845, 3846,
\pythonexternal 1908	\sep 3381	3847, 3848,
\pythoninline 741, 748, 1907	\SetAlCapFnt 1974	3849, 3850,
\pythonstyle 1898,	\SetAlCapNameFnt . 1975	3853, 3854,
1905, 1907, 1908	\SetAlCapNameSty . 1977	3855, 3856,
	\SetAlCapSty 1976	3857, 3858,
	\SetAlgoCaptionSeparator	3859, 3860,
Q	 1973	3861, 3862,
\quadroautorefname 1351	\SetKwFor 1978, 1979	3863, 3864,
\quadroname 1347, 1350, 1351	\setlist 1389, 1390	3865, 3866,
	\setmathfont 11, 820, 1053	3867, 3868,
R	 2322	3870, 3871,
\raggedright 2416, 2439,	\setstretch 1257	3872, 3873,
2683, 2809,	\settowidth 3533, 3542,	3874, 3875,
2845, 2887, 2944	3637, 3641,	3876, 3877,
\recuolongquote 3150, 3151, 3155	3644, 3645,	3878, 3879,
\refstepcounter 1156	3646, 3647,	3881, 3882,
\renewbibmacro 3530, 3539, 3622	3648, 3649,	3883, 3884,
\RequireBibliographyStyle	3650, 3651,	3885, 3886,
. 3495, 4218	3652, 3666,	3887, 3888,
\RequireCitationStyle	3668, 3672,	3889, 3890,
. 3920, 4254	3682, 3683,	3891, 3892,
\resizebox 347, 353	3686, 3701,	3893, 3894,
\restylefloat 1342, 1343	3704, 3707,	3895, 3896,
\Resultado 755	3708, 3719,	3897, 3898,
\Retorna 760	3721, 3723,	3899, 3900,
\right 797	3725, 3730,	3901, 3902,
\rightmargin 3156	3736, 3737,	3903, 3904,
\rightskip 1289	3756, 3760,	3905, 3906, 3907
\rmdefault 25	3761, 3783,	\stretch 2113,
	3789, 3805, 3811	2356, 2358, 2361
	\sfdefault 24	\stripzeros . 3553, 3555
	\sigla 904, 3061	\Subject 3397
		\subsection 324,
		645, 649, 653,

661, 672, 681, 1195	\title 29, 1732,	3691, 3699,
\subsubsection	2051, 3326, 3334	3700, 3702,
1198, 1210	\titleformat . . 1121,	3714, 3715,
\subtitle . . . 2060, 2064	1134, 1184,	3728, 3729,
\subtitledin . . 2061, 2068	1192, 1195,	3731, 3733,
\syml 193, 194, 884,	1198, 1207, 1211	3743, 3744,
885, 886, 887,	\titlein . . . 1730, 1734	3745, 3746,
888, 889, 890, 3014	\titlespacing	3747, 3748,
	. . . 1125, 1138,	3749, 3754,
	1186, 1206, 1209	3755, 3757,
T	\today 969, 2327	3765, 3766,
\tablename 396	\topmargin . . 1070, 1077	3780, 3781,
\tableofcontents . .	\toprule 301,	3782, 3784,
119, 1612	336, 355, 376, 622	3786, 3793,
\text 795, 796	\topsep 3160	3794, 3803,
\textcite	\ttb 1861, 1900, 1901,	3804, 3806,
583, 643, 3924, 3943	1912, 1914,	3813, 3814,
\textcolor . . 1885, 1897	1920, 1927,	3815, 3816, 3817
\textendash	1928, 1936, 1937	\usebox . . 1904, 1916,
. . . 1123, 1136,		1923, 1931, 1940
1149, 1152, 1973	\ttfamily 1863, 1899,	\usecoppellanguage .
\textsc 2323	1910, 1919,	 1495, 1723
\thechapter . . . 1123,	1926, 1935, 1953	\usedriver 3798, 3799,
1136, 1157,	\ttm 1862, 1898, 1909,	3818, 3819,
1160, 1162, 1165	1918, 1925, 1934	3820, 3821,
\theFancyVerbLine 1953		3822, 3823, 3824
\thefield	U	
. . 3553, 3554, 3555	\uppercase 3104	
\theparagraph 1208	\url 131, 843, 2330, 3625	V
\thesection 1193	\usebibmacro . . 3534,	\varnothing 8, 765, 1050
\thesubsection . . . 1196	3543, 3634,	\volume 2107
\thesubsubsection . .	3635, 3636,	\volumes 2106
1199, 1211	3638, 3656,	
\thetable 396	3657, 3666,	
\times 2725, 2727	3670, 3671,	X
\tipoproducao	3673, 3675,	\XeLaTeX 998
. . . 55, 2197, 2736	3676, 3688,	\xmlstyle . . 1909, 1917
\Title 3391	3689, 3690,	\xspace 867

Change History

v0.0		v3.1	
General: Creation Date.	1	\department: Included a sort key in symb	54
v0.1		v3.2	
General: Documentation:		General: Fixed version problem between cls and dtx.	1
bibliography fixed, title translation.	1	v3.3	
Sourceforge submission.	1	General: Extend abbrev to work like symb and accept a sorting key .	1
v0.2		v3.4	
General: Unification of the code for the list of symbols and abbreviations.	1	General: Some examples for figures, tables, longtables, etc. .	1
v0.3		v3.4.1	
General: Added 'draft' option. . . .	1	General: Fix english option to consider brazilian a second language, describe it in document	53
\maketitle: Added number of examiners test.	64	v3.5	
Generalization.	64	General: New command annex . .	100
v0.4		v3.5.1	
General: Beta documentation. . . .	1	General: New command annex now supports brazilian or english	100
v0.5		v4.0	
abstract: Changed from macro to environment.	71	General: Catalog-label writers (coppe@mainBegin, coppe@bibBegin, coppe@bibEnd, coppe@hasLof) made language-agnostic: detokenize wrap on label names protects ':' and '.' from babel shorthand activation, and the raw etex primitives romannumeral/number replace LaTeX's roman/arabic to bypass spanish-babel's small-caps @roman redefinition that would otherwise crash pdftex with "Unbalanced write command" on every es-main document.	1
\backmatter: Added mainmatter pages counter.	60	Class-wide string dispatcher (copperdefstring, coppestring, coppemainstring, coppeforeignstring) replaces scattered iflanguage and if english branches; pt/en outputs remain byte-identical. .	1
foreignabstract: Changed from macro to environment.	72	Multilingual model: three fixed slots (main, foreign, optional third); five built-in language options (brazilian, english,	
v1.0			
General: First COPPETEX release. .	1		
v2.0			
General: COPPETEX release 2.0. . .	1		
v2.1			
General: COPPETEX release 2.1:			
Matching the new rules.	1		
\advisor: Advisors, co-advisors, co-co-advisors, etc., all of them are simply considered advisors.	56		
v2.2			
General: Matching new guidelines, including new logo.	1		
v2.2.2			
General: Fixed some text constants in .bib and documented it here.	1		
v3.0			
General: Added support for monographs in English.	53		
New approval page layout. . . .	66		
\department: Added new course on Nanotechnology.	54		
\examiner: Examiners expansion without degree.	56		

spanish, french, italian); plug-in mechanism for any other Babel language via coppe-lang-lang.def plus lang-coppe.lbx.	1	doublespacing, no folio is printed on a pre-textual sheet when something does overflow, the approval sheet holds a board of eight, and es-nosectiondot removes the period babel-spanish put in the sumario indicative.	1
New user API: titlein, usecoppelanguage, brazilianabstract environment. .	1		
\coppetexfinalpage: Initial coppetexfinalpage colophon page.	61	New options: pdfa (PDF/A-2b via pdfx, XMP built from the document), assinaturas, coorientador, rascunhoficha, listasnosumario, morewrites/semmorewrites. New command coadvisor. Under pdfTeX the class loads morewrites by itself when installed, because a work that asks for every list needs seventeen of TeX's sixteen output streams.	1
v4.1 General: Conformance with the revised 9th edition (2026) of the UFRJ/SiBI manual: continuous pagination from the folha de rosto, folio in 10pt at 2cm from the top and right edges, one-sided layout with a 3cm left margin, additional sheet with the catalog card and the CAPES collection fields, approval sheet of 3.1.2.1.3, keywords closing the three abstracts, pre-textual lists out of the sumario, Apendice and Anexo headings centred, fifth heading level, and Latin Modern instead of the bitmap fonts.	1	Single source: pdflatex coppe.ins generates everything distributed – class, biblatex styles, language packs, .bib bases, .ist, the five per-language examples, example_pdfa, the cover montage and latexmkrc. The regression suite, the adversarial documents and the harness prove the class and are NOT distributed.	1
Fixes found by the adversarial review: cover and folha de rosto compose in single spacing so they do not overflow under			